

AN INTRODUCTION TO STATISTICS USING R



BY

ALESSANDRO GUBBIOTTI

Department of Mathematics
Higher School of Economics

September 15, 2026



*Illustration from Alexander Pushkin's "Eugene Onegin."
The poem has been object of the first statistical study
involving Markov Chains.*

0.1 INTRODUCTION

0.1.1 Scope of the Lecture Notes

The first part of the present lecture notes is an introduction to probability theory. Many texts that undertake this task exist and it would be pretentious to think that these notes are particularly better than other existing ones. However, if the notes are oriented towards a mathematical audience, they often omit to explain the reason why objects are introduced and why measure theory is used as a framework to model probability theory. As an example, the notion of sample space is considered as a god given notion, while from a probabilistic point of view it is simply the set of some particular events, the elementary events. The goal of the first part of lecture notes is to fill this gap by providing a mathematically sound introduction (though only in the discrete setting) to probability theory from the point of view of probability theory, rather than from the measure theoretic point of view. Therefore, the target audience for this part are novice to the field but also those who, having taken a probability theory course, or, better said, a measure theory course, would like to take this notes as a starting point to rethink at the already known notions from a more probabilistic perspective. The second part consist in a brief and rigorous introduction to statistics. Statistics is a wide subject whose claims are often misinterpreted due to lack of rigor. The aim of these notes is to introduce simple models, and to fully develop them pointing out the critical points and statements. The target audience are novice to the field and mathematician wishing to see some applications of statistics without delving into the general theory.

0.1.2 Models and claims of empirical sciences

ity' motivation

One of the pillars of any science is the scientific method. To validate its claims, the measurable outcomes of a scientific theory must be compared with the data that can be collected in the real world. While there are discussions on how to infer information from the collected data, there is no discussion on the fact that a theory must prove its correctness through quantitative considerations.

When dealing with more quantitative, low level sciences such as physics, claims are very precise (think at $f = ma$ or $f = Gm_1m_2/r^2$). Surprisingly (see [wigner]), those kind of claims involve few variables and, up to a certain degree of precision of the measuring instrument, do not involve uncertainty and are universally valid. To validate the formula $f = ma$ it is sufficient to measure the force, the mass and the acceleration of an object and to check whether the relation $f = ma$ holds.

The situation changes completely when dealing with more high level, involved and empirical sciences, such as economics, linguistics or social sciences. As the next example shows, most of the times, constitutive equations such as $f = ma$ - the ultimate goal of any science - have to be abandoned in favour of empirical relations which are true only to some extent and do

involve probability theory.

Indeed, consider a linguistic experiments in which respondents are subjected to a stimulus specifically studied to make them say a word. Suppose that this word can be spelled in two different ways, a formal standard variant and a dialectal variant, and the experiment aims to study the factors determining the choice. What would correspond to a deterministic, fundamental relation, and a complete solution to the research problem would be a set of variables $v_1, v_2, \dots$ and a function that relates the variables to the choice made by the respondent. Concretely, examples of such variables could be v_1 , the age of the respondent, v_2 , his/her degree of education, v_3 , the city where he/she was born, and other variabls could indicate the context in which the respondent is found when saying the word, his/her family education. A fundamental relation would then be a function f that associates to each possible values the variables $v_1, \dots$, can attain the variant used by a respondant for which the variables $v_1, \dots$, namely

$$\begin{aligned} f : \Omega &\rightarrow \{\text{Formal, Dialectal}\} \\ v_1, v_2, \dots &\rightarrow f(v_1, v_2, \dots) = \text{The variant the respondent will use,} \end{aligned} \quad (1) \quad \boxed{\text{e:impossible}}$$

where Ω is the set of all the possible values for the variables $v_1, v_2, \dots$.

Even assuming that such relation could in principle be found, it is in practice impossible to obtain and useless, since most of the variables would be unknown to us and the function f would be too complicated. A possible alternative description is given by the following probabilistic model. We loose track of all the variables with the exception of the ones we consider more relevant regard for the study, say v_1, v_2 and v_3 . The respondenti will then use the formal spelling with probability $p = p(v_1, v_2, v_3)$, that is.

$$f_{\text{eff}}(v_1, v_2, v_3) = \begin{cases} \text{Formal} & \text{with probability } p = p(v_1, v_2, v_3) \\ \text{with probability } 1 - p = 1 - p(v_1, v_2, v_3) \end{cases} \quad (2) \quad \boxed{\text{e:possible}}$$

The difference between the two models is clear. In the first case we can make deterministic claims such as "a respondent for which $v_1 = 39$, $v_2 = 0.3$ (on a scale between 0 and 1 computed in some way), $v_3 = \text{Paris}$, $v_4 = \text{Formal}$,... will use the Dialectal variable". The second model allows to make probabilistic claims such as "If the respondent has an higher degree of education (v_2), it is more probable that he/she will use the formal variant(that is, p is increasing in v_2)". We refer to Section (??) for a discussion of causality in a simple effective model known as linear regression model. With effective models such as (2), we loose the notion of causality.

Lastly we note that the relation between (1) and (2) is very complicated and is the object of study in many specific models. We simply note here that the probability present in (2) stems up?? from the uncertainty or lack of knowledge we have in the variables $v_4, ..$ and on the relation of f in (1).

0.1.3 The role of Probability and Statistics

The role of probability is used to model the uncertainty and the lack of knowledge and is the framework within which statistics works. In the In-

roduction ?? we introduce the notion of probability model and we point out that in order to infer information from the real world data we need a probabilistic model. Here we anticipate the main It allows to see the data obtained from the real word, say the experiment introduced before where a respondent says a word which has two possible spellings, as an instance between many possible and legit outcomes: other results could have been possible but we observe only one (this is the idea of the sample space in Definition 1.4 and Section 3.1.2 for more details). Moreover, it The goal of probability is to introduce a framework where data generated in 1a random way (that is, in such a way that other results could have been possible) can be rigorously defined, see Definition ?. The role of statistics is the converse one, and, once assumed that the real world data is generated as models given by probability theory, to infer from the data information about the model. As an example on what statistics does think about trying to estimate the parameter $p = p(v_1, v_2, v_3)$ in the probabilistic model introduced before.

CONTENTS

0.1	Introduction	3
0.1.1	Scope of the Lecture Notes	3
0.1.2	Models and claims of empirical sciences	3
0.1.3	The role of Probability and Statistics	4
1	PROBABILITY	1
1.1	Introduction	2
1.2	Events and Sample Space	3
1.2.1	Definition of an event	3
1.2.2	Operations and relations between events	3
1.2.3	Algebra of events	4
1.2.4	Elementary events and sample space	5
1.2.5	Events as subsets of the sample space	9
1.2.6	Relevant Examples	12
1.2.7	Exercises	14
1.3	Probability on Discrete Sample Spaces	16
1.3.1	Introduction	16
1.3.2	Definition of a probability on a finite sample space	16
1.3.3	Uniform probability	19
1.3.4	Basic properties	22
1.3.5	An Implementation Well Modelled by Probability	23
1.4	Conditional probability	25
1.4.1	Definition and first examples	25
1.4.2	Independence of Events	32
1.4.3	Sampling Interpretation of Conditional Probability	40
1.4.4	Total probability Formula	40
1.4.5	Odds and Bayes Formula	42
2	STATISTICS	43
3	STATISTICS	45
3.1	Introduction: Data and Models	46
3.1.1	Data	46
3.1.2	Models	48
3.2	Testing	55
3.2.1	A simple example: Deciding which distribution is more likely to have generated the data	55
3.2.2	Ideas and Terminology	55
3.2.3	Binomial Test	59
3.2.4	Homework	62
3.3	Framework for testing and problematics	65
3.3.1	Framework	65
3.3.2	Reviewing some examples in this setting	66
3.3.3	problematics	66
3.3.4	Reviewing some examples	68
3.3.5	Homework: the magician	68
3.4	Law of Large Numbers	70

3.4.1	Independent Samples	70
3.4.2	Statement of the law of large numbers	71
3.4.3	Dependence from the Sample Size	72
3.5	Empirical distribution	74
3.5.1	Estimation of the parameters of the Bernoulli trials model	74
3.5.2	Histograms and empirical distributions	77
3.6	Exercises	82
3.6.1	Application to testing	82
3.7	Eugene Onegin Central Limit Theorem for Markov Chains	83
3.7.1	Homework: The Magician Strikes Back	84
3.8	Central Limit Theorem	87
3.8.1	The Standard Deviation	87
3.9	Statement	88
3.10	T-test	93
3.10.1	Standard Deviation	93
3.10.2	Intuition on the t-statistic	94
3.10.3	Z-test	96
3.11	T-test	97
3.11.1	t-test	97
3.11.2	Sample mean as a minimum	99
3.11.3	Mean as a predictor	99
3.12	Some precautions to take when testing	100
3.12.1	Perform tests after looking at the data	100
3.12.2	Multiple Comparisons	100
3.12.3		100
3.12.4	Multivariate Regression	100
3.13	Some other interesting references	102
3.13.1	Eugene Onegin	102
3.14	Central limit theorem for Markov chains	108
3.15	Central limit theorem for the Markov Sequence	113
3.16	Eugene Onegin	114
3.16.1	Eugene Onegin	114

1

PROBABILITY

1.1 INTRODUCTION

s:intro

Probability Theory is a mathematical framework introduced for quantifying uncertainty. Example of sentences with which it works are

e:stat1

1. The probability that tomorrow will rain is 0.75.
2. The probability that the coin that I am about to toss will show its head is $1/2$.

For the rest of the notes, we take a subjective stance and consider the above sentences as expressing a *subjective* degree of belief that an individual has for the events.

Many ways have been tried to give a rigorous meaning of probabilistic statements as the above ones in many different context. Those interesting discussions, however, are not needed when studying probability theory. Indeed, probability theory gives a formal framework to work with sentences such as the above ones, without the need for them to have a precise meaning nor to be true. For instance, statement 2 is not universally true since one can easily cheat with coin tosses and make heads always appear. This is not peculiar to probability theory alone: geometry works with points without never defining them and logic is concerned on how to work with sentences without never being concerned on the veracity of them.

To get an idea of which objects we need to formalise, we now proceed to decompose and to analyse the above statements. Both of them are of the form "The probability of *something* is a number". That *something* in the first case is $E = \text{"Tomorrow will rain"}$, and in the second case $F = \text{"The coin that I am about to toss will show heads"}$, and it will be called an *event*. Events are the objects to which probability refers and the probability of an event is a number, 0.75 in the first statement and 0.5, in the second. We now make an intermediate step and rewrite the above statements as

- The probability of E is 0.75
- The probability of F is 0.5

Therefore, the probability of an event is a number and the probability is an operation that associates to each event a number. Denoting "The probability of" by $\mathbb{P}()$, the above statements can be rewritten as

- $\mathbb{P}(E) = 0.75$
- $\mathbb{P}(F) = 0.5$.

and $\mathbb{P}$ can be seen as a function that takes in input an event and whose output is a number p between 0 and 1

$$\begin{array}{ccc} \mathbb{P} : \text{Events} & \mapsto & [0, 1] \\ E & \rightarrow & \mathbb{P}(E) \in [0, 1]. \end{array}$$

Therefore a typical probabilistic statement assumes the form of

- The probability of the event E is p , or, $\mathbb{P}(E) = p$,

where $p \in [0, 1]$.

1.2 EVENTS AND SAMPLE SPACE

Motivated by the Introduction 1.1 we first introduce the notion of events. We then see that events can be modelled using set theory, after introducing the sample space

1.2.1 Definition of an event

In light of the subjective perspective on Probability Theory we will refer to the person who is evaluating the probabilities of events using "you". Following [definetti] we can define events as

d:event

Definition 1.1. An event is an unambiguous statement which can be either true or false and about which you are uncertain.

Informally, unambiguous statement means that a possible bet or insurance based upon it can be decided without question. Examples of events are: "The next ChatGPT sentence will be incorrect" or "Napoleon was born on March" (to decide this last statement, I should open Wikipedia). Events will be usually denoted by capital letters, $A, B, \dots$, and the impossible event will be usually denoted by $\emptyset$.

1.2.2 Operations and relations between events

ss:operations

From two events E and F it is possible to build new events using the operations "and", "or" and "not", namely, to consider " E and F ", " E or F " and "not E ". We will write them as

$$E \wedge F = \text{"E and F"}, \quad E \vee F = \text{"E or F"}, \quad \neg E = \text{"not E"}.$$

e:and'or'not

The event $E \wedge F$ is true when both E and F are true, $E \vee F$ is true when at least one of the two is true (possibly both), and $\neg E$ is true when E is false. To make a simple example, let $E = \text{"Tomorrow it will rain"}$ and $F = \text{"The coin I am about to toss will show heads"}$, the two events of the Introduction. Then

- $E \wedge F = \text{"Tomorrow it will rain and the coin will show heads"}$.
- $E \vee F = \text{"Either tomorrow it will rain or the coin will show heads (or both)"}$.
- $\neg E = \text{"Tomorrow it will not rain"}$, $\neg F = \text{"The coin will not show heads"}$.

Given two events E and F we say that E *implies* F , and we write $E \rightarrow F$, if F is true whenever E is. If $G = \text{"Tomorrow it will rain heavily"}$, then $G \rightarrow E$. Note that in general

- $E \wedge F \rightarrow E$ and $E \wedge F \rightarrow F$;
- $E \rightarrow E \vee F$ and $F \rightarrow E \vee F$;

- if E is impossible, then $E \rightarrow F$ for every event F , and if F is certain, then $E \rightarrow F$ for every event E .

The first two relations say that "and" makes events smaller (harder to be true), while "or" makes them bigger.

^a

^a Different sentences can define the same event: $E =$ "The die shows an even number" and $F =$ "The die shows 2, 4 or 6" are the same event. We identify those two propositions with the same event and we write $E = F$. In particular, any two impossible events are equal and we denote them $\emptyset$.

1.2.3 Algebra of events

There are no inherent restrictions on the events one can define and on which you could try to assign a probability. When doing a probabilistic study, however, you fix beforehand the events you want to consider, and this choice depends on what you are looking at. If we are interested in the result of a coin toss, we want to consider the event "The coin shows heads", but not the event "The coin shows heads and tomorrow it will rain in Rome".

Example 1.1: The shop

ex:shop

You are paid by a shop owner to perform a statistical analysis of the customers visiting his shop. For time and economic constraints, you have to choose what kind of events you are interested in. One possibility is to consider only the number of customers, so that the events will be of the form "More than 10 customers will visit the shop", "3 customers will visit the shop", "Between 5 and 7 customers will visit the shop", ... If the shop owner is also interested in the gender of the customers, you can do a finer analysis and consider events that take into account both the number of customers and their gender, such as "More than 3 male customers and less than 10 female customers will visit the shop" or "2 male and 1 female customers will visit the shop".

The events we want to consider are called *measurable events*, since they are the ones that our observation is able to decide (in Example 1.1, by counting the customers) and, as we will see, the ones to which we assign a probability. Their set will be denoted by $\mathfrak{F}$.

The choice of $\mathfrak{F}$ is free, up to one requirement: if we want to consider E and F , we also have to consider $E \wedge F$, $E \vee F$ and $\neg E$. One of the reasons for this is that whatever we are looking at in order to decide whether E and F are true decides also whether $E \wedge F$, $E \vee F$ and $\neg E$ are true. In Example 1.1, if by counting the customers we can decide $E =$ "More than 10 customers will visit the shop" and $F =$ "Less than 20 customers will visit the shop", by counting the customers we can also decide $E \wedge F =$ "Between 11 and 19 customers will visit the shop" and $\neg E =$ "At most 10 customers will visit the shop".

We observe that when we will be introducing a probability in Section ??, we need to introduce it in such a way that the probabilities of $E \wedge F$, $E \vee F$ and $\neg E$ are tied to the ones of E and F . For instance, the probability of $\neg E$ is one minus the probability of E : assigning a probability to E means assigning a probability to $\neg E$ as well, whether we want it or not. Finally, we ask that

the certain event belongs to $\mathfrak{F}$: it is decided by any observation whatsoever, and it is the event to which we will give probability 1 and with which all the other probabilities are compared.

d:measurable

Definition 1.2 (Measurable events). The events we want to consider are called *measurable events*. Their set $\mathfrak{F}$ is an *algebra of events*, that is,

1. the certain event Ω belongs to $\mathfrak{F}$;
2. if $E \in \mathfrak{F}$, then $\neg E \in \mathfrak{F}$;
3. if $E, F \in \mathfrak{F}$, then $E \wedge F \in \mathfrak{F}$ and $E \vee F \in \mathfrak{F}$.^b

^b If we also ask that $E_1 \vee E_2 \vee E_3 \vee \dots$ belongs to $\mathfrak{F}$ whenever $E_1, E_2, E_3, \dots$ do, $\mathfrak{F}$ is called a σ -algebra of events. This is the notion used in measure theory.

Example 1.2: The algebra of a single event

ex:cointoss

Suppose that we want to consider only one event E . By Definition 1.2, $\mathfrak{F}$ has to contain also $\neg E$, Ω and $\emptyset$, and

$$\mathfrak{F} = \{\emptyset, E, \neg E, \Omega\} \quad (1)$$

e:cointoss

is an algebra of events: $E \wedge \neg E = \emptyset$, $E \vee \neg E = \Omega$, $\neg(\neg E) = E$, and "and", "or" and "not" of events of $\mathfrak{F}$ give again events of $\mathfrak{F}$, as you can check with the rules (??). It is the smallest algebra containing E . For instance, if we toss a coin and we are interested only in the face it shows, then E = "The coin shows heads", $\neg E$ = "The coin shows tails", Ω = "The coin shows either heads or tails", and (1) is the algebra of the coin toss.

Example 1.3: The shop, continued

ex:shop'algebra

In the first choice of Example 1.1, the measurable events are all the statements about the number of customers, such as "More than 10 customers will visit the shop", "Between 5 and 7 customers will visit the shop" or "An even number of customers will visit the shop", and their set is an algebra of events: "and", "or" and "not" of statements about the number of customers are again statements about the number of customers. We denote this algebra by $\mathfrak{F}_1$. In the second choice, the measurable events are all the statements about the number of male customers and the number of female customers, and their set $\mathfrak{F}_2$ is again an algebra of events.

1.2.4 Elementary events and sample space

ss:elementary

Once the events we want to consider are fixed and collected in $\mathfrak{F}$, it is possible to define a particular kind of events, the *elementary events*, thanks to which we can model events using set theory. Informally, elementary events

are the smallest among the events we want to consider, and can be regarded as the outcomes of the “experiment” we are observing. It is indeed possible to define a notion of smallness using the relation “implies” introduced in Section 1.2.2: an event E is smaller than F if $E \rightarrow F$, since E is true in fewer circumstances than F . The impossible event is the smallest of all, since $\emptyset \rightarrow F$ for every F , and the elementary events are the smallest ones after it.

d:elementary

Definition 1.3 (Elementary events). An event $E \in \mathfrak{F}$ is *elementary* if it is not impossible and the only events of $\mathfrak{F}$ that imply it are $\emptyset$ and E itself: if $F \in \mathfrak{F}$ and $F \rightarrow E$, then $F = \emptyset$ or $F = E$.

For instance, if $\mathfrak{F}$ consists of all the statements about the result of a die, “The result is 6” is elementary, while “The result is an even number” is not, since “The result is 2” implies it.

Ex. 1.2.1 — Show that two different elementary events are mutually exclusive, that is, that if E and F are elementary and $E \neq F$, then $E \wedge F = \emptyset$.

Answer (Ex. 1.2.1) — By Definition 1.2, $E \wedge F \in \mathfrak{F}$. Since $E \wedge F \rightarrow E$ and E is elementary, either $E \wedge F = \emptyset$ or $E \wedge F = E$. In the second case $E \rightarrow F$ and, since F is elementary and $E \neq \emptyset$, $E = F$, which is excluded. Hence $E \wedge F = \emptyset$.

Ex. 1.2.2 — In the algebra $\mathfrak{F}_1$ of Example 1.3, which of the following events are elementary? “No customer will visit the shop”, “At most one customer will visit the shop”, “Exactly 3 customers will visit the shop”.

The notion of elementary event is fundamental since it allows to encode the events of $\mathfrak{F}$ as sets, see Section 1.2.5.

d:sample space

Definition 1.4 (Sample space). Given $\mathfrak{F}$, the *sample space* is the set of the elementary events of $\mathfrak{F}$. We denote it by Ω , the same symbol used for the certain event.^c

Elements of the sample space, that is, elementary events, are usually denoted by ω . The sample space can be regarded as the set of the possible outcomes of the system you are looking at, with the degree of precision that you want to impose. From now on we assume that every event of $\mathfrak{F}$ which is not impossible is implied by at least one elementary event.^d Some examples are in order.

^c The two uses of Ω are reconciled in Section 1.2.5, where the certain event is identified with the set of all the elementary events.

^d This is always the case when $\mathfrak{F}$ is finite, and it is the case in all the examples of these notes.

There are algebras of events with no elementary event at all: the statements about an infinite sequence of coin tosses that involve only finitely many tosses, see Example 1.4, form one, since a condition on one more toss makes any such statement strictly smaller.

Example 1.4: Sample space of a coin toss

If $\mathfrak{F}$ is the algebra of the coin toss (1), the elementary events are $\omega_1 = E = \text{“The coin shows heads”}$ and $\omega_2 = \neg E = \text{“The coin shows tails”}$, and the sample space is $\Omega = \{\omega_1, \omega_2\}$. Note that we can use a different notation and set $1 = \omega_1$ and $0 = \omega_2$. However, 0 and 1 must not be mistaken for numbers: they are symbols indicating elementary events.^e

ex:die'sample

Example 1.5: A die is rolled

If a die is rolled and $\mathfrak{F}$ consists of all the statements about the result, such as "The result is an even number", the elementary events are the ones that cannot be decomposed just by looking at the die, $1 =$ "The result is 1", $\dots$, $6 =$ "The result is 6". The sample space is therefore $\Omega = \{1, 2, 3, 4, 5, 6\}$ and, again, the elements $1, \dots, 6$ must not be mistaken for numbers, since they are elementary events and, for instance, $1 + 3$ doesn't make sense.

Example 1.6: The shop, continued

ex:shop'sample

In the algebra $\mathfrak{F}_1$ of Example 1.3 the elementary events are $E_n =$ "Exactly n customers will visit the shop", for $n = 0, 1, 2, \dots$: a statement about the number of customers that implies E_n is either E_n itself or impossible. The sample space $\Omega = \{E_0, E_1, E_2, \dots\}$, which we can also write as $\{0, 1, 2, \dots\}$, has infinitely many elements. In the algebra $\mathfrak{F}_2$ the elementary events are $E_{m,f} =$ "Exactly m male and f female customers will visit the shop", and the sample space is the set of the pairs (m, f) .

Example 1.7: A chess match

If you observe a chess match, the events you want to consider are those that involve the game alone, such as "The game will end before 10 moves are taken" or "Black wins", while the elementary events are the possible single games. Therefore, the sample space is the set of all possible games. A full description of a game is usually given in terms of the so called algebraic notation; for instance, the fifth game of the 1972 World Championship between Spassky and Fischer is

1.d4 Nf6 2.c4 e6 3.Nc3 Bb4 4.Nf3 c5 5.e3 Nc6 6.Bd3 Bxc3+ 7.bxc3 d6 8.e4 e5 9.d5 Ne7 10.Nh4 h6 11.f4 Ng6 12.Nxg6 fxg6 13.fxe5 dxe5 14.Be3 b6 15.O-O O-O 16.a4 a5 17.Rb1 Bd7 18.Rb2 Rb8 19.Rbf2 Qe7 20.Bc2 g5 21.Bd2 Qe8 22.Be1 Qg6 23.Qd3 Nh5 24.Rxf8+ Rxf8 25.Rxf8+ Kxf8 26.Bd1 Nf4 27.Qc2 Bxa4 0-1

The algebraic notation can be seen as a way to encode elementary events and the sample space. In Exercises 1.2.12 and 1.2.10 examples of sample spaces defined by simple games are given.

Ex. 1.2.3 — From the 2019-2020 course "Mathematics and Statistics" given by professors Khovanskaya, Bubilin, Shurov, Filimonov and Sonin at HSE. A six faced die is rolled. Find the elementary events that imply the event

1. "The result is 6"

2. "The result is a number less or equal 2"
3. "The result is an even number"
4. "The result is a number strictly greater than 4"
5. "The result is seven"

Answer (Ex. 1.2.3) — 1. It is already an elementary event

2. "The result is a number less or equal than 2" = "The result is 1" or "The result is 2".
3. "The result is a number strictly greater than 4" = "The result is 5" or "The result is 6".
4. The event is the impossible event, also denoted by $\emptyset$.

Ex. 1.2.4 — A coin is tossed twice. We are interested in the face which the coin shows when it lands: heads or tails. The set of elementary events is $\{HH, HT, TH, TT\}$, where the event HH corresponds to "the first toss gave heads, the second heads", the event HT corresponds... Write the elementary events that imply the events

1. "We obtained two heads".
2. "The first toss gave heads".
3. "We obtained 1 tails "
4. "At least one toss gave tails"

Answer (Ex. 1.2.4) — 1. HH.

2. TH, TT.
3. TH, HT.
4. HT, TH, TT.

Ex. 1.2.5 — A coin is tossed 4 times. We are interested in the face that it shows when it lands: heads or tails. How many elementary events are there? which elementary events are contained in the events

1. The first result was heads
2. The second result was tails
3. The first result was heads and the second tails
4. All the 4 tosses gave the same result

Answer (Ex. 1.2.5) — The elementary events are HHHH, HHHT, HHHT, ..., TTTT. There are $2^4 = 16$ of them.

1. HHHH, HHHT, ..., HTTT
2. HTHH, HTHT, HTTH, ..., TTTT
3. HTHH, HTHT, HTTH, HTTT
4. HHHH, TTTT

exercise:marbles

Ex. 1.2.6 — ([Ross] Chapter 2 Exercise 1) A box contains 3 marbles: 1 red, 1 green, 1 blue. Consider the experiment that consists in taking 1 marble from the box and then replacing it in the box and then drawing a second marble.

Describe the sample space.

Answer (Ex. 1.2.6) — The sample space can be thought as the set of the outcomes of the experiment. In this case the possible outcomes are "The first marble is red and the second marble is red", "The first marble is red and the second is green",... A more efficient way to write an outcome is to write an array with two entries, i.e. (r,g) , where the first entry represents the colour of the first marble, and the second entry represents the colour of the second marble. That is, we write the elementary event "The first marble is red and the second is blue" by (r,b) . The sample space becomes

$$\Omega = \{(r,r), (r,g), (r,b), (g,r), (g,g), (g,b), (b,r), (b,g), (b,b)\}$$

ex1

Ex. 1.2.7 — Repeat Exercise 1 when the second marble is drawn without replacing the first marble.

Answer (Ex. 1.2.7) — The outcomes (r,r) , (g,g) and (b,b) are no longer possible. The sample space is

$$\Omega = \{(r,g), (r,b), (g,r), (g,b), (b,r), (b,g)\}$$

ex2

Ex. 1.2.8 — You toss a coin 3 times. Describe the elementary events.

Answer (Ex. 1.2.8) — $\Omega = \{(H,H,H), (H,H,T), \dots, (T,T,T)\}$, where the event (H,T,H) is "The result of the first toss was head, the result of the second..."

1.2.5 Events as subsets of the sample space

ss:subset

d:favour

Definition 1.5 (Elementary events in favour of an event). We say that an elementary event $\omega \in \Omega$ is *in favour* of $E \in \mathfrak{F}$ if $\omega \rightarrow E$ (ω implies E).

Remark 1.6. The minimality of the elementary events implies that for each $E \in \mathfrak{F}$ and $\omega \in \Omega$ either ω is in favour of E or the two are mutually exclusive, $\omega \wedge E = \emptyset$: indeed $\omega \wedge E \rightarrow \omega$, so that $\omega \wedge E = \emptyset$ or $\omega \wedge E = \omega$, and in the second case $\omega \rightarrow E$.

We can associate to each event the subset of Ω whose elements are the elementary events in its favour:

$$\begin{aligned} \Phi : \mathfrak{F} &\rightarrow \{A, A \subset \Omega\} \\ E &\mapsto \{\omega \in \Omega, \omega \text{ is in favour of } E\}. \end{aligned}$$

e:abstract to concrete

Example 1.8: Die

ex:die'subset

We have seen in Example 1.5 that the sample space of a rolled die can be identified with $\Omega = \{1, 2, 3, 4, 5, 6\}$. The elementary events in favour of $E = \text{"The result is an odd number"}$ are 1, 3, 5. Therefore we identify E with the subset $\Phi(E) = \{1, 3, 5\}$ of Ω .

The fundamental fact about Φ is that the operations "and", "or" and "not" correspond to the operations $\cap$, $\cup$ and $()^c$ between subsets of Ω , that "implies" corresponds to "is a subset of", and that different events correspond to different subsets.

p:subset

Proposition 1.7. *Let $E, F \in \mathfrak{F}$. Then*

1. $\Phi(E \wedge F) = \Phi(E) \cap \Phi(F)$, $\Phi(E \vee F) = \Phi(E) \cup \Phi(F)$ and $\Phi(\neg E) = \Phi(E)^c$, the set of the elementary events that are not in $\Phi(E)$;
2. $E \rightarrow F$ if and only if $\Phi(E) \subset \Phi(F)$;
3. $E = F$ if and only if $\Phi(E) = \Phi(F)$.

Moreover, $\Phi(\emptyset)$ is the empty set and $\Phi(\Omega) = \Omega$, the whole sample space.

Proof. Since $E \wedge F$ is true exactly when E and F are both true, $\omega \rightarrow E \wedge F$ if and only if $\omega \rightarrow E$ and $\omega \rightarrow F$, which is the first identity. For the third one, if $\omega \rightarrow E$ and $\omega \rightarrow \neg E$, then $\omega \rightarrow E \wedge \neg E = \emptyset$, and ω would be impossible: hence $\omega \in \Phi(\neg E)$ implies $\omega \notin \Phi(E)$. Conversely, if $\omega \notin \Phi(E)$, then $\omega \wedge E = \emptyset$ and, by the rules (??), $\omega = \omega \wedge (E \vee \neg E) = (\omega \wedge E) \vee (\omega \wedge \neg E) = \omega \wedge \neg E$, that is, $\omega \rightarrow \neg E$. The second identity follows from the other two, since $E \vee F = \neg(\neg E \wedge \neg F)$ by de Morgan's laws. As for point 3, suppose that $\Phi(E) = \Phi(F)$ but $E \neq F$. Then one between $E \wedge \neg F$ and $F \wedge \neg E$ is not impossible, since $E \wedge \neg F = \emptyset$ gives $E = (E \wedge F) \vee (E \wedge \neg F) = E \wedge F$, that is, $E \rightarrow F$, and in the same way $F \wedge \neg E = \emptyset$ gives $F \rightarrow E$. Say that $E \wedge \neg F \neq \emptyset$. By the assumption made after Definition 1.4, there is an elementary event $\omega \rightarrow E \wedge \neg F$, so that $\omega \in \Phi(E)$ and $\omega \notin \Phi(F)$, which contradicts $\Phi(E) = \Phi(F)$. Point 2 follows: $E \rightarrow F$ if and only if $E \wedge F = E$ (see the exercises of Section 1.2.3), if and only if $\Phi(E) \cap \Phi(F) = \Phi(E)$ by points 1 and 3, if and only if $\Phi(E) \subset \Phi(F)$. Finally, no elementary event is in favour of $\emptyset$, since elementary events are not impossible, and every elementary event is in favour of Ω . $\square$

From now on we will consider $\mathfrak{F}$ as a set of subsets of Ω , that is, we will not distinguish between an event E and the set $\Phi(E)$ of the elementary events in its favour. This allows us to depict events, their unions and their intersections with Venn's diagrams, as Figure 1.1 shows.

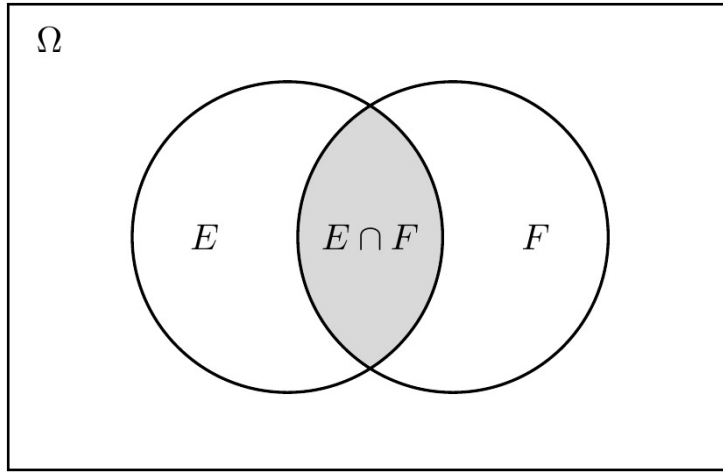


Figure 1.1: Venn's diagram

f:Venn

Ex. 1.2.9 — ([Feller], Chapter 1.8) Let Ω be a sample space and A, B, C be three arbitrary events. Find the expressions for the events that of A, B, C :

1. Only A occurs.
2. Both A and B , but not C occur.
3. All three events occur.
4. At least one occurs.
5. At least two occur.
6. One and no more occurs.
7. Two and no more occur
8. None occurs.
9. Not more than two occur.

Answer (Ex. 1.2.9) — 1. $A \setminus (B \cup C) = A \cap B^c \cap C^c$.

2. $(A \cap B) \setminus C = A \cap B \cap C^c$.
3. $A \cap B \cap C$.
4. $A \cup B \cup C$.

ex:set1

5. At least two occur if occur one between $A \cap B$, $A \cap C$ or $B \cap C$. Thus $(A \cap B) \cup (A \cap C) \cup (B \cap C)$.
6. It has to happen one between A , B and C , that is, it has to happen $A \cup B \cup C$, but it does not have to happen two of them, that is, it cannot happen the event described in 5:
7. $(A \cap B) \cup (A \cap C) \cup (B \cap C)$. Thus the event is $A \cup B \cup C \setminus ((A \cap B) \cup (A \cap C) \cup (B \cap C))$.
8. $((A \cap B) \cup (A \cap C) \cup (B \cap C)) \setminus A \cap B \cap C$ which is the same as $((A \cap B) \cup (A \cap C) \cup (B \cap C)) \cap (A^c \cup B^c \cup C^c)$.
9. $(A \cup B \cup C)^c = A^c \cap B^c \cap C^c$.
10. This event happens when the event all three events occur does not happen. Thus the event is $(A \cap B \cap C)^c = A^c \cup B^c \cup C^c$.

1.2.6 Relevant Examples

In this examples we provide the basic examples of sample spaces to which we will come back throughout the notes: the *Bernoulli trial* space and the sample space of a random number.

Model 1.1 (A Coin Toss). If in a coin toss we are interested only in the face the coin shows, as we always are when we speak about coin tosses, then the only two non-trivial events we want to consider are $0 =$ "The result is tails" and $1 =$ "The result is heads", see also Example 1.2 and 1.8. Therefore

$$\Omega_1 = \{0, 1\},$$

and we will often refer to 1 as to a success. The set of events that we want to consider can be represented by

$$\mathfrak{F} = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}.$$

ex:n'coin

Model 1.2 (two, three and n coin tosses). The sample space of two coin tosses can be encoded by $\Omega_2 = \{00, 01, 10, 11\} = \{x_1 x_2, x_i \in \{0, 1\} i = 1, 2\}$, where, for example, If we toss twice a coin and we are only interested in the faces they show, then we are interested in events of the form "The first coin shows heads" or "The second coin shows tails". The smallest events we can make by intersecting this kind of events are of the form $01 =$ "The first toss gave tails and the second tails". Therefore the sample space is constituted by the set

$$\Omega_2 = \{00, 01, 10, 11\} = \{x_1 x_2, x_i \in \{0, 1\}, \text{ for } i = 1, 2\}$$

of two digit string where every digit represents the outcome of the respective coin. Similarly, for 3 coin tosses

$$\Omega_3 = \{000, 001, 010, 011, 100, 101, 110, 111\} = \{x_1 x_2 x_3, x_i \in \{0, 1\} i = 1, 2, 3\},$$

and, in the general case of n coin tosses, the sample space is the space of strings of n digit with values in $\{0, 1\}$, namely

$$\Omega_n = \{x_1 x_2 \dots x_n, x_i \in \{0, 1\}, i = 1, \dots, n\}. \quad (2)$$

e:ntoss

Events that we will often use when introducing the Binomial distribution, see Example ?? are the events $E_k =$ "There has been exactly k successes", for $k \in \mathbb{N}$, $k < n$. For instance, if $n = 3$ and $k = 1$, is $E_1 = \{001, 010, 100\} = \{x_1 x_2 x_3, x_1 + x_2 + x_3 = 1\}$. The particular encoding given by (2) allows us to write in simple terms the condition imposed by E_k :

$$E_k = \{x_1 \dots x_n \in \Omega_n, x_1 + \dots + x_n = k\}.$$

e:ntoss'ksucce

Model 1.3 (Finite but Undetermined Amount of Coin Tosses). If we are observing a finite amount of coin tosses, but we do not know how many of

them there will be, we can define the set of possible outcomes, that is, the sample space, via the following encoding

$$\begin{aligned}\Omega &= \bigcup_{i=1}^{\infty} \Omega_i = \{0, 1, 00, 01, 10, 11, 000, \dots\}, \\ &= \{x_1 \dots x_n, n \in \mathbb{N}, x_i \in \{0, 1\}, \text{ for } i = 1, \dots, n\}\end{aligned}$$

e:finite'coin

where Ω_i has been defined 2.

The above examples had the important property that their elements could be written as a list $\Omega = \{\omega_1, \dots, \omega_n\}$ for some $n \in \mathbb{N}$, in the finite case, or $\Omega = \{\omega_1, \dots, \omega_n, \dots\}$ in the infinite case. In other words, we have decided a rule to count its elements, and, we have managed to come across all the elements of Ω with this rule. This is no longer possible in the next examples (try yourself to enumerate the elements of Ω). In mathematical terms, in the next two examples Ω is uncountably infinite.

x:infinite'coin

Model 1.4 (Infinite Coin Tosses). If we are observing an infinite amount of coin tosses, then the sample space is the set of strings whose digits denote the result of the corresponding coin.

$$\Omega = \{x_1 \dots x_n \dots, x_i \in \{0, 1\}\}.$$

e:infinite'coin

Possible events that we would like to take into account are of the form "The first toss gave a_1 , the second $a_2, \dots$, the n th a_n ", where $n \in \mathbb{N}$ and $a_i \in \{0, 1\}$ for $i = 1, \dots, n$. This event corresponds to the subset of Ω consisting on strings whose first n elements are a_i , for $i = 1, \dots, n$

$$A_{a_1, \dots, a_n} = \{x_1 \dots x_n \dots \in \Omega \mid x_1 = a_1, \dots, x_n = a_n\}.$$

e:infinite'coin'set

ex:unif

Model 1.5 (Random Number Between 0 and 1). By typing on R the command `runif`, the computer gives us a number X between 0 and 1.

```
runif(1)
## [1] 0.5413302
```

Some events in which we are usually interested are " $X < 1/2$ ", " $X \in [1/4, 1/2]$ ". By intersecting the above events we can see, at least intuitively, that the smallest events one can make are $X = x$. The sample space is therefore composed by the events " $X = a$ " for each $a \in [0, 1]$, and, by denoting a the event $X = a$,

$$\Omega = \{X = a, a \in [0, 1]\} = [0, 1]. \quad (3)$$

e:unif1

The set of events we want to consider $\mathfrak{F}$ contains the events of the form $X \in [a, b]$, where $0 \leq a < b \leq 1$, if we are using the notation in the centre of (3), and $[a, b]$ if we are using the other notation for Ω .

Example 1.4 and 1.5 constitute are different from the previous ones. In both cases the spaces are too big to be written as a list of elements, that is, in the

form $\Omega = \{\omega_1, \omega_2, \dots\}$. In jargon, Ω is not discrete. The difficulty is that now the specification of $\mathfrak{F}$ becomes important since, for good mathematical reasons, it is in general not possible to consider each $A \subset \Omega$.

1.2.7 Exercises

exercise:game'2

Ex. 1.2.10 — ([Ross] Chapter 2 Exercise 2) In an experiment, die is rolled continually until a 6 appears, at which point the experiment stops. What is the sample space of this experiment? Let E_n denote the event "The experiment is completed before n rolls". Write it as a subset of the sample space. Describe $(\bigcup_{i=1}^{\infty} E_i)^c$.

Answer (Ex. 1.2.10) — The sample space is the space of outputs of the experiment. We encode the event "The first roll gave 4, the second 2, the third 6 and then stops" in the string 426. That is, we encode the output of the experiment in strings of digits 1,2,...,5,6, where the digit correspond to the result of the die, and the position of the digit corresponds to when has the digit appeared. The string must stop at after the first 6, since the game stops when the die gives 6.

We have to keep in mind that we can observe events in which we never stop rolling the die, which correspond to infinite strings of 1,2,...,5, for example 1212121.... or 121233...34453236....

$$\Omega = \{6, 16, 26, 36, 46, 56, 116, \dots\} \cup \{x_1 x_2 \dots x_n \dots : x_i \in \{1, 2, 3, 4, 5\}\} = E \cup G.$$

The set E is the event that the game stops in a finite number of rolls. The set G is the event that the game goes on without ending. For any reasonable probability, the event G has probability 0, and it is usually neglected. The events E_i can be written as subsets of the state space in the following way: $E_1 = \{6\}$, $E_2 = \{6, 16, 26, \dots, 56\}$, $E_3 = E_2 \cup \{116, 126, \dots, 556\}$, ... (In a more compact form $E_i = \{x_1 \dots x_n 6 : x_i \in \{1, \dots, 5\} n \leq i\}$).

The event

$$\bigcup_{i=1}^{\infty} E_n = E_0 \cup E_1 \cup E_2, \dots = "E_0 \text{ or } E_1 \text{ or } \dots E_n \text{ or } \dots"$$

is, by definition of "or", the event that E_n happens for at least one n . That is, is the event "The game ends before n rolls at least one n " = "The game ends after n rolls for some n ", which coincides with E . The set event E^c corresponds to G .

Ex. 1.2.11 — ([Ross] Chapter 2 Exercise 3) Two dice are thrown. Let E be the event that the sum of the dice is odd, and F the event that at least one of the two dice lands in 1 and let G be the event that the sum is 5. Describe the events $E \cap F$, $E \cup F$, $F \cap G$, $E \cap F^c = E \setminus F$ and $E \cap F \cap G$

Answer (Ex. 1.2.11) —

$$E = \{(1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), \dots (6, 5)\}$$

$$F = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), \dots, (6, 1)\}$$

$$G = \{(1, 4), (2, 3), (3, 2), (4, 1)\}$$

$$E \cup F = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 3), (2, 5), (3, 1), (3, 2), (3, 4), \dots, (6, 5)\}$$

$$E \cap F = \{(1, 2), (1, 4), (1, 6), (2, 1), (4, 1), (6, 1)\}$$

$$F \cap G = \{(1, 4), (4, 1)\}$$

$$E \cap F \cap G = F \cap G = \{(1, 4), (4, 1)\}$$

$$E \cap F^c = \{(2, 3), (2, 5), (3, 2), (3, 4), (3, 6), (4, 3), \dots, (6, 5)\}$$

e:game

Ex. 1.2.12 — ([Ross] Chapter 2 Exercise 4) Anton, Barbara and Carlo take turns flipping a coin. The first one to get head wins. The sample space to this experiment can be defined as $\Omega = \{1, 01, 001, 0001, \dots\} \cup \{00000\dots\}$.

1. Interpret the sample space.
2. In terms of the sample space, write the following events: A ="Anton wins", B ="Barbara wins" and $(A \cap B)^c$.

Answer (Ex. 1.2.12) — 1. 1 ="First toss is heads"(So that Anton wins) 01 ="The first toss is tails, the second heads"(So that Barbara wins)... The event $00000\dots$ ="Head never arises, the game never stops".

2. As it is easy to seen, Anton wins whenever the digit 1 appears in the 1st, 4th, 7th.. position. This means that

$$A = \{1, 0001, 0000001, \dots\}.$$

In the same way, the event that Barbara wins corresponds is

$$B = \{01, 00001, 00000001, \dots\}$$

where the digit appears in the after $3 \times k + 1$ zeros, for some $k \in \mathbb{N}$.

$(A \cup B)^c = \Omega \setminus (A \cup B) = \{001, 000001, \dots\} \cup \{0000\dots\}$. The event $(A \cup B)^c$ is the event "Carlo wins or the game doesn't end"

1.3 PROBABILITY ON DISCRETE SAMPLE SPACES

1.3.1 Introduction

In this section we are going to formally define what is a probability. As already said in the Introduction 1.1, a probability associates to each $E \in \mathfrak{F}$, the set of events that we want to consider, its probability, $\mathbb{P}(E)$. A probability is therefore a function

$$\begin{aligned} \mathbb{P} : \quad \mathfrak{F} &\mapsto [0, 1] \\ E \in \mathfrak{F} &\mapsto \mathbb{P}(E) \end{aligned}$$

e:prob`def

Not every function from $\mathfrak{F}$ to $[0, 1]$ has the right to be called a probability, and some relations must be satisfied. Consider the example of the two events $E = \text{"Tomorrow will be cloudy"}$ and $F = \text{"Tomorrow it will rain"}$ and of its associated probabilities $\mathbb{P}(E)$ and $\mathbb{P}(F)$. If F happens, also E happens, and the intuition tells us that the numbers $\mathbb{P}(E)$ and $\mathbb{P}(F)$ have to verify $\mathbb{P}(F) \leq \mathbb{P}(E)$. This monotonicity property, for instance, will be proved in Corollary 1.16

In this notes we will not introduce the definition of probability in the full general context. We rather restrict to finite sample space $\Omega = \{\omega_1, \dots, \omega_n\}$, for some $n \in \mathbb{N}$. Up to replacing finite sums with infinite sums, all the definitions and the results below generalise immediately to the discrete infinite setting $\Omega = \{\omega_1, \dots, \omega_n, \dots\}$, where the sample space can be written as a list of elements. Unfortunately, many sample spaces that are used in practice, such as $\Omega = [0, 1]$, see Example 1.5 or $\Omega = \{x_1 x_2 \dots, x_i \in \{0, 1\} \text{ for } i \in \mathbb{N}\}$ defined in 1.4, are simply too big to be written as a list of elements. In such cases the theory gets more involved and invokes mathematical results from measure theory. The reason is that the usual requirements that a function $\mathbb{P} : \mathfrak{F} \rightarrow [0, 1]$, the *Kolmogorov axioms*, are too strict to allow each $E \subset \Omega$ to be in $\mathfrak{F}$, and one has to restrict to a so-called σ -algebra of subsets of Ω , a notion that this notes carefully avoid to introduce. Nevertheless, in such cases, we will evaluate the probabilities of particular events when dealing with Bernoulli trials, see Definition ?? and with continuous random variables, see Definition ??.

1.3.2 Definition of a probability on a finite sample space

Hereafter we consider a finite sample space $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$, for some $n \in \mathbb{N}$, and we consider every event that can be written in terms of the elementary events $\omega_1, \dots, \omega_n$, namely

$$\mathfrak{F} = \{A, A \subset \Omega\}$$

e:f`all

consists in every subset of Ω (including the impossible event $\emptyset$ and the certain event Ω). The particularity of the discrete case we are considering is that in order to define a probability on Ω it is sufficient to define the probabilities of the elementary events

ob'elementary

Definition 1.8 (Probability of the elementary events). A probability for the elementary events $\omega_i, i = 1, \dots, n$ is a collection numbers $p_1, \dots, p_n$ verifying that $p_i \in [0, 1]$ for each $i = 1, \dots, n$ and

$$p_1 + p_2 + \dots + p_n = 1 \quad (4) \quad \text{e:normalization}$$

The condition (10) has the intuitive meaning that we regard the event in which none of the ω_i happens as impossible, see Proposition 1.12.

Example 1.9

ex:definition

The sample space of a rolled die is $\Omega = \{1, 2, 3, 4, 5, 6\}$. Definition 1.8 allows for different choices of probabilities on elementary events, but a typical choice is

$$p_1 = \dots = p_6 = 1/6,$$

e:die'uni

which corresponds to a perfectly balanced die or a die about which we do not have any information. A probability corresponding to an unfair die could look like

$$p_1 = p_2 = p_3 = 1/4, \quad p_4 = p_5 = p_6 = 1/12. \quad (5) \quad \text{e:die'unfair}$$

The numbers p_i , for $i = 1, \dots, n$, are the probability of the elementary event ω_i , namely $\mathbb{P}(\{\omega_i\}) := p_i$. We will sometimes, for instance in (6), use the different notation $\mathbb{P}(\{\omega\}) = p_\omega$, which avoids to use the index i with which the elements of Ω are ordered and that, therefore, does not depend on the particular way the elements are ordered. If $\omega = \omega_i$, then the two notations are related by $p_i \equiv p_{\omega_i}$.

Example 1.10

coin'toss'notation

This example illustrates the difference in the two possible notations we have introduced. Consider Ω_2 , the sample space of two coin tosses given in Example 1.2. Ordering the elements $\Omega = \{00, 01, 10, 11\}$ by $\omega_1 = 00, \omega_2 = 01, \omega_3 = 10$ and $\omega_4 = 11$, then a possible probability using the notation introduced in Definition 1.8 is given by

$$p_1 = 1/8, p_2 = 1/8, p_3 = 1/2, p_4 = 1/4$$

and the same probability, using the equivalent notation p_ω , is given by

$$p_{00} = 1/8, p_{01} = 1/8, p_{10} = 1/2, p_{11} = 1/4$$

Note that the possible ways of choosing p_i , for $i = 1, \dots, n$, are quite arbitrary and the only restrictions involved in the choice is that $p_i \in [0, 1]$ and (10), which basically means that there are $n - 1$ numbers (which have to be positive and whose sum has to be less than 1) to fix. However, in the discrete

setting, this is all the freedom at our disposal, and once the probabilities of the elementary events are assigned, the probability of every event $E \in \mathfrak{F}$ is determined by the sum of the probabilities of the elementary events that compose it, according to the following definition

d:prob

Definition 1.9. Let $p_1, p_2, \dots, p_n, n \in \mathbb{N}$ be a probability for the elementary events, see Definition 1.8, and let $A \subset \Omega$ be an event. The probability of A is the sum of the probabilities of the elementary events that compose it

$$\mathbb{P}(A) = \sum_{\omega \in A} p_{\omega} \quad (6)$$

e:prob

Using a different notation, any subset A of Ω can be written as

$$A = \{\omega_{i_1}, \dots, \omega_{i_k}\}, \quad (7)$$

e:notation

for some $k \leq n$ and set of indexes $1 \leq i_1 < i_2 < \dots < i_k \leq n$. Its probability is then given by

$$\mathbb{P}(A) = p_{\omega_{i_1}} + \dots + p_{\omega_{i_k}} \equiv p_{i_1} + \dots + p_{i_k} \quad (8)$$

e:prob2

Example 1.11

ex:dice`prob

We come back to Example 1.9 in order to see Definition 1.9 in action and to get better acquainted with the two different notations (6) and (8). The event "The result is an odd number" corresponds to the subset $\{1, 3, 5\}$ and (6) tells that the two probabilities $\mathbb{P}$ and $\mathbb{Q}$ defined by the choices (??) and (5), respectively, evaluate the probability of this event by

$$\mathbb{P}(\{1, 3, 5\}) = \mathbb{P}(\{1\}) + \mathbb{P}(\{3\}) + \mathbb{P}(\{5\}) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

e:die`prob`uni

and

$$\mathbb{Q}(\{1, 3, 5\}) = \mathbb{Q}(\{1\}) + \mathbb{Q}(\{3\}) + \mathbb{Q}(\{5\}) = \frac{1}{4} + \frac{1}{4} + \frac{1}{12} = \frac{7}{12},$$

e:die`prob`unfair

respectively. In order to use the notation in (8), we rewrite the set $\{1, 3, 5\}$ as in (7) by setting $k = 3$ and $i_1 = 1, i_2 = 3$ and $i_3 = 5$. Observe also that $\mathbb{P}(\Omega) = \mathbb{Q}(\Omega) = 1$ thanks to (10).

We finally collect the definitions of the present section

d:prob3

Definition 1.10. A *probability* over Ω is a function

$$\begin{aligned} \mathbb{P} : \mathfrak{F} &\rightarrow [0, 1] \\ E &\mapsto \mathbb{P}(E) \end{aligned}$$

such that $p_i := \mathbb{P}(\{\omega_i\})$ forms a probability on elementary events in the sense of Definition 1.8 and (6) (or (8)) holds.

1.3.3 Uniform probability

In the case in which you evaluate each elementary event equally likely to occur, or in the case in which you don't have any knowledge to say that any particular elementary event is more likely to occur, a possible choice is to assign to each elementary event the same probability $p \in [0, 1]$. To determine p we use (10)

$$p_1 + \dots + p_n = \overbrace{p + \dots + p}^{n \text{ times}} = np = 1, \quad \text{e:uniform1}$$

from which we obtain that $p = 1/n$. To compute the probability of an event composed by k elementary events $A = \{\omega_{i_1}, \dots, \omega_{i_k}\}$, with $k \leq n$ and $1 \leq i_1 < \dots < i_k \leq n$ we use (8) to obtain

$$\mathbb{P}(A) = p_{i_1} + \dots + p_{i_k} = \overbrace{\frac{1}{n} + \dots + \frac{1}{n}}^{k \text{ times}} = \frac{k}{n}, \quad (9) \quad \text{e:uniform2}$$

which depends only on k , the number of elementary events that compose A . Denoting by $|E|$ the number of elements of the set $E \subset \Omega$, one can rewrite (9) as the number of elements that compose A divided by the total number of events

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

To get acquainted with the notation independent from the ordering used in (6), we repeat the same computations

$$\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\}) = \sum_{\omega \in A} p = p \left(\sum_{\omega \in A} 1 \right) = p|A| = \frac{|A|}{|\Omega|},$$

and obtain

d:uniform

Definition 1.11 (Uniform Probability). Let Ω be a finite sample space. The uniform probability is defined by

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|}, \quad \text{e:uniform}$$

or each $E \subset \Omega$.

Basically, computing the uniform probability reduces to counting configurations. Unfortunately, despite being conceptually simple, it is often computationally unfeasible. The same holds for general discrete probabilities, which can be seen as a way to count objects giving different weights to them. The uniform probability is often mistakenly taken as the definition of probability. This has led to some abuse of notation such as the following

Example 1.12: Extracting Randomly a Person

ex:person

Consider extracting a person between 20, and denote $\Omega = \{1, \dots, 20\}$ the sample space, where i denotes the elementary event "the i -th person is extracted". Usually what is meant by the sentence "a person is extracted randomly" is that a person is selected with uniform probability, namely $\mathbb{P}(\{i\}) = 1/20$. However, from our point of view, a Nobel prize is also chosen randomly among the possible candidates, but there is no reason why the probability should be the uniform one. Another that can be ambiguous are "we choose a person as randomly as possible", by meaning as fairly or as uniform as possible.

As a side note, despite being natural to model the choice of a person with uniform probability, when doing a survey or making a poll statisticians struggle to make their choice adhere to this model. When mobile phones were not so widespread, doing surveys by phone call excluded a portion of the population, for instance. See Example 3.1.

Sometimes probabilities that are not uniform can be seen as arising from a uniform probability space (in the sense of Definition ??, see Example ??)

Example 1.13: Sum of two dice

d:two dice sum

Consider the sample space of two dice

$$\Omega' = \{11, 12, \dots, 66\} = \{x_1 x_2 : x_1, x_2 \in \{1, \dots, 6\}\}$$

and endow it with the uniform probability. We will discuss in ?? the reason of this choice, but we anticipate that it is due to the fact that the dice are *fair* and they are rolled *independently*, see Subsectionss:ind?????. Assume we can only look at the sum of the results. This leads to a *coarse grained* sample space, see Example ?? where the elementary events, that is, the elements of the new sample space Ω , are the events described in Ω' by $\{(x_1, x_2) \in \Omega' : x_1 + x_2 = k\}$, and whose probabilities are

$$\mathbb{P}(A_2) = \mathbb{P}(A_{12}) = \frac{1}{36}$$

$$\mathbb{P}(A_3) = \mathbb{P}(A_{11}) = \frac{1}{18}$$

$$\mathbb{P}(A_4) = \mathbb{P}(A_{10}) = \frac{1}{12}$$

$$\mathbb{P}(A_5) = \mathbb{P}(A_9) = \frac{1}{9}$$

$$\mathbb{P}(A_6) = \mathbb{P}(A_8) = \frac{5}{36}$$

$$\mathbb{P}(A_7) = \frac{1}{6}.$$

sum

Therefore, the probability in Ω is not the uniform one.

Example 1.14: Extractions with replacement

ex:replace

Consider the sample space given by extractions with replacements, which are the object of Exercise 1.2.6. It makes sense to consider each outcome equally probable, that is

$$p_{(a,b)} = 1/9 \quad \text{for each } (a,b) \in \Omega.$$

e:extraction'replacement

This choice of a probability is a consequence of two properties, the fact that the extractions are *fair* and *independent*, see also ??? for a more general example. The fact that the extractions are independent is sometimes hidden in the extraction procedure: between two extractions we mix the balls.

Example 1.15: Extractions without replacement

ex:no'replace

If in the extractions from the bowl of Example 1.14 and Exercise 1.15 we do not reinsert the ball the sample space is smaller or, equivalently, we assign $p_{rr} = p_{nn} = p_{gg} = 0$. Nonetheless the symmetry between the colours of the marbles suggests that we might regard each other outcome as equally probable. This is again a consequence of the two following facts, which will be discussed in a more general setting in (??). Each single extraction is fair and, since we mix the marbles before drawing again, the second extraction depends on the first one only because of the fact that the composition of the bowl changes.

Ex. 1.3.1 — A coin is tossed 4 times. Recall that the state space is $\Omega = \{0000, 0001, \dots, 1111\}$, where a 1 means that the associated coin gave heads. Suppose that the probability on Ω is uniform (we will see in Section ?? that this is due to independence of each coin toss). Calculate the probability of the events

1. The first toss gave 1
2. There are exactly two heads
3. There has been a head after a tail (that is, we observed a 01)
4. The event is obtained by fixing the first digit to 1 and allowing the other to take arbitrarily the values 0 or 1, that is, $E = \{1x_2x_3x_4, x_i \in \{0,1\}\}$. Since $|E| = 2^3$ and $|\Omega| = 2^4$, by formula (??) we obtain

$$\mathbb{P}(E) = 2^3/2^4 = \frac{1}{2}.$$

Ex. 1.3.2 — We throw two dice, and we suppose that the each output has the same probability (Section 1.4.2 will justify this choice with the hypothesis of independence and fairness of the dice)

1. Calculate the probability of obtaining a 2 and a 4
2. Calculate the probability of obtain 2 with the first die and 4 with the second

1.3.4 Basic properties

Let Ω be a sample space and $\mathbb{P}$ a probability defined on it. The first property tells us that there are no possible outcomes outside the ω_i .

p1

Proposition 1.12.

$$\mathbb{P}(\Omega) = 1 \quad (10)$$

e:normalization

Proof. It is a direct consequence of (10):

$$\mathbb{P}(\Omega) = \mathbb{P}(\{\omega_1, \omega_2, \dots, \omega_n\}) = \mathbb{P}(\omega_1) + \dots + \mathbb{P}(\omega_n) = 1.$$

□

We now formulate the *additivity* property of the probability.

p:monotone

Proposition 1.13 (Additivity). *Any two mutually exclusive events A, B , that is, events that satisfy $A \cap B = \emptyset$, verify*

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B). \quad (11)$$

e:add

More generally,

p:exclusive

Proposition 1.14. *Any two events $A, B \subset \Omega$, it holds*

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

e:union

The proof looks more involved than what really is. The idea is simple and is the following up of the graphycal picture of the events as subset of the sample space using the Venn diagram in Figure 1.1. A probability can be seen as a measure of the areas covered by the events, and we can see that $\mathbb{P}(A) + \mathbb{P}(B)$ counts twice the area of $A \cap B$, while only once the area in $A \cup B$ that is not in $A \cap B$.

Proof. Let $\Omega = \{\omega_1, \dots, \omega_n\}$ be the sample space. Without loss of generality, we assume that $A = \{\omega_1, \dots, \omega_s, \omega_{s+1}, \dots, \omega_l\}$, and that $B = \{\omega_1, \dots, \omega_s, \omega_{l+1}, \dots, \omega_{l+m}\}$. In this way, $A \cap B = \{\omega_1, \dots, \omega_s\}$, and $A \cup B = \{\omega_1, \dots, \omega_{l+m}\}$. Using the Definition 1.9,

$$\mathbb{P}(A) = \mathbb{P}(\omega_1) + \dots + \mathbb{P}(\omega_s) + \mathbb{P}(\omega_{s+1}) + \dots + \mathbb{P}(\omega_l),$$

$$\mathbb{P}(B) = \mathbb{P}(\omega_1) + \dots + \mathbb{P}(\omega_s) + \mathbb{P}(\omega_{l+1}) + \dots + \mathbb{P}(\omega_{l+m})$$

and

$$\mathbb{P}(A \cap B) = \mathbb{P}(\omega_1) + \dots + \mathbb{P}(\omega_s).$$

In this way,

$$\mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) = \mathbb{P}(\omega_1) + \dots + \mathbb{P}(\omega_s) + \mathbb{P}(\omega_{s+1}) + \dots + \mathbb{P}(\omega_{l+m}) = \mathbb{P}(A \cup B).$$

□

- **Corollary 1.15.** *Let $A, B \subset \Omega$ be two mutually exclusive events, that is, two events such that $A \cap B = \emptyset$. Then*

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B). \quad (12) \quad \boxed{\text{e:addc}}$$

Proof. Use (11) and note that $\mathbb{P}(A \cap B) = \mathbb{P}(\emptyset) = 0$. □

monotonicity

Corollary 1.16 (Monotonicity). *Let A and B be two events $A, B \subset \Omega$. If $A \subset B$, then*

$$\mathbb{P}(A) \leq \mathbb{P}(B).$$

The monotonicity is a very intuitive property. The probability that today's weather is bad is greater than the probability that today there is a storm, since if there is a storm there is also bad weather.

Proof. The intuition of the proof is that when we sum over the elementary events in B we are summing over the elementary events in A and the elementary events in $B \setminus A$. Therefore we write B as $A \cup (B \setminus A)$ and using (12) we obtain that $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$. □

complementary

Corollary 1.17. *Let $A \subset \Omega$ be an event. The*

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \quad \boxed{\text{e:compl}}$$

Proof. Write $\Omega = A \cup A^c$, apply (12) with $B = A^c$ and use (10). □

This property is also very natural. The probability that I will pass the test is 0.6. Thus, the probability that I will not pass the test is 0.4.

1.3.5 An Implementation Well Modelled by Probability

ss:r'probl

In this section we give some possible

Given an n and some probabilities on elementary events $p_1, \dots, p_n$, see Definition ??, the computer is able to generate random outcomes with those probabilities. The following chunk of code, for instance, generates a natural number between 1 and 6 with probabilities proportional to p_{aux} , that is, with probabilities p .

```

n <- 6 # The number of elements of our sample space
p_aux <- c(1,2,3,0.5,1,5) # Here I take the $p_i$ to be positive numbers whose sum needs not to be 1
p <- p_aux/sum(p_aux) # Now I normalise them so that they sum 1
sample(1:n, 1, prob = p ) # The output of the sample

## [1] 5

```

This means that, defining $\omega_i = \text{"The output is } i \text{"}$, $\Omega = \{\omega_1, \dots, \omega_n\}$ and $p_1, \dots, p_n$ is a good model for the above output. Other systems for which the probability models work well that we have introduced before are Example 12 and the random extractions of balls from an urn in Example 3.2.1.

This is to be put later, after introducing independent sampling or maybe Extractions with or without replacement. The R command `sample` allows to implement the model of extractions with or without replacement.

1.4 CONDITIONAL PROBABILITY

In general, our evaluation of the probabilities changes if our state of information changes. If we observe that the sky is cloudy, our evaluation of the probability that it will rain will become greater. Given an event E , if $\mathbb{P}(\cdot)$ denotes the probability before the knowledge of E , we will denote the new probability, given the knowledge of E , by $\mathbb{P}(\cdot|E)$. With this notation, the previous example reads $\mathbb{P}(\text{"Today it will rain"}|\text{"It is cloudy"}) > \mathbb{P}(\text{"It will rain today"})$. In case the *conditioning* event E can be written in terms of the sample space, that is, in case $E \subset \Omega$, $\mathbb{P}(\cdot|E)$ can be described in simple terms.

1.4.1 Definition and first examples

Let $\Omega = \{\omega_1, \dots, \omega_n\}$ be the sample space, $\mathbb{P}$ a probability over Ω , see Definition 1.10, and let $B \subset \Omega$ be an event. The probability *conditioned to* B is a new probability denoted by the symbol $\mathbb{P}(\cdot|B)$. Intuitively, it has to satisfy that

$$\mathbb{P}(\{\omega\}|E) = 0 \quad \text{if } \omega \notin E \quad \text{e:cond1}$$

and, since we do not have information on which event $\omega \in E$ caused E , we would like to maintain the proportions

$$\mathbb{P}(\{\omega\}|E) = C\mathbb{P}(\{\omega\}) = Cp_\omega, \quad \text{e:cond2}$$

for some constant C . The constant C is to be determined by (10):

$$1 = \sum_{\omega \in \Omega} \mathbb{P}(\{\omega\}|E) = \sum_{\omega \in E} C\mathbb{P}(\{\omega\}) = C\mathbb{P}(E) \quad (13) \quad \text{e:cond3}$$

so that $C = \mathbb{P}(E)$. For the sake of concreteness we rewrite (13) using a different and extended notation. Let $E = \{\omega_{i_1}, \dots, \omega_{i_k}\}$, for some $k \in \mathbb{N}$ and $1 \leq i_1 < \dots < i_k \leq n$ be an event, then

$$\begin{aligned} 1 &= \mathbb{P}(\{\omega_1, \dots, \omega_n\}|\{\omega_{i_1}, \dots, \omega_{i_k}\}) \\ &= \sum_{i=1}^n \mathbb{P}(\{\omega_i\}|\{\omega_{i_1}, \dots, \omega_{i_k}\}) = C \sum_{j=1}^k \mathbb{P}(\{\omega_{i_j}\}) = C\mathbb{P}(E) \end{aligned} \quad \text{e:cond4}$$

Therefore $C = \frac{1}{\mathbb{P}(E)}$ and the probability of an ev

that is, we denote the probability of A conditioned to B by $\mathbb{P}(A|B)$. Of course, the new probability has to satisfy $\mathbb{P}(\omega|B) = 0$ if $\omega \notin B$. If $\omega \in B$, what should the new value $\mathbb{P}(\omega|B)$ be? Since the new information tells us only that B is true, but it does not give us information about the single ω s in B , the proportions given by $\mathbb{P}$ in B should be maintained by $\mathbb{P}(\cdot|B)$. Therefore we set $\mathbb{P}(\omega|B) = c\mathbb{P}(\omega)$ if $\omega \in B$, for some constant c . c which

can be determined by requiring $\mathbb{P}(\cdot|B)$ has to satisfy (??). If $B = \{\omega_1, \dots, \omega_k\}$, we obtain that

$$\begin{aligned} 1 &= \mathbb{P}(\omega|B) + \dots + \mathbb{P}(\omega_k|B) + \mathbb{P}(\omega_{k+1}|B) + \dots + \mathbb{P}(\omega_n|B) \\ &= c\mathbb{P}(\omega_1) + \dots + c\mathbb{P}(\omega_k) = c(\mathbb{P}(\omega_1) + \dots + \mathbb{P}(\omega_k)) = c\mathbb{P}(B). \end{aligned}$$

Therefore $c = 1/\mathbb{P}(B)$ and

$$\mathbb{P}(\omega|B) = \begin{cases} \frac{\mathbb{P}(\omega)}{\mathbb{P}(B)} & \text{if } \omega \in B \\ 0 & \text{if } \omega \notin B, \end{cases}$$

e:condi

Considering an event $A = \{\omega_1, \dots, \omega_s, \omega_{k+1}, \dots, \omega_{k+m}\}$, where $s \leq k$ formula (1.9) leads to

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(\omega_1) + \dots + \mathbb{P}(\omega_s)}{\mathbb{P}(B)} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

conditional

Definition 1.18. Let Ω be the sample space and let $\mathbb{P}$ be a probability. Consider an event $B \subset \Omega$ such that $\mathbb{P}(B) > 0$. The conditional probability $\mathbb{P}(\cdot|B)$ given B is the probability measure defined by

$$\Omega \ni A \mapsto \mathbb{P}(A|B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)},$$

that is, is the probability that associates to each $A \subset \Omega$ the value $P(A|B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$.

As shown in Figure ??, there is an easy interpretation of the conditional probability. Recall from ??? that one can interpret a probability as a way of measuring areas (in the discrete setting as a way to count which gives different weights to different events). If we observe B , our new evaluation of the area doesn't change in E and outside E is updated set zero. Since one of the conditions imposed to a probability is that the total area is 1, see Proposition ??, to maintain this property we need to divide the new evaluation by E . In this sense we maintain the proportions of our evaluations inside E (since we do not have any information on which event caused E) and we set them to 0 outside, since we know that E happened. Alternative, one can think that we are redistributing the mass of E^c , that is $\mathbb{P}(E^c)$, to E , in such a way that the proportions do not change. One can think at a probability as a distribution of a mass, which in total is 1, over Ω . If one takes the distribution of mass given by $\mathbb{P}$ and sets the mass of B^c to be zero, one has a distribution of mass whose total is not 1. In order to obtain the distribution of mass given by the conditional probability $\mathbb{P}(\cdot|B)$, one has therefore to redistribute the mass it has removed, $\mathbb{P}(B^c)$, in such a way to not change the relative masses of the singletons in B . Thus $\mathbb{P}(\omega|B) = \mathbb{P}(\omega) + \mathbb{P}(\omega)\mathbb{P}(B^c)/\mathbb{P}(B)$. This way of thinking can be useful for the Exercise ??. Note also that Example (1.4.1) and Exercise (??) show that probabilities can vary a lot if conditioned to the an event B . This is schematically shown in Figure (1.3)

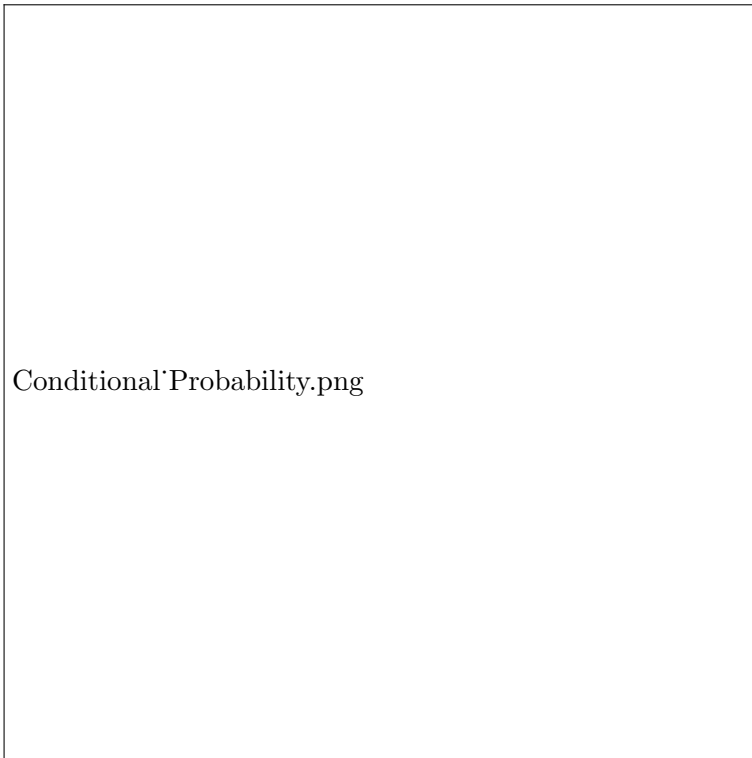
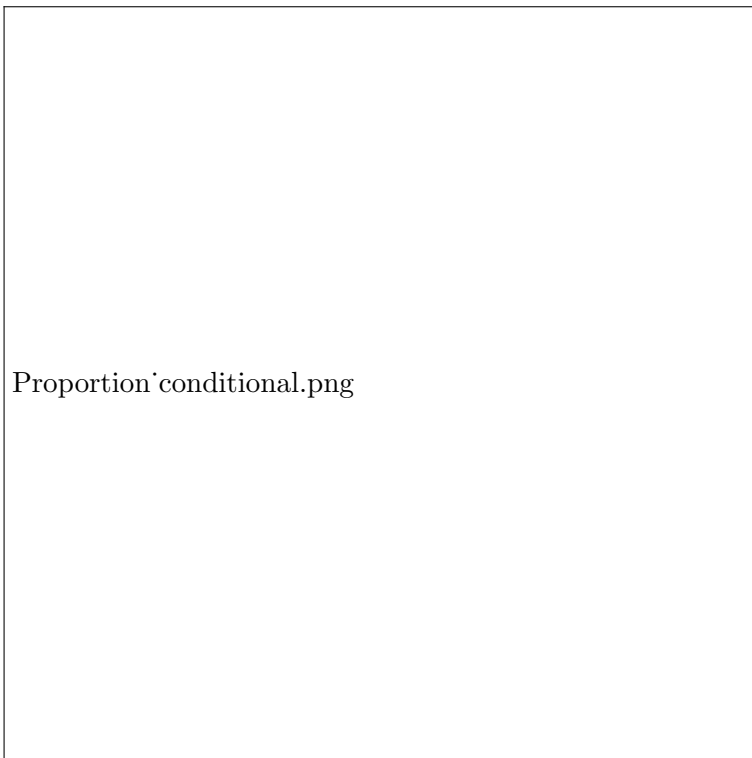


Figure 1.2: We forget about B^c and we consider $\mathbb{P}$ restricted to subsets of $\mathbb{B}$. However, to obtain total mass 1 without favouring any of the $\omega \in B$, we multiply by $\frac{1}{\mathbb{P}(B)}$.

f:cond`prob



Proportion`conditional.png

Figure 1.3: $\mathbb{P}(E)$ is much greater than $\mathbb{P}(E|B)$.

fig:prop

^f In other words, on Ω we assume that $\mathbb{P}$ is the uniform one. Why this is the case investigated in Subsection 1.4.2

Ex. 1.4.1 — We throw two dice and assume that each of numbers is equally probable ^f what is the probability conditioned on "The sum of the two dice is 7"?

Ex. 1.4.2 — You toss a fair coin independently 4 times ¹. Show that, knowing that only one head has been observed, the head is equally probable to have appeared in the 1st, 2nd, 3rd and 4th position.

Answer (Ex. 1.4.2) — The sample space is $\Omega = \{0000, 0001, 0010, \dots, 1111\}$. Let $B = \{x_1 x_2 x_3 x_4, x_1 + x_2 + x_3 + x_4 = 1\} = \{1000, 0100, 0010, 0001\}$ be the event that only one heads result has shown up. The probability of the singleton $x_1 x_2 x_3 x_4$, where $x_i \in \{0, 1\}$ for $i = 1, 2, 3, 4$, conditioned to B is given by

$$\mathbb{P}(x_1 x_2 x_3 x_4 | B) = 0$$

if $x_1 + x_2 + x_3 + x_4 = 1$ and is equal to

$$\mathbb{P}(x_1 x_2 x_3 x_4 | B) = \frac{1/2^4}{4/2^4} = 1/4,$$

which is to say that there is not favourite position in which the heads might appear.

Ex. 1.4.3 — Let $\Omega = \{\omega_1, \dots, \omega_n\}$ be a sample space and let $\mathbb{P}$ be a probability assigned to it. Let $B \subset \Omega$ be an event, and assume $\mathbb{P}(B) > 0$.

1. Prove that $\mathbb{P}(\cdot | B)$ is a probability on the sample space B .
2. Assume that $\mathbb{P}$ is the uniform probability defined in (??) on Ω . Show that $\mathbb{P}(\cdot | B)$ is the uniform probability on the new sample space B .

Answer (Ex. 1.4.3) — 1. In order to show that $\mathbb{P}(\cdot | B)$ is a probability measure, we have to show that

- $\mathbb{P}(\omega | B) \in [0, 1]$
- ((??) holds) $\sum_{\omega \in B} \mathbb{P}(\omega | B) = 1$
- ((1.9) holds) For each $F \subset B$, $\mathbb{P}(F | B) = \sum_{\omega \in F} \mathbb{P}(\omega | B)$.

The first condition is true since, if $\omega \in B$, $\mathbb{P}(\omega) \leq \mathbb{P}(B)$, and therefore $\mathbb{P}(\omega | B) = \mathbb{P}(\omega) / \mathbb{P}(B) \leq 1$. The second condition, which can be rewritten by Proposition (1.12) as

$$\mathbb{P}(B | B) = 1$$

is true since

$$\mathbb{P}(B | B) = \frac{\mathbb{P}(B \cap B)}{\mathbb{P}(B)} = 1.$$

¹ The same remark applies here: the probability is the uniform one and in Subsection 1.4.2 we will see why

The last condition is true since

$$\begin{aligned}\mathbb{P}(F|B) &= \frac{\mathbb{P}(F \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(F)}{\mathbb{P}(B)} \\ \sum_{\omega \in F} \frac{\mathbb{P}(\omega)}{\mathbb{P}(B)} &= \sum_{\omega \in F} \mathbb{P}(\omega|B).\end{aligned}$$

2. As for the second question, since we have just proved that $\mathbb{P}$ is a probability measure on B , it suffices to note that every elementary event has the same probability (recall how we have derived formula (??): simply by assuming that the probability was equal on each elementary event). In fact $\forall \omega \in B \mathbb{P}(\omega|B) = \mathbb{P}(\omega)/\mathbb{P}(B)$ does not depend on ω , since by assumption $\mathbb{P}$ does not depend on ω .

Ex. 1.4.4 — ([Feller], Chapter V.8) Three dice are rolled. If no two show the same face, what is the probability that one is an ace?

Answer (Ex. 1.4.4) — The sample space is $\Omega = \{(i, j, k), i, j, k \in \{1, \dots, 6\}\}$, and the event "No two dice show the same face" is $A = \{(i, j, k) \in \Omega, i \neq j, j \neq k, i \neq k\}$. The event B = "The sum of the two dice is 7" is such that $A \cap B$ is the disjoint union of the events "A and the 1 is in 1st position", "A and the 1 is in 2nd", "A and the 1 is in 3rd position". This union is disjoint since if A happens, a 1 in the 1st position implies that there is no 1 neither in the 2nd nor in the 3rd. In formulas $A \cap B = (A \cap B_1) \cup (A \cap B_2) \cup (A \cap B_3)$, where $B_1 = \{(1, j, k) \in \Omega\}$, $B_2 = \{(i, 1, k) \in \Omega\}$ and $B_3 = \{(i, j, 1) \in \Omega\}$. Thus

$$\begin{aligned}\mathbb{P}(B|A) &= \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} = \frac{\mathbb{P}(A \cap B_1)}{\mathbb{P}(A)} \\ &+ \frac{\mathbb{P}(A \cap B_2)}{\mathbb{P}(A)} + \frac{\mathbb{P}(A \cap B_3)}{\mathbb{P}(A)} = 3 \frac{\mathbb{P}(A \cap B_1)}{\mathbb{P}(A)}.\end{aligned}$$

The event $A \cap B_1 = \{(1, j, k), j \neq k\}$ is composed by $6 \times 5 = 30$ elements. The event A is composed by $6 \times 5 \times 4$ elements. Thus

$$\mathbb{P}(B|A) = \frac{6 \times 5 / 216}{6 \times 5 \times 4 / 216} = \frac{6 \times 5}{6 \times 5 \times 4} = 1/4$$

Ex. 1.4.5 — Consider an urn containing 2 blue marbles, 1 green marble and 1 red marble. Draw randomly (with uniform probability) one of them.

1. What is the probability of extracting a red marble if you know that the extracted marble is not green?
2. We do the following procedure. We draw a marble uniformly. If it is red we draw another marble independently, if not we finish the procedure. We are interested in the colour of the extracted marble: what is the probability of extracting a red marble?
3. Note that it is not necessary to know the probability of drawing in one draw the red ball, that is $\mathbb{P}(r)$. Note also that in the second exercises

gives you a procedure to draw from a bowl where you have removed the red marbles, even if you are drawing from a bowl that has them.

Answer (Ex. 1.4.5) — 1. The sample space is $\Omega = \{r, b, g\}$, $\mathbb{P}(r) = 1/4$, $\mathbb{P}(b) = 1/2$ and $\mathbb{P}(g) = 1/4$. Thus

$$\mathbb{P}(g^c | r^c) = \frac{\mathbb{P}(b)}{\mathbb{P}(b, g)} = \frac{2}{3}$$

2. Now the sample space is more complicated, and resembles the game of Exercise (1.2.12)

$$\Omega = \{r \dots r c_n, n \in \mathbb{N}, c_n \in \{b, g\}\} \cup \{rr \dots\}$$

The probability that we place in Ω is the following: The infinite sequence has probability 0: $\mathbb{P}(r \dots)$, while the finite sequences have probability

$$\begin{aligned} \mathbb{P}(rr \dots r c_n) &= \mathbb{P}(rr \dots r c_n | \text{"first extraction } r") \mathbb{P}(r) = \mathbb{P}(r \dots c_{n-1}) \mathbb{P}(r) \\ &= \dots = \mathbb{P}(r)^{n-1} \mathbb{P}(c_n) \end{aligned}$$

(It will be maybe more clear once the geometric random variable has been obtained) and we take $\mathbb{P}(c) = (\text{number of balls of colour } c) / (\text{total number of balls})$. The event I draw a blue ball is the event $\{b, rb, rrb, \dots\}$ which has probability

$$\begin{aligned} \mathbb{P}(\text{"I draw a blue ball"}) &= \mathbb{P}(b) + \mathbb{P}(b)\mathbb{P}(r) + \mathbb{P}(b)\mathbb{P}(r)^2 + \dots \\ &= \mathbb{P}(b) \left(\sum_{i=0}^{+\infty} \mathbb{P}(r)^i \right) = \frac{\mathbb{P}(b)}{1 - \mathbb{P}(r)} = \frac{\mathbb{P}(b)}{\mathbb{P}(b) + \mathbb{P}(g)} = 2/3 \end{aligned}$$

3. The above procedure can be used to draw on a bowl of. In a more realistic situation, maybe you want to interview families with children, but you don't have at disposal the "bowl", that is, the list with of the families of two children. This exercises tells that it suffices to contact each family until a family with children is contacted. In more complex situations (i.e. in montecarlo simulations), a similar method is used to get to know the conditional probabilities

Example 1.16: Sampling interpretation of the Conditional Probability

Suppose that the Italian population is composed by N individuals, that N_B of them are taller than 175 cm and that N_A of them come from Sardinia. Let the events that a person chosen at random (i.e. with uniform probability) is taller than 175 cm and a comes from Sardinia be A and B , respectively. Then $\mathbb{P}(A) = N_A/N$ and $\mathbb{P}(B|A) = N_{AB}/N_A$, where N_{AB} is the number of people coming from Sardinia being taller

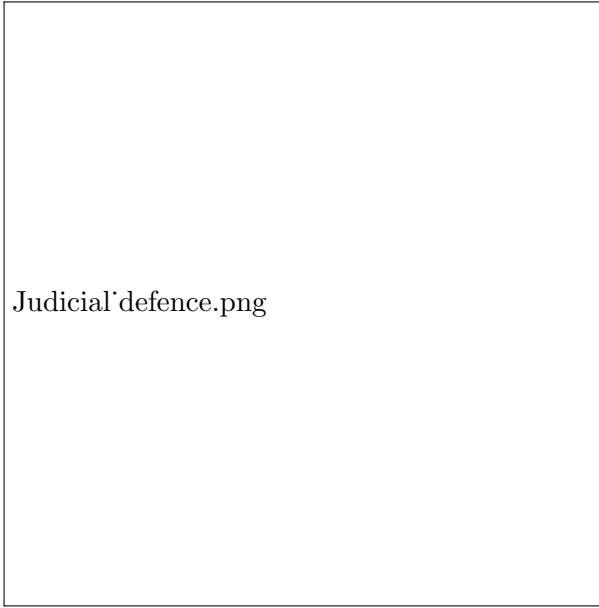


Figure 1.4: Defense argument: The proportion between the two areas is small

than 175 cm. The interpretation of the probability measure $\mathbb{P}(\cdot|A)$ is clear: we are extracting a random person from the subpopulation N_A (which is a population on its own right), the one composed by people from Sardinia .

e:Judicial

A judicial case A woman had been killed, and the principal suspect was his husband. During the investigations, the police discovered that the husband had beaten his wife more than once. His lawyer used some data that showed that only 1/10000 men beating their wives end up killing them, thus claiming that there were no sufficient clues to judge him guilty. Fortunately, the prosecutors pointed a fallacy in the defense argument. 1/10000 gives an estimate of the probability that a man kills his wife knowing that he beats her. Here much more is known: the wife actually died.

Denoting the events

$A = \text{"The husband beat the wife"}$

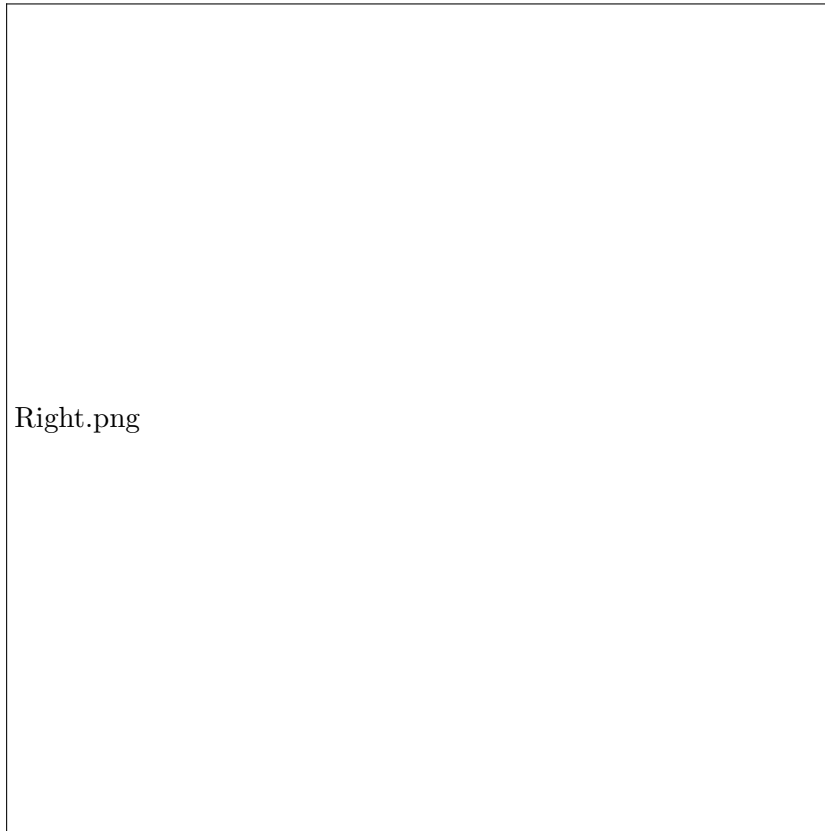
$B = \text{"The woman has been assassinated by his husband"}$

$C = \text{"The woman has been assassinated by a person different from her husband"}$

The claim of the defence is depicted in Figure 1.4 We wish to compute $\mathbb{P}(B|A \cap (B \cup C))$, which is depicted in Figure 1.5.

$$\begin{aligned} \mathbb{P}(B|A \cap (B \cup C)) &= \frac{\mathbb{P}(B \cap A \cap (B \cup C))}{\mathbb{P}(A \cap (B \cup C))} = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}((B \cup C) \cap A)} \\ &= \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(B \cap A) + \mathbb{P}(C \cap A)} = \frac{1}{1 + \frac{\mathbb{P}(C \cap A)}{\mathbb{P}(B \cap A)}}. \end{aligned} \quad (14)$$

e:proportions



Right.png

fig:right

Figure 1.5: What we need to compute: the proportion of the blue area with respect to the total area. Why this is what we need to compute is shown in (14)

Thus it is not $\mathbb{P}(B|A)$ that matters, but only its relative value $\mathbb{P}(B|A)/\mathbb{P}(C|A) = \mathbb{P}(C \cap A)/\mathbb{P}(B \cap A)$, as shown in Figure 1.5. We assume that C and A are independent, so that $\mathbb{P}(C|A) = \mathbb{P}(C)$ and (14) becomes

$$\mathbb{P}(B|A \cup (B \cap C)) = \frac{1}{1 + \frac{\mathbb{P}(C)}{\mathbb{P}(B|A)}}.$$

We estimate the $\mathbb{P}(C)$ by the probability that a woman gets killed: $\mathbb{P}(C) \leq \mathbb{P}(B \cup C)$ which they estimated by $\mathbb{P}(B \cup C) = 1/100000$. Therefore

$$\frac{\mathbb{P}(C|A)}{\mathbb{P}(B|A)} \leq 1/10.$$

Plugging in this value, one obtains that

$$\mathbb{P}(B|A \cap (B \cup C)) \geq 10/11$$

At the end, the man has been judged guilty.

1.4.2 Independence of Events

ss:independence

In view of the definition of conditional probability in Definition ?? the following definition is natural to say that the events A and B are independent if $\mathbb{P}(A) = \mathbb{P}(A|B)$. In a more symmetric form

Definition 1.19. Two events $A, B \subset \Omega$ are *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

e:ind

It is important to notice that the word “independent” can be misleading, since it could be wrongly associated to the absence of a causal relation between A and B . Example ?? shows that there are events not related by any causal relation that are dependent and that, conversely, there are independent events that are causal. The events A and B are independent if from the knowledge of one of them you cannot infer anything on the evaluation of the probability of the other one: the probability remains the same. For this reason the independence property is sometimes stochastic independence. Another way to see that the notions of causality and independence are unrelated is by noting that the notion of independence depends from the probability $\mathbb{P}$, while the notion of causality does not.

Note that the notion of independence is unrelated with the notion of causality. The notion of causality is a notion that takes into account only the relation between events, while the notion of independence takes into account the notion of probability. Independence means that, given $\mathbb{P}$, that is, having an evaluation of the probabilities of the events, we can infer no information from the realization of B that causes to change the evaluation of the probability of A . Thus is a probability dependent hypothesis and, for instance, Exercise (??) shows that coin tosses are not independent if we don’t know exactly the parameter of the coin, since we can infer some information useful to determine the coin parameter by tossing the coin, which, in turn, will change the probability of the next toss. It is the same situation of (1.4.13), which makes clear the striking fact that the number of shark attacks and the number of ice creams sold are numbers with a strong dependence, even if, of course, not related by any causal relation.

Example 1.17: Two independent uniform samplings

Consider a bowl where 3 marbles, 1 red, 1 blue and 1 green, are placed and draw two marbles with replacement. The state space of the system is

$$\Omega = \{(r, r), (r, b), \dots, (g, g)\},$$

We want to prove that the following are equivalent:

- The probability measure on Ω is the uniform one.
- In each single extraction you can draw each marble with the same probability and the extractions are independent

To see what the second condition means precisely, let’s assume that $\mathbb{P}$ is uniform and define the events

$$F_c = \text{“The colour of the first marble is } c\text{”} = \{(c, r), (c, b), (c, g)\},$$

for $c = r, b, g$, and

$$S_c = \text{"The colour of the second marble is } c" = \{(r, c), (b, c), (g, c)\},$$

for $c = r, g, b$. By saying that each extraction is a "random" extraction (that is, is uniform) we mean that

$$\begin{aligned}\mathbb{P}(F_c) &= 1/3 \\ \mathbb{P}(S_c) &= 1/3\end{aligned}\tag{15}$$

1

for each $c = b, g, r$, which is easily seen to hold true. By the fact that the extractions are independent we mean that for each c and c' in $\{r, b, g\}$

$$\mathbb{P}(F_c \cap S_{c'}) = \mathbb{P}(F_c)\mathbb{P}(S_{c'}).\tag{16}$$

2

The event on the left hand side is the singleton $\{(c, c')\}$, so that its probability is $1/9$. The events on the right hand side have probability $1/3$, so that the above equality holds. Let's assume that now (15) and (16) hold. Then

$$\begin{aligned}\mathbb{P}(c, c') &= \mathbb{P}(F_c \cap S_{c'}) \\ &= \mathbb{P}(F_c)\mathbb{P}(S_{c'}) = 1/3 \times 1/3 = 1/9,\end{aligned}$$

so that $\mathbb{P}$ is uniform. This example will be generalised in Section ?? to the case where the two extraction procedure can be performed jointly in an arbitrary way. Examples (1.18) and Exercise (??) and (??) will show examples in which you perform two dependent extractions and how this can lead to surprising results.

Example 1.18: Mouty Hall Problem

MoutyHall

In a game show called the Mouty Hall, the participants had to choose between three doors. Behind 1 door there was a car while behind the other two there was a goat. The participant had to guess which door contained the car. After the first choice, the host disclosed a door which had behind a goat and then asked the participant whether or not he would like to change the choice of the door. The result is at a first sight surprising. Let's call Strategy 1 the strategy in which we don't change the door, and Strategy 2 the strategy in which we decide to change the door. Then $\mathbb{P}(\text{"win"}) = 1/3$ if we play following Strategy 1, and $\mathbb{P}(\text{"win"}) = 2/3$ if we play by Strategy 2. To prove the above fact, assume that the participant choses the door 1 and that he plays following Strategy 2. The result is clear if one observes that the participant wins if and only if he chooses a goat at

the beginning. **Exercise:** Consider the Mouny Hall problem with n doors and k cars. What is the probability of winning under Strategy 2?

Solution The sample space should encode the doors, the thing that is behind the door, the first door chosen by the participant and its second choice. However, we can give a reduced description since we are interested only on the choices the participant makes. The sample space can be taken to be $\Omega = \{CC, GG, GC, CG\}$ where the first digit denotes what is there behind the first door and similarly for the second. Let's denote the event "The first choice is a car" by F , and the event "The second choice is a car" = "win" by S . Therefore

$$\mathbb{P}(S) = \mathbb{P}(S|F)\mathbb{P}(F) + \mathbb{P}(S|F^c)\mathbb{P}(F^c),$$

but, given F , after the goat is revealed, the participant chooses between $n-2$ doors with $k-1$ cars. Given F^c , then, after the host reveals the goat, has to choose between $n-2$ doors and k cars. Therefore

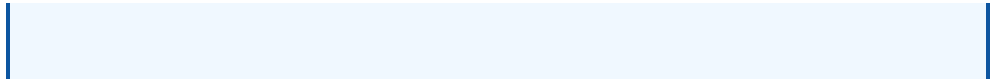
$$\begin{aligned}\mathbb{P}(S) &= \mathbb{P}(S|F)\mathbb{P}(F) + \mathbb{P}(S|F^c)\mathbb{P}(F^c) \\ &= \frac{k-1}{n-2} \frac{k}{n} + \frac{k}{n-2} \frac{n-k}{n}\end{aligned}$$

So, it is not difficult to understand the Mouny Hall problem: what happens is that the different extractions, the first door and the second one are dependent.

The next "paradox" has interest on its own and shows that it is important to clearly state the conditioning event

Example 1.19: Boy or Girl Paradox

Consider a family with two children and consider their genders. The state space is $\Omega = \{MM, MF, FM, FF\}$ and consider the uniform probability on it. ⁸



1. What is the probability that the second child of a couple is a boy, knowing that the first one is a boy?
2. Assume that we get to know that one of the children is a boy. What is the probability that both children are males?
3. We have chosen a child randomly (and independently on the gender, of course) and have seen that he is male. What is the probability that both children are boys? (Note that the first point is a particular case of this one, when the chosen child is surely the first one).

For the first question, look at the sample space $\Omega = \{(M, M), (M, F), (F, M), (F, F)\}$ and endow it with the uniform probability (see Example 1.17). The event $A = \text{"The first child is male"}$ is given by $\{(M, M), (M, F)\}$. A basic computation shows that

$$\mathbb{P}((M, M)|A) = \mathbb{P}(M, M)/\mathbb{P}(A) = \frac{1/4}{2/4} = 1/2.$$

As for the second question, maybe counterintuitively, the answer is $1/3$. In this case the conditioning event A is $\{(M, M), (M, F), (F, M)\}$, and

$$\mathbb{P}((M, M)|A) = \mathbb{P}(M, M)/\mathbb{P}(A) = \frac{1/4}{3/4} = 1/3.$$

In the last case, the probability is $1/2$! Let's take as state space $\Omega = \{MM1, MM2, MF1, MF2, FM1, FM2, FF1, FF2\}$, where the last number means that you have observed the gender of the child 1 or child 2. The event $A = \text{"you observed a male child"} = \{MM1, MM2, MF1, FM2\}$, while the event "The couple has two male children" is $B = \{MM1, MM2\}$. Since we assume that the choice is done independently, we have that if we denote $1 = \text{"I observed the first child"}$ $\mathbb{P}(MM1|1) = \mathbb{P}(MF1|1) = \mathbb{P}(FF1|1) = 1/4$. Similarly for the event $2 = \text{"I observed the second child"}$. We assume that $\mathbb{P}(1) = p$ and we compute

$$\mathbb{P}(A) = \mathbb{P}(\{MM1, MF1\}|1)\mathbb{P}(1) + \mathbb{P}(\{MM2, FM2\}|2)\mathbb{P}(2) = p/2 + (1-p)/2 = 1/2,$$

and

$$\mathbb{P}(\{MM1, MM2\}) = \mathbb{P}(MM1|1)\mathbb{P}(1) + \mathbb{P}(MM2|2)\mathbb{P}(2) = 1/4.$$

$$\mathbb{P}(\{MM1, MM2\}|A) = \mathbb{P}(A \cap B)/\mathbb{P}(A) = 1/2.$$

Even if the second point of the example might come as a surprise, if one interprets the probability with a wrong frequentist definition, if one were to obtain the list of all the families with two children such that one is a male, then approximately $1/3$ of the families in those lists

will have 2 male children. The procedure 3 would have the following equivalent in the frequentist approximation. From the list of all the families with two children, you toss an unfair coin to decide which children to take into account and to check. So there is a

Ex. 1.4.6 — You toss a fair coin 4 time. Assume that every outcome is equally probable (we will rigorously see why). Show that the events "The fourth toss gave heads" and the event "The first three tosses gave head" are independent.

Ex. 1.4.7 — In the setting of Example 1.17 you draw two marbles without replacement.

1. What are the probabilities of each single extraction?
2. Are extractions independent?

ex:3indep

Ex. 1.4.8 — $\Omega = \{1, 2, 3, 4\}$ and $\mathbb{P}(i) = 1/4$ for every $i \in \Omega$. Show that $A = \{1, 2\}$ $B = \{1, 3\}$ and $C = \{2, 3\}$ are pairwise independent but not independent.

Ex. 1.4.9 — 3 prisoners, (A,B,C) are condemned to death. They decide to save one of them, but the name will be given only some minutes before the executions. A asks to a guard if he will be saved, but gets no answer. A asks then which one between B and C will be executed, and the guard says C. At this point A says. "Now only one between me and B will be executed; I have 50% of chances to survive ". Is it correct his reasoning?

Ex. 1.4.10 — There are three cards one of them has both faces black, one with both red faces, and the last one with one black face and the other face red. You draw a card and observe a red face. What is the probability that the other face is red?
This example shows that is important to clearly state the conditioning event.

1. What is the probability that the second child of a couple is a boy, knowing that the first one is a boy?
2. We know that one of the child is a boy. What is the probability that both children are males?
3. We have chosen a child randomly (and independently on the gender, of course) and have seen that he is male. What is the probability that both children are boys? (Note that the first point is a particular case of this one, when the chosen child is surely the first one).

For the first question, look at the sample space $\Omega = \{(M, M), (M, F), (F, M), (F, F)\}$ and endow it with the uniform probability (in section 3.2.1 we will see that this is due to the fact that we regard the gender of the two child as independent, and the probability of each single child to be either a boy or a girl $1/2$).

The event $A = \text{"The first child is male"}$ is given by $\{(M, M), (M, F)\}$. We are interested in

$$\mathbb{P}((M, M)|A) = \mathbb{P}(M, M)/\mathbb{P}(A) = \frac{1/4}{2/4} = 1/2.$$

Maybe counterintuitively, the answer is $1/3$. But this is clear since the conditioning event A is $\{(M, M), (M, F), (F, M)\}$, and

$$\mathbb{P}((M, M)|A) = \mathbb{P}(M, M)/\mathbb{P}(A) = \frac{1/4}{3/4} = 1/3.$$

Using a frequentist approach, if one had the list of all the couples with two children, one of which male, approximately $1/3$ would have two male children.

In the last case, the probability is $1/2$! Let's take as state space $\Omega = \{MM1, MM2, MF1, MF2, FM1, FM2, FF1, FF2\}$, where the last number means that you have observed the gender of the child 1 or child 2. The event $A = \text{"you observed a male child"} = \{MM1, MM2, MF1, FM2\}$, while the event $B = \text{"The couple has two male children"} = \{MM1, MM2\}$. Since we assume that the choice is done independently, we have that if we denote $1 = \text{"I observed the first child"}$ $\mathbb{P}(MM1|1) = \mathbb{P}(MF1|1) = \mathbb{P}(FF1|1) = 1/4$. Similarly for the event $2 = \text{"I observed the second child"}$. We assume that $\mathbb{P}(1) = p$ and we compute

$$\mathbb{P}(A) = \mathbb{P}(\{MM1, MF1\}|1)\mathbb{P}(1) + \mathbb{P}(\{MM2, FM2\}|2)\mathbb{P}(2) = p/2 + (1-p)/2 = 1/2,$$

and

$$\mathbb{P}(\{MM1, MM2\}) = \mathbb{P}(MM1|1)\mathbb{P}(1) + \mathbb{P}(MM2|2)\mathbb{P}(2) = 1/4.$$

$$\mathbb{P}(\{MM1, MM2\}|A) = \mathbb{P}(A \cap B)/\mathbb{P}(A) = 1/2.$$

This apparent paradox is due to the fact that the event A is different from the event $B = \text{"There is at least one male child"}$. If you repeat the same calculations with B instead of A , you should obtain $1/3$.

You can interpret this example as the fact that it is important to know how to get the information and it is important to specify the conditioning event rigorously.

Definition 1.20. Three events A, B and C are independent if they are pairwise independent

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B),$$

$$\mathbb{P}(A \cap C) = \mathbb{P}(A)\mathbb{P}(C),$$

$$\mathbb{P}(B \cap C) = \mathbb{P}(B)\mathbb{P}(C),$$

and

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C).$$

An example three events that are pairwise independent but not independent is given in Exercise 1.4.8.

1.4.3 Sampling Interpretation of Conditional Probability

ss:r'cond

We now come back to the example of Subsection ??, where $\Omega = \{\omega_1, \dots, \omega_6\}$ and we assume that we observe $E = \{\omega_1, \omega_4, \omega_5, \omega_6\}$. We can define the conditional probability by setting $p_2 = p_3 = 0$ and normalising the thus obtained vector so that the sum is 1.

Looking at Example ??, if we know how to sample from $\mathbb{P}$, and indeed we were able to do it in Subsection ??, we can sample from the $\mathbb{P}()$

1.4.4 Total probability Formula

When speaking about events many times one reasons by cases in the following sense. Assume you don't know whether to go out this night, then, in order to think at the event "I will go out this night" one thinks at this event first in the case it rains, then in the case it is good weather. Since the probability is simply a quantitative version of the logic we use when speaking about events, we might hope to reproduce the same reasoning at the quantitative level of probabilities. This is indeed true and it is done in the Total Probability Formula.

Proposition 1.21 (Total Probability Formula for two conditioning events). *Let $B \subset \Omega$ be an event such that $\mathbb{P}(B) \in (0, 1)$. Then*

$$\mathbb{P}(A) = \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^c)\mathbb{P}(B^c) \quad (17)$$

e:totaleasy

Proof.

$$\mathbb{P}(A) = \mathbb{P}(A \cap \Omega) = \mathbb{P}(A \cap (B \cup B^c)) = \mathbb{P}((A \cap B) \cup (A \cap B^c)).$$

Since $A \cap B$ and $A \cap B^c$ are disjoint, we can use the (11) to develop the above expression into

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} + \frac{\mathbb{P}(A \cap B^c)}{\mathbb{P}(B^c)} \\ &= \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^c)\mathbb{P}(B^c), \end{aligned}$$

which is what we wanted to prove. $\square$

Ex. 1.4.11 — We toss a fair coin. If the coin gives heads we draw a ball from a bowl that contains a blue ball, a red ball and a white ball. If the coin gives tails, we draw a ball from a bowl that contains a blue ball or a red ball. What is the probability of drawing a red ball?

Answer (Ex. 1.4.11) — Let's call B ="The coin gave heads". We know that $\mathbb{P}(B) = \mathbb{P}(B^c) = 1/2$. We also know that the event r =" We draw a red ball" has the following probabilities conditioned on B :

$$\mathbb{P}(r|B) = 1/3, \quad \mathbb{P}(r|B^c) = 1/2.$$

We can apply formula (17) to obtain

$$\mathbb{P}(r) = \mathbb{P}(r|B)\mathbb{P}(B) + \mathbb{P}(r|B^c)\mathbb{P}(B^c) = 1/2 \cdot 1/3 + 1/2 \cdot 1/2 = 5/12.$$

We now give the total probability formula in full generality. The proof is almost the same, but we need to introduce the concept of partition of the sample space Ω :

Definition 1.22. The events $B_1, \dots, B_n$ are said to be mutually disjoint if $B_i \cap B_j = \emptyset$ whenever $i \neq j$. A partition is a set $\{B_1, \dots, B_n\}$ of mutually disjoint events B_i , such that $\Omega = \bigcup_{i=1}^n B_i$.

Examples of partitions are

- In 4 coin tosses, a partition of $\Omega = \{0000, 0001, \dots, 1111\}$ in two events is, for example, $\{B_1, B_2\}$, where B_1 ="The first toss gave tail" and $B_2 = (B_1)^c$ ="The first toss was head". Another partition would be $\{C_0, C_1, C_2, C_3, C_4\}$ where C_i = "There have been i heads".
- In the roll of a die, an example of partition of $\Omega = \{1, 2, 3, 4, 5, 6\}$ is $\{B_1, B_2\}$, where $B_1 = \{2, 4, 6\}$ = "The result is an even number", and $B_2 = \{1, 3, 5\}$ = "The result is an odd number".
- If we roll two dice, a partition is $\{B_2, \dots, B_{12}\}$, where B_i is the event "The sum of the dice is i ".
- We observe some coin tosses but we don't know how many of them there will be. Thus Ω is the one given by (??) and it can be partitioned into $\Omega = \bigcup_n \Omega_n$, where Ω_n =" There have been n tosses " = $\{x_1 \dots x_n, x_i \in \{0, 1\}\}$.

In other words a partition is a family of mutually disjoint events (or hypotheses) that together they include all the possible eventualities, conditions intuitive and necessary to make reasonings of the form " If B_1 , then and you can reason separately on each element of the partition, both at the level of sets and at the level of probabilities thanks to the next Note that if a partition consists of two events B_1 and B_2 , then $B_2 = (B_1)^c$.

Proposition 1.23 (Total probability formula). *Let $\{B_1, \dots, B_n\}$ be a partition of Ω . Then*

$$\mathbb{P}(A) = \mathbb{P}(A|B_1)\mathbb{P}(B_1) + \dots + \mathbb{P}(A|B_n)\mathbb{P}(B_n).$$

We omit the prove, which is similar to the previous and we remark that, up to some basic knowledge of series, one could take a partition to be an infinite family of the form $\{B_i\}_{i \in \mathbb{N}}$, and the total probability formula would still be true.

Ex. 1.4.12 — Consider an infection that is contagious and which is suffered by 1 person out of 1000. There is a test used to determine whether a person has the infection and the test has a 5% of possibility of giving a false positive: You don't have the infection but the test is positive, while the probability of having a false negative is zero.

1. If you are tested positive, what is the probability of actually having the infection?
2. You have some symptoms and the doctors estimate around 50% of probability that you have the infection. What is the probability that you have the infection provided that the test gives a positive result?

ex:indep

Ex. 1.4.13 — We have two urns, one containing 3 red balls, 1 blue ball, and one white ball, the other one containing 3 blue balls, 1 red ball and 1 white ball. We toss a fair coin to choose one of the two urns.

1. We draw a ball from the chosen urn. What is the probability that you extract a red ball? (Hint: Use the total probability formula)
2. Now we extract a ball, reinsert it in the urn, and draw again another ball from the same urn. Are the event "The first ball extracted is red" and the event "The second ball extracted is red" independent?

1.4.5 Odds and Bayes Formula

2 | STATISTICS

3 | STATISTICS

3.1 INTRODUCTION: DATA AND MODELS

3.1.1 Data

ss:data

One of scopes of statistics is to infer knowledge from data, which often comes in the form of many different instances of the same variables. Next we briefly introduce the kind of data we will deal with and the corresponding types in R. First we differentiate between numerical variables and non-numerical variables. In between numerical variables we distinguish integer variables and real (continuous) variables. Integer variables usually come from counting procedures, e.g. the number of children, and real variables can be seen as those that require to fix a precision to truncate the decimal expansion, e.g. height in cm, which otherwise would go on indefinitely.

```
typeof(2)

## [1] "double"

class(2)

## [1] "numeric"

typeof(2L)

## [1] "integer"

class(2L)

## [1] "integer"

sample_size <- 10000
# We generate artificially a sample of integers
Age <- sample(15:70, sample_size, replace = T)
typeof(Age)

## [1] "integer"

class(Age)

## [1] "integer"

# We generate artificially a sample of continuous data
Income <- rexp(sample_size)
typeof(Income)

## [1] "double"

class(Income)

## [1] "numeric"
```

Among non-numerical variables we are particularly interested in categorical variables, that is, in those variables that can assume only a finite number of values, e.g., the gender. The set of possible choices for the variable is denoted the set of levels, and this can be ordered or unordered. Ordered categorical variables are stored as an ordered factor in R and unordered categorical variables in factors

```

levels j- c('Formal', 'Informal')
Variant`char j- sample( levels, sample`size, replace = T, prob = c(0.7,0.3) )
Variant j- factor( Variant`char, levels = levels)
typeof(Variant)

## [1] "integer"

class(Variant)

## [1] "factor"

typeof(Variant`char)

## [1] "character"

class(Variant`char)

## [1] "character"

Education`levels j- c('Diploma', ' Bachelor\'s Degree', 'Master\'s Degree', 'Ph.D. ')
Education`char j- sample(Education`levels, sample`size, replace = T )
Education j- factor( Education`char, levels = Education`levels, ordered = T )
typeof(Education)

## [1] "integer"

class(Education)

## [1] "ordered" "factor"

```

The data in the previous chunk of code can be seen as the result of a linguistic experiment that consists in collecting data on the frequency with which one of the two variants, one formal and the other one informal, is used. The data structure that we use to collect the data is the *data frame*, whose columns consist on different variables and whose rows are the different observations of the same variable. Therefore, the output of a linguistic experiment might look like

```

library(uuid)
Informant`id =replicate(sample`size, UUIDgenerate( ))
Experiment`result j- tibble( Informant`id = Informant`id, Age = Age, Income = Income, Education = Education)

## Error in tibble(Informant_id = Informant_id, Age = Age, Income = Income,
: non trovo la funzione "tibble"

```

```
head(Experiment$result )
## Error in head(Experiment$result): oggetto 'Experiment$result' non trovato
```

The different columns represent the different variables we are observing, while different rows correspond to different observations

3.1.2 Models

ss:models

Once the data into the tibble

```
head(Experiment$result)
## Error in head(Experiment$result): oggetto 'Experiment$result' non trovato
```

the crucial question becomes how to use it to infer knowledge about the variables. This is at its core a complicated question of epistemology. Narrowing our scopes, recall from Subsection 0.1.2 that an empirical, probabilistic model for the data is the most we can hope for. Statistics uses the probability theory framework to see the observed data as an instance of all the different possible tibbles that we could have obtained, which constitutes the sample space

Ω = The set of all possible tibbles.

The observed tibble is therefore an element of Ω , and we can evaluate the probability, or degree of belief, of other possible tibbles. Denote by $\mathcal{P} = \{\mathbb{P}, \mathbb{P} \text{ is a probability over } \Omega\}$ the set of probability measures on Ω . Rather counterintuitively, to make any use of the data we first need to make assumptions on it, as outlined in the following scheme

- We define our *model*, that is, we make assumptions on the data Ω . Formally, we fix a subset $\mathcal{Q} \subset \mathcal{P}$, and we assume that the obtained tibble is extracted according to some distribution in $\mathcal{Q}$. In other words, we restrict to consider not all the possible probabilities on Ω , but only subset $\mathcal{Q}$ of them. Typically the subset $\mathcal{Q}$ is of the form $\mathcal{Q} = \{\mathbb{P}_\theta, \theta \in \Theta\}$, where Θ is a set of indexes. In this case θ is said to be the parameter of the model, see Example 3.2 for an example where $\theta \equiv p$ and $\Theta = [0, 1]$.
- Once we observe the data, we can try to guess which of the $\theta \in \Theta$ is more likely to have generated the data. That is, we *infer* information on the parameter θ .

The above abstract framework requires an example

Example 3.1: Sampling

ex:sampling`model

The model that it is used for sampling k individuals from a population of n individuals $\{a_1, \dots, a_n\}$ is the one of extractions with or without replacements. We consider for simplicity the case where replacement

occurs, and the model consists in considering the people from the population you are extracting from as balls in an urn, see also Example ?? and Example ?. The model consists on one single probability $\mathcal{Q} = \{\mathbb{P}\}$, the uniform probability on each one of the outcomes, collected in the sample space

$$\Omega = \{x_1 \dots x_n\}$$

We recall that this choice is due to the independence of the choice of each single person and on the fair (uniform) choice of each person, see Example 12. However, this conditions are not so straightforward to achieve and statisticians organizing polls have to find ways to choose a person among the others in a way as uniform as possible. For instance, polls made by cell phone calls in the early 2000 would miss a good portion of the population. In Example ?? that this sampling model when we restrict to look not at the individual but at a feature that could be common between individuals, say the eye color, gives rise to different models.

Example 3.2: Bernoulli trials

x:Bernoulli model

Consider the vector *Variants* defined in Subsection 3.1.1, which consists in position i the variant of the word used by the i -th informant. In non mathematical terms, the Bernoulli trials model for the vector *Variants* is the following. Each informant, independently one from another, tosses a coin of parameter $p \in [0, 1]$ and uses the formal variant if the coin shows heads. To put this model in a more formal way, consider the vector $\underline{X} = (X_1, \dots, X_n)$, where $n \in \mathbb{N}$ is the number of informants and X_i indicates whether the formal variant has been used :

$$X_i = \begin{cases} 1 & \text{if } i\text{-th informant used the Formal variant} \\ 0 & \text{otherwise,} \end{cases}$$

e:Bernoulli

for $i = 1, \dots, n$. In the next chunk of code we define the vector $\underline{X}$

```
Variant'numeric = as.numeric( Variant == "Formal" )
Varian'numeric[1:10]
```

```
## Error: oggetto 'Varian_numeric' non trovato
```

The model consists in assuming that the real world data $\underline{X}$ has been extracted according to one of the distributions contained in the set $\mathcal{Q} = \{\mathbb{P}_p, p \in [0, 1]\}$, where $\mathbb{P}_p$ is the Bernoulli trial measure defined in ??, which, for the sake of readability, we recall next. Under this

model, the probability of seeing the result $\underline{x} = (x_1, \dots, x_n)$, where $x_i \in \{0, 1\}$ for each $i = 1, \dots, n$, is

$$\mathbb{P}_p(\underline{X} = \underline{x}) = p^{x_1 + \dots + x_n} (1 - p)^{n - (x_1 + \dots + x_n)}, \quad (1)$$

e:Bernoulli1

Now, and only now, the question of which one of the $\mathbb{P} \in \mathcal{Q}$ is more likely to have generated the data is well posed. In the present case one is tempted to say that the data has been generated according to $\mathbb{P}_{\hat{p}}$, for $\hat{p}$ = "The proportion of times the formal variant has been used". This intuitive claim, if H_0 is true, is a consequence of Theorem ??, see Corollary 3.2.1

```
mean(Variant`numeric)
```

```
## [1] 0.6933
```

```
# This is a good guess for the coin parameter $p$.
```

Note that, in this case, the data has been actually generated using a Bernoulli trial measure of parameter $p = 0.7$ (compare it with the estimated one!) and, therefore, the model $\mathcal{Q}$ contains the actual distribution that has generated the data.

Despite being false in general, models are often useful. In some cases, as in the sampling case Example 3.1 and the following, are even close to reality

Example 3.3: Sampling 2

ex:sampling`features

Assume we are sampling from a population, see Example 3.1, and that we are looking at some features rather than at the exact person extracted. If the feature has only two possibilities, 0 or 1, then the Bernoulli measure model, see Example 3.2 is a good model. As an example, consider we are performing a statistical analysis on the number of languages known by the Moscow population. The whole population of Moscow is saved in the factor *Moscow_population*

```
levels j- c('Bilingual', 'Not Bilingual')
```

```
Moscow`population`char j- sample( levels, size = 1000000, replace = T, prob = c(0.7, 0.3) )
```

```
Moscow`population j- factor(Moscow`population`char, levels = levels)
```

```
# This is the whole Moscow population
```

The whole population is inaccessible to us, and, in particular, we cannot see how many people are bilingual by the simple command

```
real_p <- mean( Moscow_population == 'Bilingual')
real_p

## [1] 0.700764
```

However we can take samples (where we allow to select the same person twice, even if this happens with small probability)

```
sample_size = 100
Sampled_results <- sample(Moscow_population, sample_size, replace = T)
head(Sampled_results)

## [1] Bilingual    Bilingual    Not Bilingual Not Bilingual Bilingual
## [6] Bilingual
## Levels: Bilingual Not Bilingual
```

Each extraction is independent from another, and we extract a Bilingual citizen with probability $real_p$, so that, denoting as usual a bilingual by 1 and a non bilingual by 0, the corresponding vector of 0s and 1s follows a Bernoulli trial distribution $\mathbb{P}_{p_real}$, see Definition ?? . The parameter p_real is unknown to us, but we know that the distribution is one from the Bernoulli model $\mathcal{Q} = \{\mathbb{P}_p, p \in [0, 1]\}$. In more computational terms, by saying that the Bernoulli model is a good model for this experiment, we mean that the two samples generated by the following code

```

# The two commands
sample.size j- 100
sample( Moscow`population, size = sample.size , replace = T)

## [1] Not Bilingual Bilingual Bilingual Bilingual Not Bilingual
## [6] Bilingual Bilingual Not Bilingual Not Bilingual Not Bilingual
## [11] Not Bilingual Not Bilingual Bilingual Not Bilingual Not Bilingual
## [16] Bilingual Bilingual Bilingual Bilingual Bilingual
## [21] Bilingual Bilingual Bilingual Bilingual Bilingual
## [26] Not Bilingual Bilingual Not Bilingual Not Bilingual Bilingual
## [31] Bilingual Not Bilingual Bilingual Not Bilingual Not Bilingual
## [36] Not Bilingual Bilingual Not Bilingual Not Bilingual Bilingual
## [41] Bilingual Bilingual Not Bilingual Not Bilingual Bilingual
## [46] Not Bilingual Bilingual Not Bilingual Bilingual Bilingual
## [51] Bilingual Bilingual Bilingual Not Bilingual Bilingual
## [56] Bilingual Bilingual Not Bilingual Not Bilingual Bilingual
## [61] Bilingual Not Bilingual Bilingual Not Bilingual Bilingual
## [66] Bilingual Not Bilingual Bilingual Bilingual Bilingual
## [71] Bilingual Bilingual Bilingual Bilingual Bilingual
## [76] Not Bilingual Bilingual Bilingual Bilingual Bilingual
## [81] Not Bilingual Bilingual Bilingual Bilingual Not Bilingual
## [86] Not Bilingual Bilingual Bilingual Not Bilingual Not Bilingual
## [91] Bilingual Bilingual Not Bilingual Bilingual Bilingual
## [96] Bilingual Bilingual Bilingual Not Bilingual Not Bilingual
## Levels: Bilingual Not Bilingual

# and
rbinom( n = sample.size, size = 1, prob = mean(as.numeric(Moscow`population == 'Bilingual'))

## [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 0 0 0 1 1 1 1 1 1 0 1 1 0
## [38] 0 1 1 1 1 1 1 0 1 0 1 0 1 0 1 0 1 1 1 1 0 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1
## [75] 1 1 0 1 0 1 1 1 1 1 1 1 1 1 1 1 0 0 0 1 0 1 1 1 1 1 1 1 1

# have the same distribution. rbinom samples from an abstract population

have the same distribution. Since the sampling model from the whole
population in Example ?? is a good model for, then also, the

```

Note the similarity between the two examples. In both cases we do not have access to the object we are sampling from, either if it is a real population or a generalized population (a distribution), but we can take samples from them. To summarise this section, what statistics does is to consider data as a sample from a set of possible distributions ($\mathcal{Q}$), and try to infer knowledge on the distribution.

Example 3.4

Eugene Onegin 2 In Example ?? we have introduced a sequence of os and 1s coming from drama texts, replacing the vowels therein by o and the consonants by 1, and we have saved it into the variable `text_01`. In the same Section ?? we also have introduced the following two models

1. H_0 The sequence is extracted from a Bernoulli measure, that is, $\mathcal{Q} = \{\mathbb{P}_p, p \in [0, 1]\}$, see Definition ??.
2. H_1 The sequence is extracted from a Markov measure, that is, $\mathcal{Q} = \{\mathbb{P}_{a,b}, a, b \in [0, 1]\}$, see Definition ??

Once we have defined the models in plain terms, we can proceed estimating the parameters. As will be proved in Example ?? If H_0 is true a good estimate for p , denoted by $\hat{p}$, is the proportion of 1s in the string, that is

```
hat_p <- mean(text_01)
```

```
## Error in mean(text_01): oggetto 'text_01' non trovato
```

Example ?? shows that, if H_1 is true, a good estimate for a and b are the proportion of jumps from 0 to 1 among all the jumps from 0 and the proportion of jumps from 1 to 0, among the total number of jumps from 1, respectively:

```
n_01 <- sum(from == 0 & to == 1)
```

```
## Error: oggetto 'from' non trovato
```

```
n_00 <- sum(from == 0 & to == 0)
```

```
## Error: oggetto 'from' non trovato
```

```
n_10 <- sum(from == 1 & to == 0)
```

```
## Error: oggetto 'from' non trovato
```

```
n_11 <- sum(from == 1 & to == 1)
```

```
## Error: oggetto 'from' non trovato
```

```
# Compute the proportions
```

```
a_hat <- n_01 / (n_00 + n_01) # P(1 | 0)
```

```
## Error: oggetto 'n_01' non trovato
```

```
b_hat <- n_10 / (n_10 + n_11) # P(0 | 1)
```

```
## Error: oggetto 'n_10' non trovato
```

We can also see the distribution of Y under the above two claims are, once more, a consequence of the Law of Large numbers ??, see Corollary ????. Finally, notice that models are an oversimplification of reality, so that $\mathbb{P}(H_0) = \mathbb{P}(H_1)$. However, we are not interested in absolute probabilities but in relative probabilities, such as, for instance, the odds $\mathbb{P}(H_1)/\mathbb{P}(H_0)$, which tell you which of the two models you regard as more likely.

3.2 TESTING

Let Ω be the sample space of all the possible data we could have obtained in an experiment and assume we have modelled it according probabilistically by assigning probability measure $\mathbb{P}$, which also weights the probabilities of obtaining different data. However, the same data could be possibly have arisen from various other alternative models. We denote the initial probabilistic model H_0 , and, by theoretical considerations that are behind the system of study, we formulate alternative hypotheses H_1 . Tests give you a decision rule and a framework that allow to say that there is enough evidence to reject the null hypothesis and to say that there and we wonder whether other alternative models might fit better. The same data could have been obtained even if it were sampled from a different probability measure Q . How to decide which model is more likely to be true?

3.2.1 A simple example: Deciding which distribution is more likely to have generated the data

- Assume we are observing a random variable X , so that Ω is the set of all the possible values for it. To fix the ideas, think of X as the logarithm of the reaction time to a linguistic stimulus. Imagine that researcher have long established the model which constitutes the *null hypothesis* H_0 : X distributes according to a normal of parameters $\mu = 1$ and $\sigma = 2$. Your own research suggests that X would rather follow the distribution Assume that we obser

Example 3.5: T

3.2.2 Ideas and Terminology

Every statistical test follows the same intuitive framework, which we introduce by means of a simple example. A magician claims to have magical abilities and we decide to test whether it is true or not by the following *experiment*. I will toss a fair coin n times, each time making the coin hit the table and then falling to the floor (in this way the tosses are independent). In the meantime, the magician will try to use his abilities to make the coin show more often heads than tails. The hypotheses we want to test are two, the *null hypothesis* H_0 , which is the conservative one, which states that he is not a magician and *alternative hypothesis* H_1 , that tells us that he is a magician. This, however, is unprecise, since the hypotheses are precise statements on the model, that is, the probability, with which the data obtained in the experiment distribute:

- H_0 The coin tosses follow a Bernoulli trial measure of parameter $p = 1/2$ (independent tosses of a fair coin)

- $H_1: p > 1/2$

Would the result HHH be sufficient to *reject* H_0 in favour of H_1 ? No, the result could be due to chance, that is, it is an “unlikely” result of H_0 even if it would be less unlikely under H_1 . On the other extremum, $HHHHHHHHHHHHHHHHHHHH$ would suffice. The reason is that the result seems so unlikely under H_0 that maybe H_0 is not true. A test is a sort of statistical proof by contradiction: we cannot say that the results are impossible if H_0 is true, but we can say that they are very unlikely.

However we must make precise what unlikely means, since every string has the same probability. We define the *statistic* of the number of heads as

$$H(\underline{X}) = X_1 + \dots + X_n$$

e:heads'numb

where X_i is 1 if the i -th toss gave heads and 0 otherwise and $\underline{X} = (X_1, \dots, X_n)$ denotes the vector of the coin tosses. H behaves differently whether H_0 or H_1 holds, and in the second case tends to be bigger. We recognize that in the long sequence the statistic $H(\underline{X})$ attains a value that is not typical if H_0 were true.

Having fixed a statistic and a notion of extremality (extremal results are the ones where $H \gg n/2$), the decision rule is therefore to fix a h_{crit} and say that if $H(\underline{X}) > h_{\text{crit}}$ we reject H_0 in favour of H_1 . The scheme is given in Figure ?? and the test is defined once we fix h_{crit} .

This decision rule, however, cannot lead out (by the statistical nature of the two hypotheses) the possibility of two kind of errors, see Figure ??: type I error, where we reject H_0 in favour of H_1 while H_0 is true, and type II error, where we keep H_0 , while H_1 is true. That is, if $H > h_{\text{crit}}$, while H_0 is true, we say that H is extremal due to randomness and we are committing a type I error. Otherwise, if $H < h_{\text{crit}}$ and H_1 holds, we fail to see a real effect and we speak about type II error. Since it is arguabilty more dangerous to arrogate to a normal person the condition of magician than not recognizing that a person is a magician, we would like to have a control on the type I error. We therefore fix a *significance level* α , usually $\alpha = 0.05$, and we impose that if H_0 holds, we accept that we can make an error with probability α :

$$\mathbb{P}(\text{"}H_0 \text{ rejected"}|H_0) = \mathbb{P}(H \geq h_{\text{crit}}|H_0) = \alpha \quad (2)$$

e:value

In graphical terms, we choose h_{crit} in such a way that the area in red in the H_0 part, see Figure ??, that is $H_0 \cap \{H > h_{\text{crit}}\}$ is a α of the total area of H_0 . The relation (2) defines $h_{\text{crit}} = h_{\text{crit}}(\alpha)$, so that the decision rule is defined provided once we fix a significance level (see the end of this Subsection for the solvability the relation (2) between α and h_{crit} .) We now fix $\alpha = 0.05$ and try to guess h_{crit} for $n = 8$ tosses. In the next subsection, we will see how the h_{crit} is found with simple R commands

```
# We know that if H'0 is true
# H distributes as a binomial
# of parameters n = 10, p = 1/2
# We now plot on the P( H = i — H'0) vs i
alpha = 0.05
```

test.png

f:type'errors

```
n = 10
h`guess`;- 10 # We observe 8 heads
ggplot(data = tibble(x = 0:n, y = dbinom(0:n, size = n, prob = 1/2))) +
  geom`point(aes(x,y)) +
  geom`vline(xintercept = h`guess, col = 'red')

## Error in ggplot(data = tibble(x = 0:n, y = dbinom(0:n, size = n, prob =
1/2))): non trovo la funzione "ggplot"

#  $P(H = 8 - H^0) = P(H = 10 - H^0) + P(H = 9 - H^0) + P(H = 8 - H^0)$ 
# We have to sum the height of the points on the right hand side of the
# vertical line
sum( dbinom(h`guess:n, n, prob = 1/2))

## [1] 0.0009765625

# pbinom sums the heights on teh left hand side. Therefore
1-pbinom( h`guess -1 , size = n, prob = 1/2)

## [1] 0.0009765625
```

```

# The result is smaller than alpha = 0.05, let's try
# to repeat the code with a smaller guess.
h`guess` <- 9
1- pbinom( (h`guess`-1),size = n, prob = 1/2)

## [1] 0.01074219

# It is still smaller than 0.05. Let's try to see whether
# the type I error probability that we make if h`crit` = 8 is
# still smaller than alpha
h`guess` <- 8
1- pbinom( (h`guess`-1),size = n, prob = 1/2)

## [1] 0.0546875

# No, therefore h`crit` = 9

h`guess` <- 9 # We observe 8 heads
ggplot(data = tibble(x = 0:n, y = dbinom(0:n, size = n, prob = 1/2))) +
  geom`point`(aes(x,y)) +
  geom`vline`(xintercept = h`guess`, col = 'red')

## Error in ggplot(data = tibble(x = 0:n, y = dbinom(0:n, size = n, prob =
1/2))): non trovo la funzione "ggplot"

# The quantile function finds immediately the critical value
qbinom(1-alpha, size = n, prob = 1/2 )

## [1] 8

```

You might have noted that (2) is not always solvable for discrete distributions, since usually there is no h such that $\mathbb{P}(H > h|H_0) = \alpha$ precisely. This technicality is overcome by choosing h_{crit} to be the leftmost element for which $\mathbb{P}(H \geq h_{\text{crit}}|H_0) \leq \alpha$.

We finish this section by introducing the concept of p-value. The p-value of an observed value h_{obs} is the probability of obtaining a more extremal result, **provided** H_0 **holds**. Since in this case more extremal means greater, the *p-value* of h is

$$\mathbb{P}(H > h|H_0)$$

e:p-value

```

# The p-value of h`obs` = 7
h`obs` <- 7
sum( dbinom(h`obs`:n, size = n, prob = 1/2))

## [1] 0.171875

1- pbinom(h`obs` - 1, size = n, prob = 1/2)

## [1] 0.171875

```

It is easy to prove that $h > h_{\text{crit}}$ if and only if the p-value associated to h is smaller than α . Therefore the decision rule can be equivalently restated as: " We reject H_0 in favour of H_1 if and only if the p-value of the statistic is smaller than α ".

3.2.3 Binomial Test

The binomial test is used for data collected from a categorical data with two levels, where the observations are independent and one from another and equally distributed. We come back to the Linguistic experiment where the informants will use a word which has two variants, a formal one and an informal one. and we define $X_1, \dots, X_n$ by setting $X_i = 1$ if the i -th informant said the Formal variant, 0 otherwise. A research suggested that across Russia, a good model for the X_i is the following: each person tosses, independently from the others, a coin of parameter p and say the Formal variant if the coin gives heads. They estimated $p = 0.6$. However, backed by some considerations, you think that for Moscow $p > 0.6$ and you collect some data to verify this

Therefore you reproduce the experiment with informants from the Moscow region. The data is collected in the column *Variant* of the tibble *Experiment_result*

```
head(Experiment$result$Variant)

## Error in head(Experiment$result$Variant): oggetto 'Experiment_result'
non trovato
```

The *null hypothesis*, which corresponds to the current model accepted, is

- H_0 : $X_1, \dots, X_n$ are Bernoulli trials of parameter $p = 0.6$
- H_1 : $p > 0.6$.

The statistic to use is the number of formal variants used.

Ex. 3.2.1 — Take a look at the previous chunks of code. Is H_0 true? Is H_1 true? The test tries to give a rule to decide this question in the real cases, where we cannot look at the "code" that has generated the data.

Let's check the h_{crit} for this test and compute the p-value for the observed H . The quantile function is the inverse of $h \rightarrow \mathbb{P}(H \leq h | H_0)$, and to find $h_{\text{crit}}(\alpha)$ we need to use it.

```
sample_size j- length(Experiment$result$Variant )

## Error: oggetto 'Experiment_result' non trovato

alpha = 0.05
p j- 0.6
h_obs j- sum(Experiment$result$Variant == 'Formal')

## Error: oggetto 'Experiment_result' non trovato
```

```
h`crit j- qbinom( 1- alpha,size = sample`size, prob = p )
ggplot( data = tibble(x = 0:sample`size , y = dbinom(0:sample`size, size = sample`size, prob = p)
  geom`point(aes(x, y )) +
  geom`vline(xintercept = h`obs, col = 'red') +
  geom`vline(xintercept = h`crit, col = 'blue' )

## Error in ggplot(data = tibble(x = 0:sample_size, y = dbinom(0:sample_size,
: non trovo la funzione "ggplot"

# To compute the mass on the right side of teh curve we use
```

We repeat once more that the test can be seen as a way to decide whether h_{obs} has been sampled from the distribution shown in the above graph or by some other. R has a command to perform the binomial test.

```
n`formal j- sum( Experiment`result$Variant == 'Formal')

## Error: oggetto 'Experiment_result' non trovato

n`total j- length(Experiment`result$Variant)

## Error: oggetto 'Experiment_result' non trovato

binom.test( n`formal, n`total , alternative = 'greater')

## Error in binom.test(n`formal, n`total, alternative = "greater"): oggetto
'n`formal' non trovato
```

ex:Eugene`Onegin3

Example 3.6: Eugene Onegin 3: Binomial Test

We now come back to the Eugene Onegin case, see Example ?? and ??, where we created a sequence of 0s and 1s, stored in the vector `text_01`, in which the characters of the texts of the German drama corpus were replaced by 1, if the character was a consonant, 0, if it was a vowel, and simply omitted if it was any other character. In Example ?? we introduced two models for the sequence, H_0 , in which the digits are i.i.d. with some parameter p , and we have estimated the parameter p by the proportion of 1s in the sample,

```
mean(text`01)

## Error in mean(text_01): oggetto 'text_01' non trovato
```

and the reason will be clear after having introduced the Law of Large Numbers ?? and its application in Example 3.2.1. We now want to see that by looking at the jumps from 0 to 1 and from 1 to 0 that this model is wrong.

Denote the string in `text_01` by $x_1 x_2 \dots x_N$, for some $N \in \mathbb{N}$, and let n_0 denote the number of zeros and n_1 the number of ones. At an intuitive level, we think that the digits are not independent, since it is difficult to pro-

nounce a word without consonants, and we will Now we will perform a binomial test on a different sequence related to the jumps from o. Assume that n_0 is the number of os in the sequence and denote by $\alpha(i)$ the position of the i -th zero, for $i = 1, \dots, n_0$. The define the sequence Y_i , for $i = 1, \dots, n_0$, indicating if from the o at the $\alpha(i)$ -th position the sequence jumps to a 1, namely

$$Y_i = \begin{cases} 1 & \text{if } X_{\alpha(i)+1} = 1 \\ 0 & \text{otherwise} \end{cases}$$

e:interface

Let's give an example for concreteness. For the sequence 100101, $n_0 = 3$ and $\alpha(1) = 2$, $\alpha(2) = 3$ and $\alpha(3) = 5$. $Y_1 = 0$ since its first o is followed by a o, $Y_2 = Y_3 = 1$ s since the last two zeros are followed by a 1. Graphically

$$\begin{array}{ccccccc} & & & \underbrace{1} & & & \\ & & & & & & \\ 1 & 0 & 0 & 1 & 0 & 1 & \\ & & \underbrace{} & & \underbrace{} & & \\ & & o & & i & & \end{array}$$

and the resulting sequence is $\underline{Y} = 011$. If $\underline{X}$ follows a Bernoulli measure of parameter $\hat{p}$ (the null hypothesis), then also Y_i are independent and identically distributed. Let's first check the mean

$$\mathbb{P}(Y_i = 1) = \mathbb{P}(X_{\alpha(i)+1} = 1) = \hat{p}.$$

Let's check the independence: for each choice of $k \leq n_0$, of indexes $1 \leq i_1 < i_2 < \dots < i_k \leq n_0$ and of $y_{i_1}, \dots, y_{i_k} \in \{0, 1\}$ we should have

$$\mathbb{P}(Y_{i_1} = y_{i_1}, \dots, Y_{i_k} = y_{i_k}) = \mathbb{P}(Y_{i_1} = y_{i_1}) \dots \mathbb{P}(Y_{i_k} = y_{i_k})$$

e:Bernoulli'indep

Indeed this is the case since on the left hand side we have

$$\mathbb{P}()$$

We therefore test the following assumptions

- $\underline{X}$ is distributes as a Bernoulli measure of parameter $\hat{p}$, and, in particular, $\underline{Y}$ distributes as a Bernoulli measure od parameter $\hat{p}$
- $\underline{Y}$ distributes as a Bernoulli meausre of parameter $p \neq \hat{p}$

using a binomial test

```
# Recall that we already computes
# the jumps from 0 to 0 and from
# 0 to 1, and we saved them in
# the variables n`{00} n`{11}
binom.test(n`01 , n`00 + n`01, prob = hat`p )

## Error in binom.test(n`01, n`00 + n`01, prob = hat`p): argomento
non utilizzato (prob = hat`p)
```

Ex. 3.2.2 — In the last year, in two different hospitals, one small and one big, 100 and 10000 child were born, respectively. One hospital 56% of the new borns were males, while in the other one 52%. Estimate your degree of belief that in the big hospital 56% of the new borns were males.

Answer (Ex. 3.2.2) — $\mathbb{P}(D)$

3.2.4 Homework

Ex. 3.2.3 — **Position of verbs in verses.**

The dataset “The last words in verses” contains a sample of lines taken from the RNC Corpus of Russian Poetry. Actually, there are two samples comprising the texts written in the 1820s and 1920s. We took only one line per author to keep our observations as independent as possible.

1. Decade — decade of creation: 1820s, 1920s.
2. RhymedNwords — the number of words in the rhyming position (usually one word, but there are two words in cases such as ‘I would like to get) wine’ (which is rhymed with ‘toad’, see <http://russian-poetry.ru/Poem.php?PoemId=18261>)).
3. RhymedNsyl — the number of syllables in the rhyming position.
4. UPoS — part of speech of the last word.
5. LineText — a sampled verse.
6. Author — the author of the text.

The **Research question** in which we are interested is whether we can decide that in verses written in 1820s, verbs are used in the rhyming position more often or less often than expected for verbs in general? To calculate the probability to come across a verb in written Russian texts (general expectations), use the frequency dictionary of the Russian National Corpus (Lyashevskaya, Sharoff 2009).

1. State the model in terms of Bernoulli trials and estimate the parameter p
 - 1.1.1 Read the file
2. First of all, we have to read the RNC frequency dictionary data. Read frequency dictionary file and save the result as variable `freq_rnc`.

```
read-rnc-dataset}
library(tidyverse)
freq_rnc <- read_tsv('https://raw.githubusercontent.com/LingData2019/LingData/master/data/freq_rnc.tsv')
## Error in parse(text = input): |text|:1:18: '}' inatteso
## 1: read-rnc-dataset}
## ^
```

3. Use head command to show the beginning of the table stored in variable freq_rnc.

```
head(freq_rnc)
## Error in head(freq_rnc): oggetto 'freq_rnc' non trovato
```

4. Note that verbs are coded as 'v' in the PoS field, and their frequency is shown in the Freq(ipm) field (relative frequency, number of items per million words in the corpus). Calculate the probability to see verbs dividing the sum of their frequency by the sum of frequency of all words in the dictionary. Note that due to presence of brackets in the name of column Freq(ipm), to access this column, you have to put its name into backticks, i.e.:

```
all_freqs <- freq_rnc$`Freq(ipm)` # access column Freq(ipm)
## Error: oggetto 'freq_rnc' non trovato

all_freqs[1:10] # show first 10 elements
## Error: oggetto 'all_freqs' non trovato

p0 <- freq_rnc %>%
  filter(PoS == 'v') %>%
  select(`Freq(ipm)`) %>%
  sum()/sum(all_freqs)

## Error in freq_rnc %>% filter(PoS == "v") %>% select("Freq(ipm)")
## %>% sum(): non trovo la funzione "%>%"

p0
## Error: oggetto 'p-0' non trovato
```

5. Assume that every time the author decides which word to use as a last word of a verse, they toss a coin. If they get head, they use verb, otherwise they use word of different part of speech. Denote the probability to get head (i.e. use verb) by p . Recall that we are interested in the following question: can we decide that in verses written in 1820s, verbs are used in the rhyming position more often or less often than expected for verbs in general?
How can we write the model H_0 in terms of coin tosses? The alternative hypothesis? This is usually synthetically written as $H_0: p = p_0$ $H_1: p \neq p_0$
6. Read the dataset "The last words in verses" and put it variable lwv. Filter out the relevant observations from 1820s, calculate the number of verbs observed in the sample, and the sample size. Hint. You can

use function table to calculate how many times each value occurs in a vector of factors.

```
lvw j- read_tsv('https://raw.githubusercontent.com/LingData2019/LingData/master/
## Error in read_tsv("https://raw.githubusercontent.com/LingData2019/LingData/m
non trovo la funzione "read_tsv"
```

7. **Do statistical testing** Test stated hypothesis and find p-value. Use `binom.test` that will calculate p-value for binomial test (note that it uses two-sided alternative by default). You have to put actual number of successes (i.e. heads), number of trials (i.e. coin-tossings) and expected probability of success that is used in null hypothesis.

```
successes j- lvw %i%
  filter(UPoS == 'VERB' & Decade == '1820') %i%
  nrow()

## Error in lvw %i% filter(UPoS == "VERB" & Decade == "1820")
%i% nrow(): non trovo la funzione "%i%"

size j- lvw %i%
  filter( Decade == '1820') %i%
  nrow()

## Error in lvw %i% filter(Decade == "1820") %i% nrow(): non
trovo la funzione "%i%"

binom.test(x = successes, n = size, p = p_0 )

## Error in binom.test(x = successes, n = size, p = p_0): oggetto
'successes' non trovato
```

8. **Interpreting the results** Give your interpretation of obtained p-value. Would you reject null hypothesis? Answer the initial question: can we decide that in verses written in 1820s, verbs are used in the rhyming position more often or less often than expected? Assuming we have set the significance level (the level at which we regard the hints strong enough to reject H_0 in favour of H_1) to 0.05, we do not have enough evidence to reject H_0 in favour of H_1 .

3.3 FRAMEWORK FOR TESTING AND PROBLEMATICS

ral framework

3.3.1 Framework

Every test follows the same idea of the binomial test. We need

- i:1 1. An experiment with which we collect data $\underline{X}$
- i:2 2. A statistic $t(\underline{X})$.
- i:3 3. A null hypothesis H_0 , which is the current model that specifies how the data $\underline{X}$ (or at least the statistic $t(\underline{X})$) distribute
- i:4 4. An alternative hypothesis H_1
- i:5 5. The statistic has to behave in two different ways depending on whether H_0 or H_1 holds. Usually t is more extremal under H_1 . What more extremal means depends on the cases: it might mean bigger, smaller or that $|t|$ is bigger. The decision rule is that we *reject* H_0 *in favour of* H_1 if t is more extremal than a value t_{crit} to be fixed
- i:6 6. A significance level α , usually set to $\alpha = 0.05$
- i:7 7. t_{crit} is defined, and therefore also the test, by imposing that the type II error probability **provided** H_0 **holds** is given by α

$$\mathbb{P}("t \text{ more extremal than } "t_{\text{crit}}" | H_0) = \alpha \quad (3) \quad \text{e:tcrit}$$

Note that the above quantity is computable, since under H_0 we know the distribution of t . Equivalently, we reject H_0 iff and only if the *p-value* of the observed statistic $t(\underline{X})$ is smaller than α -

For the binomial test the previous quantities are Let's see

1. The experiment of collecting $\underline{X} = (X_1, \dots, X_n)$, Bernoulli random variables
2. The statistic $H(\underline{X}) = X_1 + \dots + X_n$
3. The null hypothesis H_0 : the vector $\underline{X}$ distributes according to the Bernoulli trial measure (1) of parameter $p = p_0$. Under the null hypothesis, H is a binomial of parameters n and $p = p_0$
4. The alternative hypothesis, say H_1 : $p < p_0$.
5. H is statistically smaller if H_1 holds
6. α
7. $h_{\text{crti}} = \text{qbinom}(1 - \alpha, \text{size} = n, \text{prob} = p_0)$

3.3.2 Reviewing some examples in this setting

3.3.3 problematics

Notice that apart from deciding which statistic to use and the notion of extremality, the alternative hypothesis does not enter in the definition of the decision rule, since both the p-values and the t_{crit} are computed under H_0 . It is inevitable that even if the test gives a positive result and we reject H_0 in favour of H_1 , this doesn't imply that we control the quantity that we would like to know: the *posterior* degree of belief of H_1 :

$$\mathbb{P}(H_1 | "H_0 \text{ is rejected}")$$

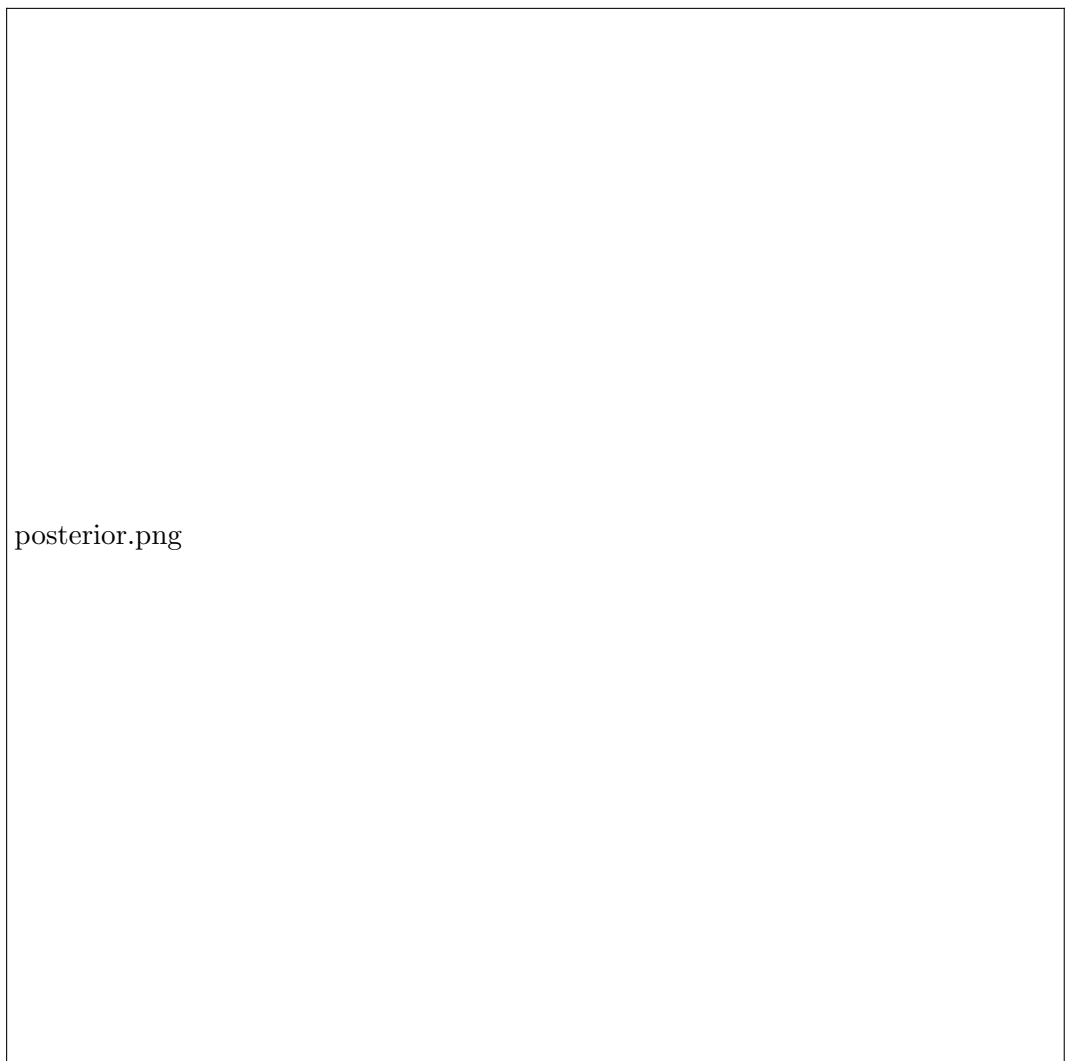
e:posterior

or, at least, with the *posterior odds*

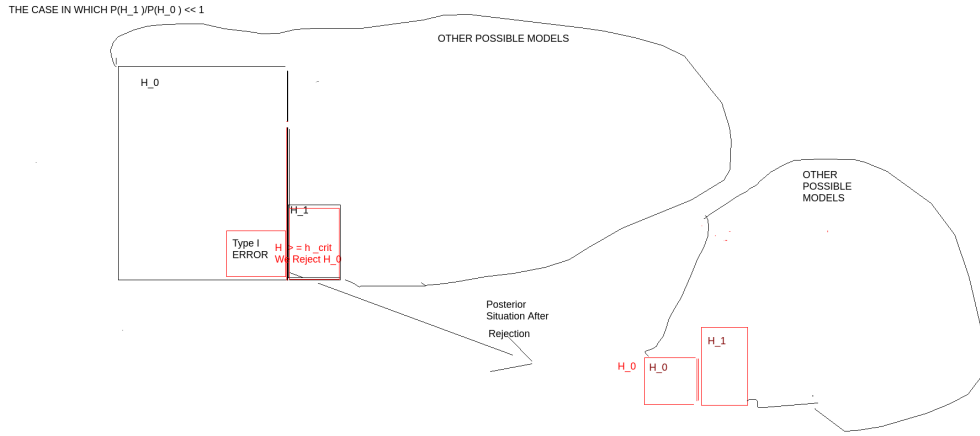
$$\frac{\mathbb{P}(H_1 | "H_0 \text{ is rejected}")}{\mathbb{P}(H_0 | "H_0 \text{ is rejected}")}$$

e:posterior'od

A graphical representation of these odds is given in Figure ?? . The Bayes



f:posterior



posterior's small

formula tells us analitically how this posterior odds update, once we observe a positive result of the test

$$\frac{\mathbb{P}(H_1 | "H_0 \text{ is rejected}"))}{\mathbb{P}(H_0 | "H_0 \text{ is rejected}"))} = \frac{\mathbb{P}("H_0 \text{ is rejected}" | H_1) \mathbb{P}(H_1)}{\mathbb{P}("H_0 \text{ is rejected}" | H_0) \mathbb{P}(H_0)}$$

e:Bayes

By the definition of t_{crit} , (3), the above quantity equals to

$$\frac{\mathbb{P}(H_1 | "H_0 \text{ is rejected}"))}{\mathbb{P}(H_0 | "H_0 \text{ is rejected}"))} = \frac{\mathbb{P}("H_0 \text{ is rejected}" | H_1) \mathbb{P}(H_1)}{\alpha \mathbb{P}(H_0)}$$

e:Bayes2

To compute the left hand side of the above quantity we need and estimate for an estimate

- The *prior* beliefs of the two hypotheses, $\mathbb{P}(H_0)$ and $\mathbb{P}(H_1)$, or at least the prior odds $\mathbb{P}(H_1)/\mathbb{P}(H_0)$.
- The distribution of the statistic under the alternative hypothesis: $\mathbb{P}(H > h_{crit} | H_1)$.

In the following links it is majestically exposed how difficult are to compute the above two quantities <https://terrytao.wordpress.com/2022/10/03/what-are-the-odds/> and <https://terrytao.wordpress.com/2024/08/02/what-are-the-odds-ii-the-venezuelan-pr>

In general, however, the trivial bound $\mathbb{P}("H_0 \text{ rejected}" | H_1) \leq 1$ gives us the estimate

$$\frac{\mathbb{P}(H_1)}{\mathbb{P}(H_0)} \leq \frac{1}{\alpha} \frac{\mathbb{P}(H_1)}{\mathbb{P}(H_0)}$$

e:just an estimate

and, if the prior odds are small, the posterior odds need not to be big, see Figure ??

3.3.4 Reviewing some examples

3.3.5 Homework: te magician

Ex. 3.3.1 — The magician. Assume that a person is claiming to have paranormal abilities and that he agrees to have his abilities tested. We prepare a box containing 3 balls, identically in every aspect but color, 2 of them being blue and one of them red. An assistant is then in charge of shuffling the box, extracting a ball randomly, recording its color and reinserting the ball again, and this procedure will be repeated a fixed number of times, and the magician will use its abilities to guess the colour of the balls.

1. **The best strategy.** What is the best strategy for the magician if he actually doesn't have magical abilities?
2. **Rephrasing of the experiment into a coin tossing problem** We decide to pick a ball 12 times. Denoting by X the number of correct guesses, how can you rephrase the problem in terms of coin tosses?
3. **Statement of the hypotheses.** State your null hypothesis and an alternative in terms of the original question and in terms of the coin-tossing model.
4. **Distribution of the statistic under H_0** Assume that H_0 holds, i.e., that he is not a magician. What is the distribution of X ?
5. **Decision rule.** To decide whether or not to reject H_0 in favour of H_1 we consider the statistic of the number of guesses. We decide that if $X \geq X_{\text{crit}}$ for a given threshold X_{crit} , we reject H_0 in favour of H_1 . We fix $X_{\text{crit}} = 12$, that is, we claim that he is a magician if and only if the he guesses correctly 12 times out of 12. Recall that a type I error is the error made when H_0 holds but the test rejects it, while a type II error is the error that is made if H_0 is not rejected but H_1 is true. Compute the type I and the type II error probabilities assuming that H_0 holds, i.e., that he is not a magician.
6. **Decision rule for a given significance level**

Answer (Ex. 3.3.1) — We now want to fix the critical level X_{crit} by fixing in advance the type I error probability provided H_0 holds. For $\alpha = 0.05$, find the smallest value of X_{crit} such that probability to make type I error is less than α .

```

# Type I error probability
pbinom(11, size = 12, prob = 2/3, lower.tail = F )

## [1] 0.007707347

# With lower.tail = F, pbinom(k) computes P( X ≥ k — H0 )
(2/3)^(12)

## [1] 0.007707347

# The type II error probability is 0, since we are assuming that $H_0$ holds
0

## [1] 0

```

Answer (Ex. 3.3.1) — Under H_0 X is a binomial of parameters $n = 12$ and $p = 2/3$. Therefore

```

size | 12
p | 2/3
## By guessing
pbinom(7:11, size = size, prob = p, lower.tail = F )

## [1] 0.631520714 0.393074678 0.181122646 0.053951426 0.007707347

## Using the quantile function
qbinom(0.95, size = size, prob = p) + 1

## [1] 12

```

Answer (Ex. 3.3.1) — Let us consider a coin that lands with “head” with probability p . Then our null hypothesis and alternative are stated as follows:

$$H_0 : p = \frac{2}{3}$$

$$H_1 : p > \frac{2}{3}$$

Note that any p that is larger than theoretically possible $2/3$ constitute violation of null hypothesis: i.e. even if the magician can predict correct result with probability $2/3 + 0.000000001$, he still a magician.

3.4 LAW OF LARGE NUMBERS

3.4.1 Independent Samples

Recall from the definition of distribution function of a random variable X in Definition 3.2.1, and the notion of independent samples from the variable X in Section 3.2.1, which we rewrite below for the sake of clarity.

d:sample

Definition 3.1. Let $n \in \mathbb{N}$ be a number, called the sample size. The random variables $X_1, \dots, X_n$ are *independent samples* of X if

- The X_i are identically distributed, and their common distribution coincides with p_X , the one of X : for each $a, b \in \mathbb{R}$ with $a < b$,

$$\mathbb{P}(X_1 \in (a, b)) = \dots = \mathbb{P}(X_n \in [a, b]) = p_X([a, b])$$

e:identically

- The X_i are independent. For each $a_i, b_i \in \mathbb{R}$, $i = 1, \dots, n$ with $a_i \leq b_i$ for every i ,

$$\mathbb{P}((X_1 \in [a_1, b_1], \dots, X_n \in [a_n, b_n]) = \mathbb{P}(X_1 \in [a_1, b_1]) \dots \mathbb{P}(X_n \in [a_n, b_n]).$$

e:independenc

We can take samples from a real population. If we want to make a statistical study on the height of citizens in Moscow we can take independent sample from the actual Moscow population

```
n`moscow j- 1000000
Moscow`heights j- 10*runif(n`moscow) + 172
### The whole population, that is, the above vector
### Is inaccessible to us
### However, we can take samples
sample`size = 1000
Sampled`heights j- sample( Moscow`heights, sample`size )
```

One can also take independent samples from a distribution, see Definition 3.1. In the following code we take samples from an exponential random variable.

```
sample`size j- 10000
Sample j- rexp(sample`size)
##### Usually we don't know that for generating the
##### sample we used an exponential random variable
```

Note that the two codes are very similar and the only difference is that `sample( Moscow`heights,)` is replaced by `rexp()`. In both cases, for real data, we would like to know the function used to generate the samples, but we have only access to the sample. A random variable can be regarded as an abstract population and what we will say applies in the same fashion to both cases, as long as we can take samples. Moreover, sampling from an actual population can be seen as a particular case of sampling from a random variable. An exponential random variable is a good model for the living time of objects

and you can think that a mobile work decides how long it will work by sampling from the population given by an exponential random variable the total amount of time

3.4.2 Statement of the law of large numbers

We are interested in computing the mean height of the Moscow population.

```
n`moscow j- 1000000
Moscow`heights j- 10*runif(n`moscow) + 176
mean(Moscow`heights)

## [1] 176.9977
```

Unfortunately, we do not have access to the whole population and, so that the vector *Moscow heights* and therefore its mean *mean(Moscow heights)* are unknown to us. However, it is possible to take independent samples, see Definition 3.1, and we can compute the *sample_mean*

```
sample`size j- 1000
sample j- sample( Moscow`heights , sample`size)
# Althought we aren't replacing, the samples are almost independent since the sample
# size is much smaller that the whole population
sample`mean j- mean(sample)
sample`mean

## [1] 177.02

mean(Moscow`heights)

## [1] 176.9977
```

Similar considerations hold when sampling from a generalised population, or a random variable. For a discrete random variable we have defined its mean in Definition ??, and the Lebesgue integration procedure allows to extend the definition of mean for generic distribution. Take for instance data on the reaction times to a linguistic stimulus collected in an experiment. They could look in the following way

```
sample`size j- 10000
reaction`times j- rlnorm(sample`size)
# We are assuming that the reaction times follow an
# exponential of rate 1.
# It's mean is 1
mean(reaction`times)

## [1] 1.683053
```

Once more, we don't know that to generate the data we used a lognormal distribution (a usual model for reaction time data) of mean, and we have only access to data. What the Law of Large Numbers tells us is that the sample mean is a good approximation of the mean of the population, in the first case, and of the random variable, in the second.

d:sample`mean

Definition 3.2 (Sample mean for a random variable). Let p_X be a distribution of a random variable X with a well defined mean μ , and let $\underline{X} = (X_1, \dots, X_n)$ be samples from X . The sample mean is

$$m(\underline{X}) = \frac{X_1 + \dots + X_n}{n}$$

e:sample`mean

Similarly, $m(\underline{X})$ can be defined also for samples from a population. The Law of Large Numbers tells us that

Theorem 3.3 (Strong Law of Large Numbers). *Let X be a distribution with a well defined mean μ and let $\underline{X} = (X_1, \dots)$ be independent samples form X . Then*

$$m(\underline{X}) := \frac{X_1 + \dots + X_n}{n} \rightarrow \mu$$

Note that the random quantity on the left hand side, is converging to a deterministic constant.

3.4.3 Dependence from the Sample Size

One could study the Law of Large Numbers in more detail and provide some *quantitative* version of it: we can tell how good, depending on the sample size, the approximation of the real mean by the sample mean is. Instead, we plot some graphs to visualize the dependence of the sample mean from the sample size

```
n`moscow j- 100000
Moscow`heights j- 10*runif(n`moscow) + 172
sample`size j- 10000
Sampled`heights j- sample(Moscow`heights, sample`size)
sample`mean j- cumsum(Sampled`heights)/1:sample`size
ggplot(data = tibble(x = 1:sample`size, y = sample`mean), aes( x, y)) +
  geom`line()+
  geom`hline( yintercept = mean(Moscow`heights), col = 'red')

## Error in ggplot(data = tibble(x = 1:sample_size, y = sample_mean), aes(x,
: non trovo la funzione "ggplot"
```

Similarly for the reaction time experiment

```
sample`size j- 10000
reaction`times j- rexp(sample`size)
# We are assuming that the reaction times follow an
# exponential of rate 1.
```

```

# It's mean is 1 )
mu = 1
sample_mean <- cumsum(reaction_times)/1:sample_size
ggplot(data = tibble(x = 1:sample_size, y = sample_mean), aes(x, y)) +
  geom_line() +
  geom_hline(yintercept = mu, col = 'red')

## Error in ggplot(data = tibble(x = 1:sample_size, y = sample_mean), aes(x,
: non trovo la funzione "ggplot"

```

3.5 EMPIRICAL DISTRIBUTION

In this section we show that, as a corollary of the Law of Large Numbers ??, it is the same to know the distribution of a random variable X and to be able to take independent samples from it. In particular, we show that from the histogram of the sampled values we are able to recover the distribution of X . This is often taken as a basis in the frequentist interpretation of probability, but it is a result and it is not always possible to take independent samples

3.5.1 Estimation of the parameters of the Bernoulli trials model

Example 3.7: Estimation of the p in Bernoulli trials

ex:estimate Bernoulli

Consider a string $\underline{X}$ whose digits X_i , for $i = 1, \dots, n$, $n \in \mathbb{N}$, are either 0 or 1, and assume we have observed $x_1 x_2 \dots x_n$. We use the model of Bernoulli trials $\mathcal{Q} = \{\mathbb{P}_p, p \in [0, 1]\}$, see Definition 3.2.1, so that we assume that $\underline{X}$ distributes according to $\mathbb{P}_p$ for some p . To estimate it, we notice that

$$p = \mathbb{E}[X_i] \quad \text{for each } i = 1, \dots, n, \quad (4)$$

e:p'as'mean

and the law of large numbers, together with the independence assumption of the digits that each $\mathbb{P}_p \in \mathcal{Q}$ possesses, guarantees that

$$\frac{X_1 + \dots + X_n}{n} \rightarrow p$$

Example 3.8: Estimation of the distribution

Let X_i , for $i = 1, \dots$, be independent samples from a distribution X . In this example we show that it is the same to know the distribution of X , $\mathbb{P}_X$, see ?? or the sample X_i , $i = 1, \dots$. Indeed, define

$$Y_{[a,b]} = \begin{cases} 1 & \text{if } X_i \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

e:indicator'set

It is intuitive, and indeed (Maybe add a proof on how to show that two Bernoulli are independent in their independence section???) it can be proved, that the Y_i are independent, since they are function of X_i alone, for each $i = 1, \dots, n$. Therefore the Y_i form a sequence of Bernoulli trials and we can apply the Law of Large Numbers in the form of (4) that guarantees

$$\frac{Y_1 + \dots + Y_n}{n} \rightarrow p$$

where p is the mean of Y_i , for each i . Since the equality of events $(Y_i = 1) = (X \in [a, b])$ holds,

$$p = \mathbb{E}[Y_i] = \mathbb{P}(Y_i = 1) = \mathbb{P}(X_i \in [a, b]) = \mathbb{P}_X([a, b])$$

so that

$$\frac{Y_1 + \dots + Y_n}{n} \rightarrow \mathbb{P}(X \in [a, b]). \quad (5)$$

e:equality1

We further note that the left hand side of the (5) is the proportion of the number of X_i that fall in the interval $[a, b]$. We can rewrite everything in the more suggestive form

$$\frac{\text{\#of } i \text{ such that } X_i \in [a, b]}{n} \rightarrow \mathbb{P}_X([a, b]).$$

The above is often mistakenly taken as the definition of probability in the frequentist setting.

Example 3.9: Estimation of the parameters of the Markov model

Consider the Markov model $\mathcal{Q} = \{\mathbb{P}_{a,b}\}$ introduced in (12) for sequences of 0s and 1s, namely we assume that the observed the string $s_1 x_2, \dots, x_n$, $n \in \mathbb{N}$, $x_i \in \{0, 1\}$ for each $i = 1, \dots, n$, is an instance of the distribution $\mathbb{P}_{a,b}$. The aim is to estimate the parameters a , the probability of jumping to 1 while at 0, and b , the probability of jumping from 0 while at 1, using the Law of Large Numbers. To this end let n_0 be the number of 0s in the string $x_1 x_2 \dots x_n$ and n_1 the number of 1s, and denote, for $i = 1, \dots, n_0$, $\alpha(i)$ the position of the i -th 0, and, for $j = 1, \dots, n_1$, by $\beta(j)$ the position of the j -th 1. For the sake of concreteness, if we observe 100111000, then $n_0 = 5$, $n_1 = 4$ and $\alpha(1) = 2$, $\alpha(2) = 3$, $\alpha(3) = 7$, $\alpha(4) = 8$, $\alpha(5) = 9$, $\beta(1) = 1$, $\beta(2) = 4$, $\beta(3) = 5$ and $\beta(4) = 6$. We then define the two variables

$$Y_i = \begin{cases} 1 & \text{if } X_{\alpha(i)+1} = 1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$Z_j = \begin{cases} 1 & \text{if } X_{\beta(j)+1} = 0, \\ 0 & \text{otherwise} \end{cases}$$

for $i = 1, \dots, n_0$ and $j = 1, \dots, n_1$. The variables Y_i indicate if after the i -th 0 there is a 1 or a 0, while the variables Z_j indicate whether after

the j -th 1 there is a 0 or a 1. They are Bernoulli random variables of parameter

$$\begin{aligned}\mathbb{P}_{a,b}(Y_i = 1) &= \mathbb{P}_{a,b}(X_{\alpha(i)+1} = 1) = \mathbb{P}_{a,b}(X_{\alpha(i)+1} = 1, X_{\alpha(i)} = 0) \\ &= \mathbb{P}_{a,b}(X_{\alpha(i)} = 1 | X_{\alpha(i)} = 0) = a\end{aligned}$$

and

$$\begin{aligned}\mathbb{P}_{a,b}(Z_j = 1) &= \mathbb{P}_{a,b}(X_{\beta(j)+1} = 0) = \mathbb{P}_{a,b}(X_{\beta(j)+1} = 0, X_{\beta(j)} = 1) \\ &= \mathbb{P}_{a,b}(X_{\beta(j)} = 0 | X_{\beta(j)} = 1) = b,\end{aligned}$$

for each $i = 1, \dots, n_0$ and $j = 1, \dots, n_1$, where we have used the strong Markov property and the fact that $X_{\alpha(i)} \equiv 0$ and $X_{\beta(j)} \equiv 1$, by definition of α and β . They are also independent (maybe in the section independence put how to check whether two Bernoulli are independent???) Let's check some of the pairwise independence.

$$\mathbb{P}(Z_i = 1, Y_j = 1) = \mathbb{P}(X_{\alpha(i)+1} = 0, X_{\alpha(i)} = 1, X_{\beta(j)+1} = 1, X_{\beta(j)} = 0) = \mathbb{P}()$$

since $X_{\alpha(i)} = 0$ and $X_{\beta(j)} \equiv 1$ (I am using the strong Markov property, the randomness here is in the functions α and β) Therefore, both Y_i and Z_j form Bernoulli trials of parameter a and b , respectively, and we can use (4) to estimate

$$\begin{aligned}\frac{Y_1 + \dots + Y_{n_0}}{n_0} &\rightarrow a \\ \frac{Z_1 + \dots + Z_{n_1}}{n_1} &\rightarrow b\end{aligned}$$

e:Markov'estimat

Finally we see that on the left hand side in the above equality there is the proportion of jumps from 0 to 1 and to 1 to 0, respectively

Example 3.10: Montecarlo

ex:montecarlo

Given $X_i, i = 1, \dots, n$, independent samples from a distribution $\mathbb{P}_X$, we can estimate $\mathbb{E}[f(X)]$, see Definition ??, using the Law of Large Numbers. The random variables $f(X_i), i = 1, \dots, n$ can be seen to be independent (since $f(X_i)$ depends only on X_i and none other, and the X_i are independent), so that the Law of Large Numbers ?? applies

$$\frac{f(X_1) + \dots + f(X_n)}{n} \rightarrow \mathbb{E}[f(X)]$$

e:montecarlo

Example 3.11: Estimation of the standard deviation of the distribution

Let $X_i, i = 1, \dots, n$, be independent samples from a distribution $\mathbb{P}_X$ of mean μ . Applying Example 3.10 to the function $f(x) = (x - \mu)^2$, we can estimate the variance of $\mathbb{P}_X$ by

$$\frac{(X_1 - \mu)^2 + \dots + (X_n - \mu)^2}{n} \rightarrow \mathbb{E}[(X - \mu)^2] = \text{Var}(X)$$

3.5.2 Histograms and empirical distributions

Assume we have collected some continuous data $\underline{X} = (X_1, \dots, X_n)$ associated to the variable X . An histogram is a way to visualise the data that requires to specify

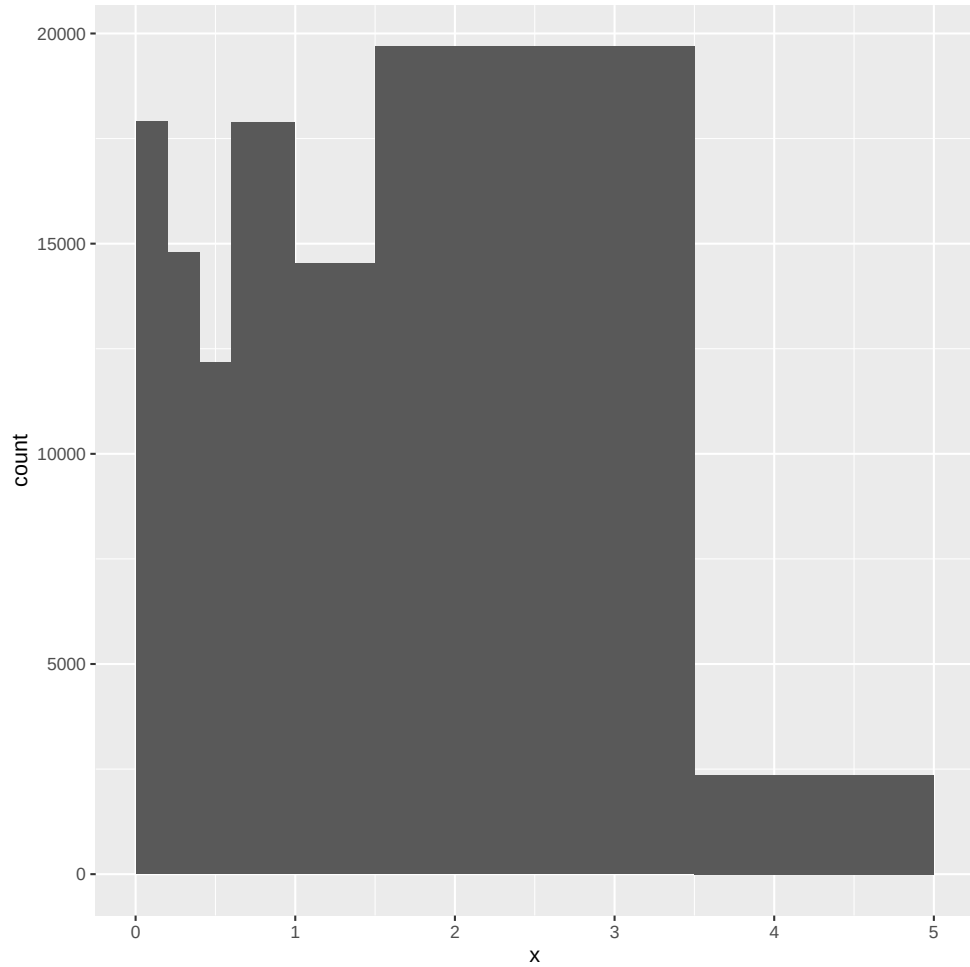
- A vector $\underline{X}$ with the data
- A vector of breaks that specify intervals in the x -axis. For instance, the vector `breaks = c(1,3,5)` specifies the intervals $(-\infty, 1]$, $[1, 3]$, $[3, 5]$ and $[5, +\infty)$.
- A specification of what needs to be plotted in the y axis. It could be the total frequency of the data, that is, the height vertical bar associated to the base segment $[a, b]$ is the total frequency of X_i falling in the interval $[a, b]$.

We can easily plot the histogram associated to samples from the exponential distribution, defined by $\mathbb{P}(X \in [a, b]) = \int_{\max\{a, 0\}}^{\max\{b, 0\}} e^{-x} dx$.

```
library('tidyverse')

## - Attaching core tidyverse packages ----- tidyverse 2.0.0 -
## v dplyr      1.1.4    v readr      2.1.5
## v forcats    1.0.0    v stringr   1.5.1
## v ggplot2    3.5.2    v tibble    3.3.0
## v lubridate  1.9.4    v tidyr     1.3.1
## v purrr      1.0.4
## - Conflicts ----- tidyverse_conflicts() -
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (http://conflicted.r-lib.org/) to force all
## conflicts to become errors

sample_size <- 100000
data <- rexp(sample_size)
## breaks is the vector of the endpoints of the segments in the xx axis
breaks <- c(0,0.2,0.4,0.6, 1, 1.5,3.5 , 5)
ggplot( data = tibble( x = data )) +
  geom_histogram( aes( x ), breaks = breaks )
```



One of the consequence of the Law of Large numbers that we wish to outline is that knowing the distribution and knowing how to take an arbitrary number of samples is the same thing. Indeed, let $[a, b]$ be an interval with $a \leq b$ and define the random variable Y_i that indicates whether or not X_i lies in the interval $[a, b]$:

$$Y_i = \begin{cases} 1 & \text{if } X_i \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

e:indicator

The Y_i are independent since the X_i are and the mean of Y_i is $\mathbb{P}(X \in [a, b])$. The Law of Large Numbers implies that

$$\frac{Y_1 + \dots + Y_n}{n} \rightarrow \mathbb{P}(X \in [a, b]) \quad (6)$$

e:sample=dist

as n gets bigger. The above fact is useful to better understand the histogram. Indeed, if $[a, b]$ is an interval in the x axis of the histogram, then $(Y_1 + \dots + Y_n)/n$ is the height of the histogram bar plotting the relative frequencies. Therefore plotting the histogram of the relative frequencies is a good way to represent the distribution of a random variable, which is at the basis of the problematic frequentist interpretation of Probability Theory

Denote $[a_i, b_i]$ the base intervals of the histogram. The approximation (6) is good as soon as $n \gg \frac{1}{\mathbb{P}(X \in [a_i, b_i])}$. Therefore, the histogram is a good

approximation of the distribution if $n \gg 1/p_X([a_i, b_i])$ for each $i = 1, \dots, k$, where k is the number of bins in the histogram. Moreover, if one wants to have small intervals on the x-axis we should take k large. Therefore to give a good representation of the distribution we should have $n \gg k \gg 1$, where $n \gg k$ is needed to apply the Law of Large Numbers to each interval, and $k \gg 1$ is needed to have small intervals.

Example 3.12: Density histogram

As the intervals $[a_i, b_i]$ shrink, the relative frequency of each interval decreases (unless we are dealing with discrete random variables, see Definition ??) Next we show that for continuous random variables it is useful to plot the density histogram, especially when $[a, b]$ is small. Continuous random variable are characterised by the fact that they have a *density* function f , that is, a function f verifying $\mathbb{P}(X \in [a, b]) = \int_a^b f(x)dx$. If the interval $[a, b]$ is small then $\int_a^b f(x)dx \sim f(a)(b - a)$ (we are also assuming that f is continuous) and we have

$$\mathbb{P}(X \in [a, b]) \sim f(a)(b - a) \quad (7)$$

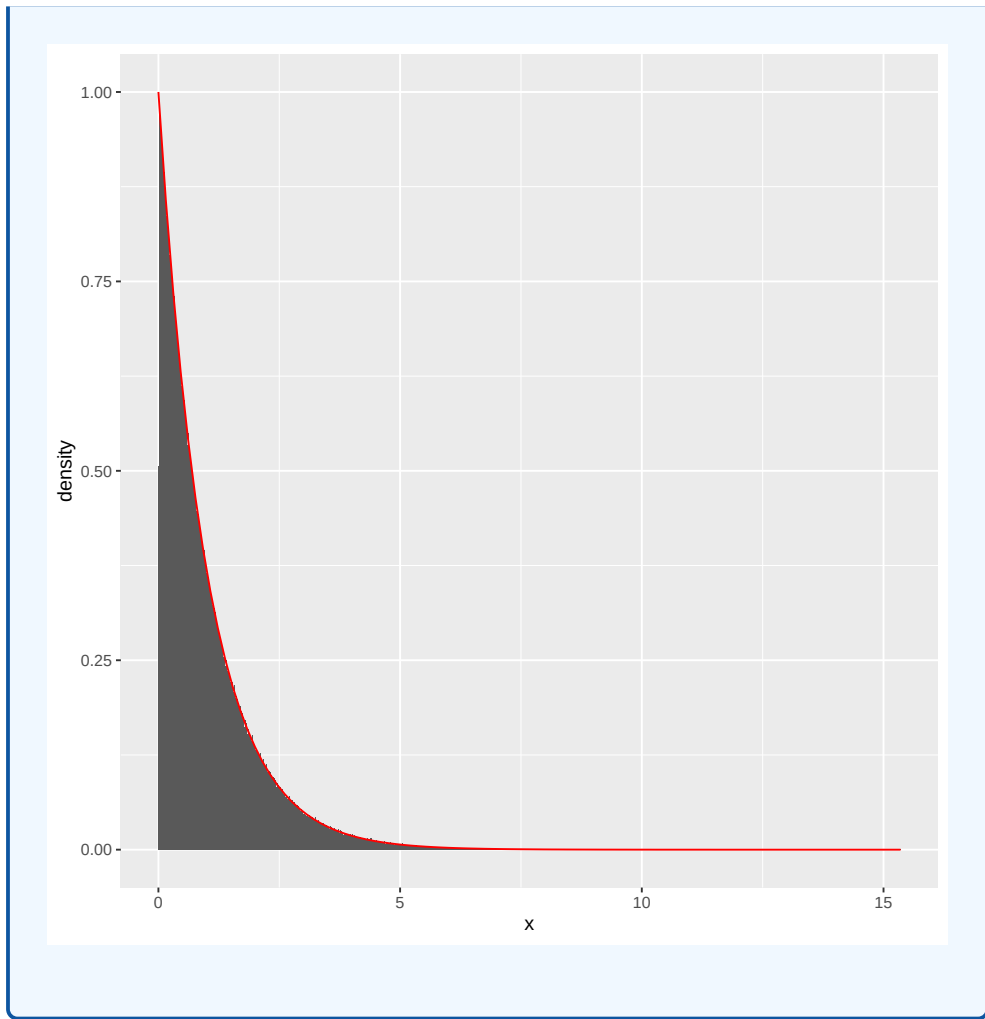
e: int' approx

(we are assuming f continuous in a) is a good approximation. Plotting the histogram of the density, which is equal to the relative frequency divided by the length of the base segment, gives a good approximation of the density function:

$$\frac{Y_1 + \dots + Y_n}{n(b - a)} \rightarrow \frac{\mathbb{P}(X \in [a, b])}{(b - a)} \sim f(a)$$

Once more, $n \gg k$ since for each interval $[a_i, b_i]$ we need $n \gg 1/p_X([a_i, b_i])$ to apply the Law of Large Numbers, and in order to apply (7) we need $b_i - a_i$ small for each i , and therefore $k \gg 1$, where k is the number of intervals

```
sample`size` j- 1000000
number`bins` j- 1000
data j- rexp(sample`size`)
ggplot( data = tibble(x = data )) +
  geom`histogram`( aes( x , y = after`stat`(density)), bins = number`bins`) +
  stat`function`( fun = dexp , args = list(rate = 1), col = 'red')
```



Here we comment of the fact that the great majority of distributions are neither continuous nor discrete. (7) can be seen as a characterization of continuous distributions: no matter how small is the interval $[a, b]$, the probability of X being in that interval is proportional to the length of the interval:

$$\mathbb{P}(X \in [c - \varepsilon, c + \varepsilon]) = \int_{c-\varepsilon}^{c+\varepsilon} f(x)dx \sim f(c)2\varepsilon$$

e:density'eps

This is of course not true for discrete random variables, see Definition ?? . Think at a Bernoulli random variable and take $c = 0$. Then the probability of obtaining a point in $[-\varepsilon, \varepsilon]$ is $\mathbb{P}(X \in [-\varepsilon, \varepsilon]) = 1 - p$, which is not small or proportional to (2ε) . However, the support of the distribution, that is, the places where the X can be found are only two points, that is, X concentrates in a set of dimension 0. In general there is a balance on the dimension of the set where the points concentrate and the how much is probable to find points on the set where they concentrate:

$$\mathbb{P}(X \in [c - \varepsilon, c + \varepsilon]) \sim f(c)(2\varepsilon)^\delta$$

and the function f is different from zero only on a set of fractal dimension $\delta \in [0, 1]$. The case $\delta = 0$ corresponds to the discrete case, while the case $\delta = 1$ corresponds to the continuous random variables. All the other cases

correspond to strange but typical distributions. An example with $\delta = 2/3$, where the function f has its support on the Cantor set, a set of fractal dimension $2/3$, is given by the "devil's staircase". In this case, for retrieving f , we should divide the relative frequencies not by the length of the base interval but by the length to the power δ (Indeed, in the discrete case we divide by $1 = (b - a)^0$). (The devil's staircase corresponds to the uniform distribution on the Cantor set, which consists on the numbers in $[0, 1]$ whose ternary expression does not contain the digit 1)

Ex. 3.5.1 — Fix a degree of precision $k \in \mathbb{N}$ and sample $(a_1, \dots, a_k)$, digits that can be either 0 or 2 with the same probability. Define the number $x = a_1 3^{-1} + a_2 3^{-2} + \dots + a_k 3^{-k}$, that is, use the digits as the digits of the ternary expression of x . The number thus obtained is sampled with a distribution that, as $k \rightarrow \infty$ is similar to the Cantor one.

3.6 EXERCISES

In this homework we are going to take a look back into the general framework, see Section 3.2.1 of testing and define one to see what it is possible to do when point . In this case we can use the Law of Large Numbers, see Theorem 3.2.1, or, more specifically, Example ??, and replace the true distributio

3.6.1 Application to testing

In many cases the statistic defined for a test, see ?? whose extremal values give an hint in favour of H_1 , is not known explicitly, but it can be simulated. We use the results in

Example 3.13: Eugene Onegin 4: Simulating the Statistic under the Null

ex:Eugene'Onegin3

3.7 EUGENE ONEGIN CENTRAL LIMIT THEOREM FOR MARKOV CHAINS

Theorem 3.4 (LLN and CLT for Markov Chain). *Let $\underline{X} = X_1 \dots$ a string of os and 1s that distributes according to $\mathbb{P}_{a,b}$. Then the Law of Large numbers*

$$\frac{X_1 + \dots + X_n}{n} \rightarrow p$$

e:LLN'Markov'Chain

holds with $p = b/(a+b)$. The Central limit theorem holds with a standard deviation

$$\sigma_{\text{Markov}} = \frac{ab(2-a-b)}{(a+b)^3},$$

e:sd'markov

namely

$$\sqrt{n} \frac{\frac{1}{n}(X_1 + \dots + X_n) - p}{\sigma_{\text{Markov}}} \rightarrow Z$$

e:CLT'markov

where $Z \sim \mathcal{N}(0,1)$ is a standard normal.

We rapidly check the above two theorems by repeating the same plots of Subsection 3.2.1 for the Law of Large Numbers and of Subsection 3.2.1 for the Central Limit Theorem, substituting the function `rbinom` with the function defined in Subsection 3.2.1 `simulate_markov_chain`, and write **The Law of Large Numbers**

The Central Limit Theorem

We want to check that the sequence of os and 1s saved in the vector `text_01` satisfies the CLT with the variance σ_{Markov} rather than the one that predicted by the CLT 3.6, $\sigma = \sqrt{p(1-p)}$, see also Example 3.2.1. In this way we conclude that H_1 is (way) more likely to be true. To this end, we choose n big enough so that

$$z(X_1, \dots, X_n) = \sqrt{n} \left(\frac{1}{n} (X_1 + \dots + X_n) - p \right) \quad (8)$$

e:CLT'statistic

approximates a Gaussian of standard deviation $\sqrt{p(1-p)}$ if $X_1 \dots X_n$ follows a Bernoulli measure or a Gaussian of standard deviation σ_{Markov} , if H_1 holds. We then separate the vector `text_01` into many chunks of length n : $a_1 = X_1 X_2 \dots X_n$, $a_2 = X_{n+1} \dots X_{2n}$... We then compute on for each of this chunks the statistic (??)

$$z_i = z(X_{i*n+1}, \dots, X_{(i+1)*n})$$

e:many'z

and we use the result of Example 12 to simulate the distribution of z . Notice that we are making the assumption that the z_i s are independent, which not true. since the different chunks are not independent. However, the dependence between z_i comes from the boundary terms, that is, from the terms X_{i*n+1} , $X_{(i+1)*n}$ which are much less than $\sqrt{n}$, and since each of the X_{i*n+j} enters in the definition of z_i divided by $\sqrt{n}$, it is not a big assumption. Therefore, the histogram and we can check graphically that

3.7.1 Homework: The Magician Strikes Back

Credits: Exercise taken from http://math-info.hse.ru/2019-20/Linguistic_Data:Quantitative_Analysis_and_Visualisation:linguistic_theory

Ex. 3.7.1 — Problem 2: the magician and the mean The magician of Subsection 3.2.1 is claiming to have psychokinetic abilities. To test them we agree to perform the following experiment. In a box we place 5 balls, identical to one another with the exception that each ball has a number on it: 1, 3, 4, 7, 8. We select random ball, record the number, put the ball again in the box, shuffle the balls and repeat the process several times obtaining a sample. The magician uses his abilities to make average of these numbers to be as large as possible.

1. **Population mean** Find population mean, i.e. average of numbers in the box.
2. **Sample mean vs. population mean**
Assume that at some experiment we obtained the following sample: 8, 3, 3, 1, 1, 7, 8, 7. Its mean (find it) is larger than the population mean. The magician says: Now you see? I has psychokinetic abilities! I said I will increase the mean and it is indeed larger than the population mean. Though
3. **More results**
Assume that you obtained sample mean that is equal to 8 and your sample size is 5. Does it look convincing now?
4. **Hypotheses** State null hypothesis and an alternative in this experiment.
5. **The decision rule** As the magician claims that he is trying to make the sample mean as large as possible, we use the following decision rule: we choose some value $\bar{x}_{crit}$ and will claim that the magician has magical abilities if sample average is equal to $\bar{x}_{crit}$ or larger. Let $\bar{x}_{crit} = 7$. Assume that the magician doesn't have magical abilities. Simulate 10000 sampling from our box, sample size = 5. How many times would you reject null hypothesis, i.e. claim that the magician indeed has magical abilities?

```
Answer (Ex. 3.7.1) — population j- c(1,3,4,7,8)
mean(population)
## [1] 4.6
```

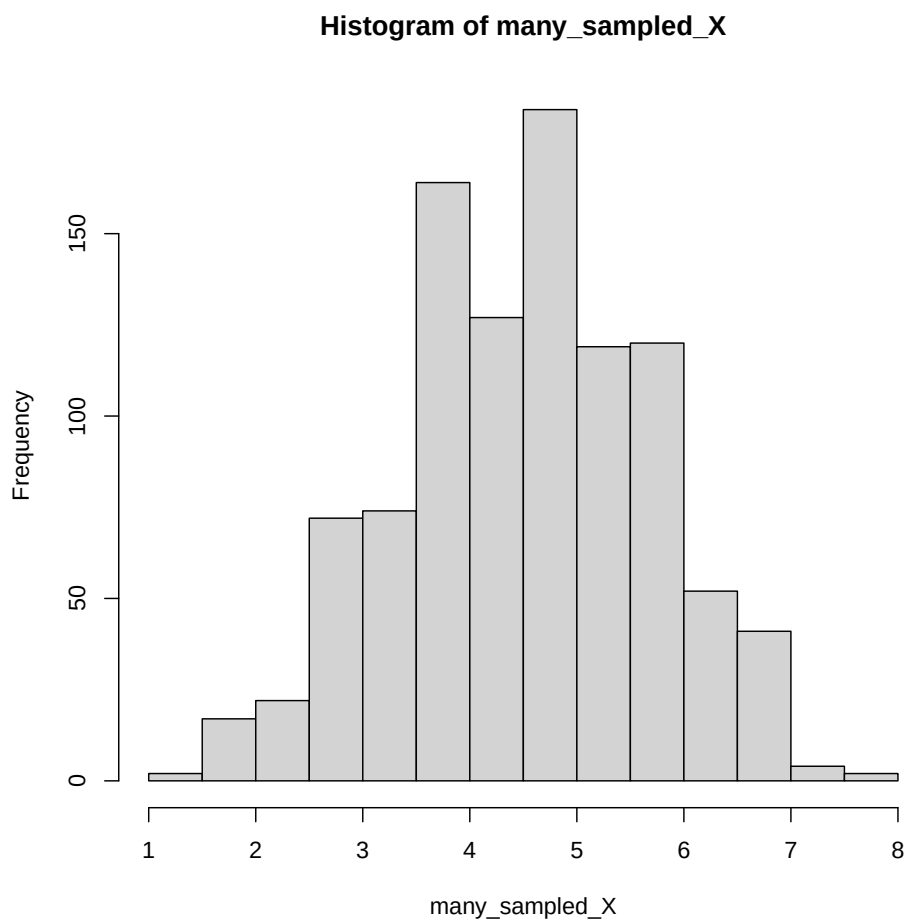
```
sample j- c(8,3,3,1,1,7,8,7)
mean(sample)
## [1] 4.75
```

the output doesn't look so different from an output that would have arisen under H_0

Answer (Ex. 3.7.1) — Yes, If H_0 were true this result would be possible only if we select 5 times out of five the ball with an 8

Answer (Ex. 3.7.1) — H_0 : X is the mean of random extractions from the urn.
 H_1 : X follows a distribution with mean $\mu > 4.6$ Note that the following step is crucial to compute the distribution of X if H_0 were true

Answer (Ex. 3.7.1) — ## Although one could theoretically compute the distribution of X given H_0 , for ins
 k j- 1000
 many'sampled`X j- replicate(k, mean(sample(size = 5, x = population, replace = T)))
 ## mean(sample(size = 5, x = population, replace = T)) computes an observation of X , and I replicate it k time
 hist(many'sampled`X)



many'sampled`X

```

## [1] 5.6 4.4 5.2 5.0 6.4 5.2 5.0 4.8 3.6 4.4 7.0 3.6 6.6 3.4 6.2 4.4 5.2 5.6
## [19] 5.8 6.2 4.6 4.4 7.6 4.0 3.8 4.8 4.0 3.8 5.0 5.6 5.4 6.0 3.4 3.6 5.0 4.4
## [37] 2.6 5.4 5.8 3.4 5.0 4.8 6.2 6.6 3.8 5.2 4.8 2.2 6.0 5.0 6.0 5.0 5.8 3.4
## [55] 5.6 5.8 4.8 6.0 4.0 5.0 6.6 3.8 4.2 5.6 4.6 3.6 4.0 6.8 3.4 6.4 5.4 4.6
## [73] 2.6 2.8 5.0 5.8 3.0 3.2 5.6 6.0 5.6 4.0 6.8 4.8 5.8 6.2 3.8 6.2 2.4 2.4
## [91] 4.0 3.2 3.6 6.6 4.2 3.2 4.4 6.4 2.6 5.8 5.4 4.2 2.8 5.2 3.4 6.4 5.4 5.8
## [109] 4.8 4.2 4.8 4.0 5.2 4.0 5.4 4.8 1.4 4.8 4.4 6.2 4.8 3.6 6.8 6.2 2.4 4.0
## [127] 3.0 4.4 5.0 4.8 5.0 3.6 2.8 3.6 3.2 2.8 2.6 3.4 3.8 2.8 5.0 3.8 3.4 5.4
## [145] 5.4 6.6 2.0 7.0 4.8 4.8 6.6 5.2 3.8 3.0 5.2 2.0 3.6 2.8 4.2 6.0 3.8 5.2
## [163] 5.2 2.6 5.2 4.6 5.6 4.8 6.2 4.8 5.4 4.4 4.2 4.2 6.6 4.2 5.4 3.6 6.0 4.0
## [181] 4.6 3.8 3.8 5.0 3.2 3.0 3.6 3.8 3.4 3.8 5.6 3.0 5.4 5.8 5.4 5.0 4.4 4.4
## [199] 6.2 4.2 5.0 3.2 2.4 5.2 5.4 6.0 3.2 5.6 4.4 3.0 5.2 3.4 4.6 2.8 6.6 5.4
## [217] 4.0 6.6 5.4 3.6 5.6 5.6 5.6 4.4 4.0 3.8 3.0 4.8 4.8 4.8 4.8 3.8 4.2 4.4
## [235] 4.0 3.2 6.2 6.8 4.8 4.2 4.2 5.0 4.6 3.8 6.8 3.0 4.4 4.4 3.2 4.0 4.0 3.4
## [253] 6.2 4.6 4.2 3.0 6.8 4.0 4.2 4.0 4.4 4.2 4.4 4.4 4.2 4.8 4.2 4.4 3.2 4.6
## [271] 3.8 5.2 3.6 5.4 6.2 5.8 5.6 2.6 3.6 4.0 2.8 3.2 5.2 6.2 2.8 4.8 5.6 3.8
## [289] 3.8 3.2 5.0 3.6 5.4 3.4 3.4 2.4 1.6 5.4 5.0 6.4 5.0 3.4 5.4 5.0 3.2 5.2
## [307] 4.0 5.4 4.0 4.4 7.0 6.0 5.0 3.4 5.2 4.8 5.8 4.4 4.2 6.2 5.6 2.8 5.8 5.0
## [325] 6.0 4.0 6.2 7.4 4.6 4.6 3.4 6.2 4.4 3.0 5.6 2.0 3.4 5.2 2.2 4.0 5.8 4.8
## [343] 3.2 4.6 4.2 3.4 5.2 4.2 4.6 5.4 2.8 5.4 2.8 4.8 4.2 4.0 4.0 3.8 6.6 4.4
## [361] 3.8 7.0 4.6 4.6 3.8 4.2 4.8 4.0 3.8 4.8 3.0 4.4 6.6 2.2 2.4 4.4 4.8 3.0
## [379] 5.8 4.2 4.2 4.8 4.4 3.4 4.6 3.4 4.6 1.8 4.4 4.8 6.6 3.8 4.0 5.2 3.8 4.8
## [397] 4.0 5.6 4.8 4.4 3.0 6.2 3.8 4.4 6.2 5.0 3.0 3.2 4.4 5.4 5.8 4.4 5.0 3.8
## [415] 6.0 4.8 4.4 6.2 5.0 3.6 4.8 4.2 6.6 2.8 4.6 5.4 4.6 4.4 4.6 3.8 4.0 5.0
## [433] 5.6 5.2 3.0 6.2 5.2 4.0 5.8 5.0 3.0 6.6 3.0 6.8 4.0 4.6 3.2 4.2 2.6 5.4
## [451] 4.8 3.8 5.8 5.2 4.4 5.0 5.2 3.4 4.8 5.6 5.0 5.6 4.4 4.8 3.8 5.6 4.2 6.2
## [469] 6.0 4.4 4.4 2.4 5.2 3.0 6.0 2.0 4.4 5.2 4.4 5.8 3.8 4.4 5.8 3.2 6.0 5.0
## [487] 4.8 3.6 5.0 5.0 5.2 1.4 6.4 5.8 6.4 7.2 4.6 4.8 5.0 6.2 3.2 4.2 4.8 5.6
## [505] 1.6 3.8 6.8 6.0 4.2 5.8 4.2 4.4 5.8 3.0 5.0 3.4 5.4 6.0 5.0 5.4 4.0 4.2
## [523] 4.2 5.2 1.8 5.6 4.8 2.4 3.2 4.6 4.4 4.6 4.4 3.0 4.2 4.6 2.6 5.8 4.6 5.0
## [541] 5.2 2.6 2.8 2.2 4.6 3.4 5.2 3.4 4.2 4.2 4.6 5.0 4.8 4.8 4.8 6.2 4.4 2.0
## [559] 5.8 6.0 6.2 3.0 3.4 3.8 4.0 6.2 5.4 4.4 4.4 5.0 5.0 4.4 5.8 2.8 6.2 3.2
## [577] 2.4 5.4 4.0 5.0 2.6 4.6 3.0 5.4 5.2 5.2 4.0 5.2 3.0 5.2 4.0 4.2 6.0 5.2
## [595] 6.6 4.0 5.0 3.8 5.2 6.4 3.8 3.2 5.2 5.4 3.6 5.0 4.0 5.2 3.4 4.6 3.6 2.8
## [613] 3.6 5.2 2.0 3.8 1.6 4.6 2.8 3.6 3.6 2.6 4.2 1.6 5.4 5.0 4.4 3.6 3.4 4.6
## [631] 3.2 4.0 3.8 4.0 5.0 3.2 5.4 4.2 6.0 2.6 5.0 5.6 3.6 2.0 5.4 5.6 5.8 3.6
## [649] 4.4 4.0 2.6 4.2 3.2 5.4 6.4 5.0 5.0 3.8 3.4 3.4 5.2 5.0 5.0 4.6 5.8 3.8
## [667] 3.8 5.6 6.8 5.0 3.8 4.0 3.2 7.2 6.2 5.8 5.4 1.8 5.2 4.6 6.8 3.4 4.2 3.2
## [685] 5.0 4.0 5.8 5.8 5.6 5.8 3.2 5.6 5.0 4.6 6.0 5.8 5.0 2.2 4.8 3.6 4.4 5.8
## [703] 6.8 6.0 3.4 2.2 5.6 2.0 4.2 4.4 4.8 3.4 6.2 4.2 4.6 5.2 5.8 6.2 5.6 4.0
## [721] 4.6 3.4 5.0 4.8 4.6 5.4 4.4 2.2 4.2 4.0 4.2 4.2 5.8 5.0 3.6 3.2 4.6 6.6
## [739] 5.2 5.2 3.4 6.0 6.2 7.4 4.6 5.4 3.6 3.2 3.6 4.2 6.4 4.4 6.0 4.6 4.8 4.6
## [757] 5.2 6.2 4.6 6.0 3.8 3.0 6.2 4.0 5.8 4.4 5.4 3.4 5.4 5.0 4.0 5.6 4.0 6.2
## [775] 4.0 4.0 5.4 5.6 5.0 6.8 4.4 3.8 5.4 4.0 2.8 3.6 3.6 4.0 3.2 4.4 4.2 5.2
## [793] 3.2 6.4 5.0 4.4 6.2 5.6 2.2 3.8 7.6 4.4 2.0 5.2 5.4 6.0 5.8 5.2 6.8 3.8
## [811] 5.0 2.6 6.0 7.0 4.4 5.0 3.4 5.6 4.2 4.0 3.0 3.6 4.8 4.2 4.6 5.2 4.2 4.2
## [829] 5.6 4.0 5.4 4.2 6.0 5.4 2.6 3.6 5.6 6.2 6.0 4.0 3.6 4.4 2.2 6.2 5.0 5.6
## [847] 4.0 4.6 5.2 4.6 3.0 5.8 5.8 5.4 2.2 4.4 4.6 5.2 5.2 3.4 5.4 4.2 4.0 5.4
## [865] 3.6 4.8 3.4 4.6 3.2 5.6 5.0 3.0 4.0 5.2 4.2 5.0 5.4 4.4 5.2 2.4 5.4 5.2
## [883] 5.2 3.4 5.8 5.8 2.4 3.6 3.0 2.0 3.0 4.8 4.4 3.6 4.6 2.8 4.4 2.6 5.0 3.6
## [901] 3.6 6.6 4.0 5.2 3.2 6.6 3.0 5.0 5.6 5.6 6.2 4.6 5.2 5.8 6.0 5.2 3.0 4.8
## [919] 6.0 5.6 4.0 2.8 6.8 3.8 5.4 6.6 4.0 2.6 3.8 4.0 4.4 5.6 5.2 5.0 6.2 3.2
## [937] 2.6 4.2 4.6 2.2 2.8 5.8 4.4 4.0 4.0 4.6 3.6 5.2 5.8 4.0 4.0 3.8 6.2 5.4

```

3.8 CENTRAL LIMIT THEOREM

3.8.1 The Standard Deviation

The sample mean defined in Definition 3.2 gives us a notion of centrality of the sampled data which, if the data is sampled from the same distribution and it is independent, is an estimate of the mean of the distribution. We now define a concept which is the analogue for the standard deviation defined in Definition 3.8. Recall that for a distribution, the standard deviation measures of how the data distributes around its mean.

standard deviation

Definition 3.5.

$$\text{sd}(\underline{X}) = \sqrt{\frac{(X_1 - m(\underline{X}))^2 + \dots + (X_n - m(\underline{X}))^2}{n - 1}}$$

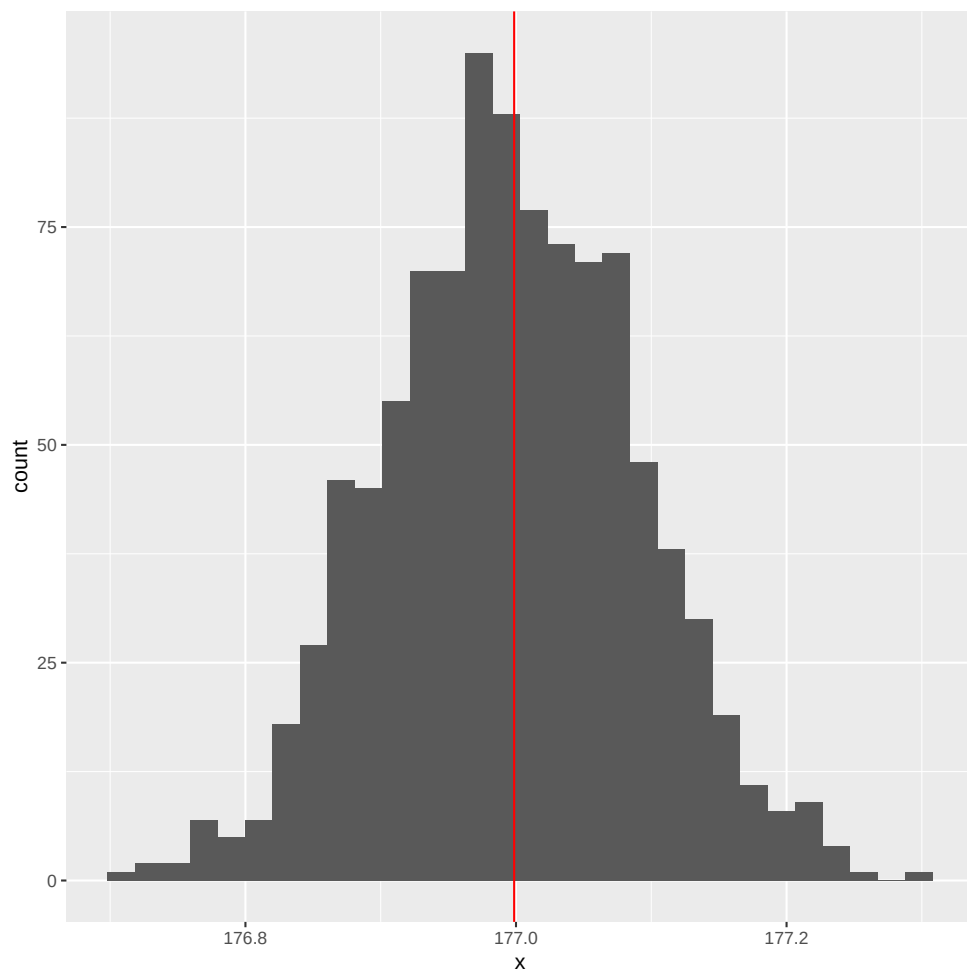
Notice that the $\sqrt{}$ means that $\text{sd}(\underline{X})$ has the same units and scales as $m(\underline{X})$. The choice of dividing by $n - 1$ rather than dividing by n is of no importance for our scopes.

3.9 STATEMENT

In Subsection 3.5.2 we have learned how histogram of independent samples can reconstruct the distribution. We can therefore have a better look at the distribution of the sample means by taking many samples

```
n <- 1000 #sample size
k <- 1000 #number of samples
ManySampleMeans <- replicate(k, mean(sample(Moscow`heights`, n)))
ggplot( data = tibble(x = ManySampleMeans) ) +
  geom_histogram( aes( x ) ) +
  geom_vline( xintercept = mean(Moscow`heights`), col = 'red' )

## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



As expected from the Law of Large Numbers, the sample means concentrate around the real mean. The Central Limit Theorem answers to the question of more precisely it distributes around the mean. Analytically, the Law of Large Numbers tells us that

$$m_n(\underline{X}) = \frac{X_1 + \dots + X_n}{n} = \mu + \text{Small Error} \quad (9) \quad \boxed{\text{e:LLN1}}$$

The Central Limit Theorem specifies further the structure of the small error:

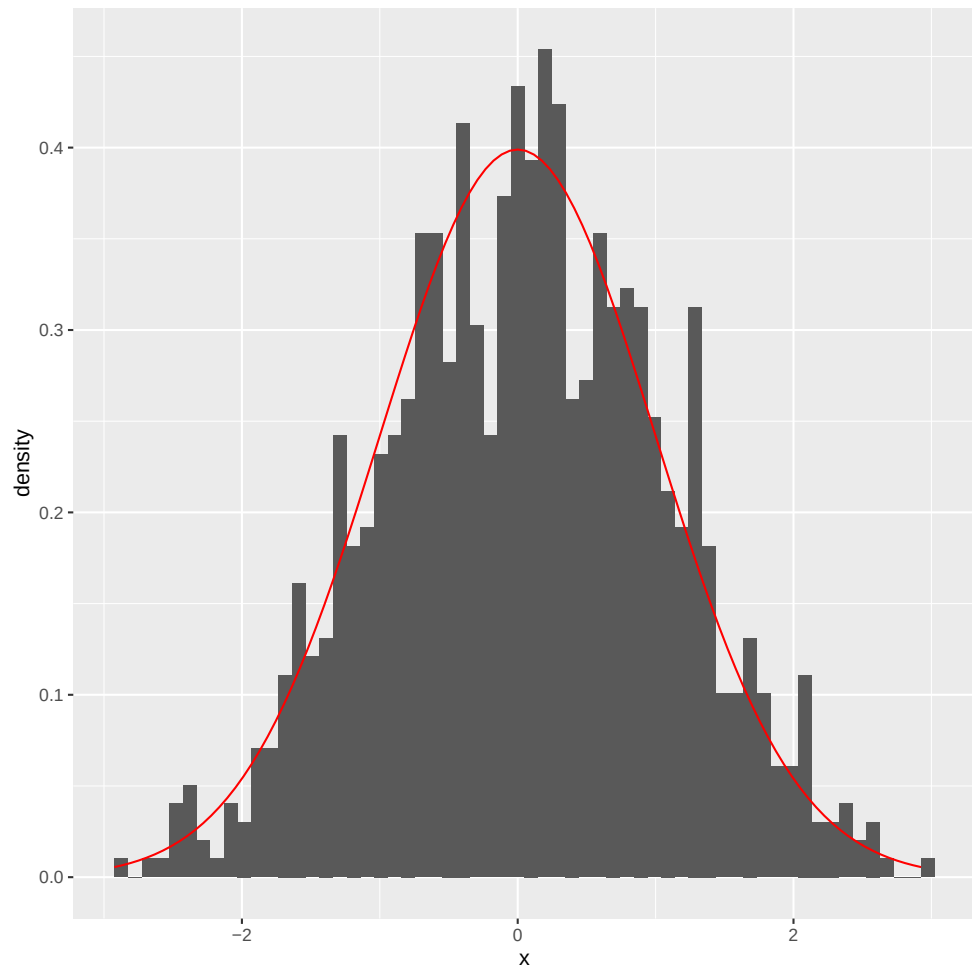
$$\text{Small error} = \frac{\sigma}{\sqrt{n}}Z + \frac{1}{\sqrt{n}}\text{Another small error}, \quad (10) \quad \boxed{\text{e:small error}}$$

where Z is a standard normal random variable. Substituting (10) into (9) we obtain

$$\frac{X_1 + \dots + X_n}{n} = m(\underline{X}) + \frac{\sigma}{\sqrt{n}}Z + \frac{1}{\sqrt{n}}\text{Another Small Error}$$

Note that the σ in (10) is necessary, since Z does not have units of measure (it is a universal object: its distribution is independent of the variables X_i !!!). Graphically, studying the quantity $\sqrt{n}\text{Small Error} = \sqrt{n}(m(\underline{X}) - \mu)$ is the quantity that we obtain from the histogram of the sample means that we have plotted above if we center it and if we zoom in by a factor of $\frac{\sqrt{n}}{\sigma}$. (In an histogram, the data lies on the x axis and, therefore, zooming in by a factor of $\sqrt{n}$ means that we dilate the axis). We plot the density histogram since Z is a continuous random variable of density $\frac{1}{\sqrt{2\pi}}e^{-x^2/2}$.

```
sample.size <- 10000
n.sample <- 1000 # The number of sample means we look at to generate the histogram
Many.Samples <- replicate(n.sample, mean(sample(Moscow.heights, sample.size)))
CLT <- sqrt(sample.size)*( Many.Samples - mean(Moscow.heights))/sd(Moscow.heights)
ggplot( data = tibble(x = CLT )) +
  geom_histogram( aes(x, after.stat(density)), bins = 60) +
  stat_function( fun = dnorm, args = list(mean = 0, sd = 1 ), col = 'red' )
```



t:CLT

Theorem 3.6. Let $\underline{X} = (X_1, \dots)$ be independent samples from a distribution that has a well defined mean μ and standard deviation σ . Then the distribution of

$$\sqrt{n} \frac{m_n(\underline{X}) - \mu}{\sigma} \quad (11)$$

e:CLT1

converges (in some sense) to a standard Gaussian random variable Z

In the case the sample comes from a population, the standard deviation of the random variable X has to be substituted by the standard deviation of the population

```
sd(Moscow'heights)
```

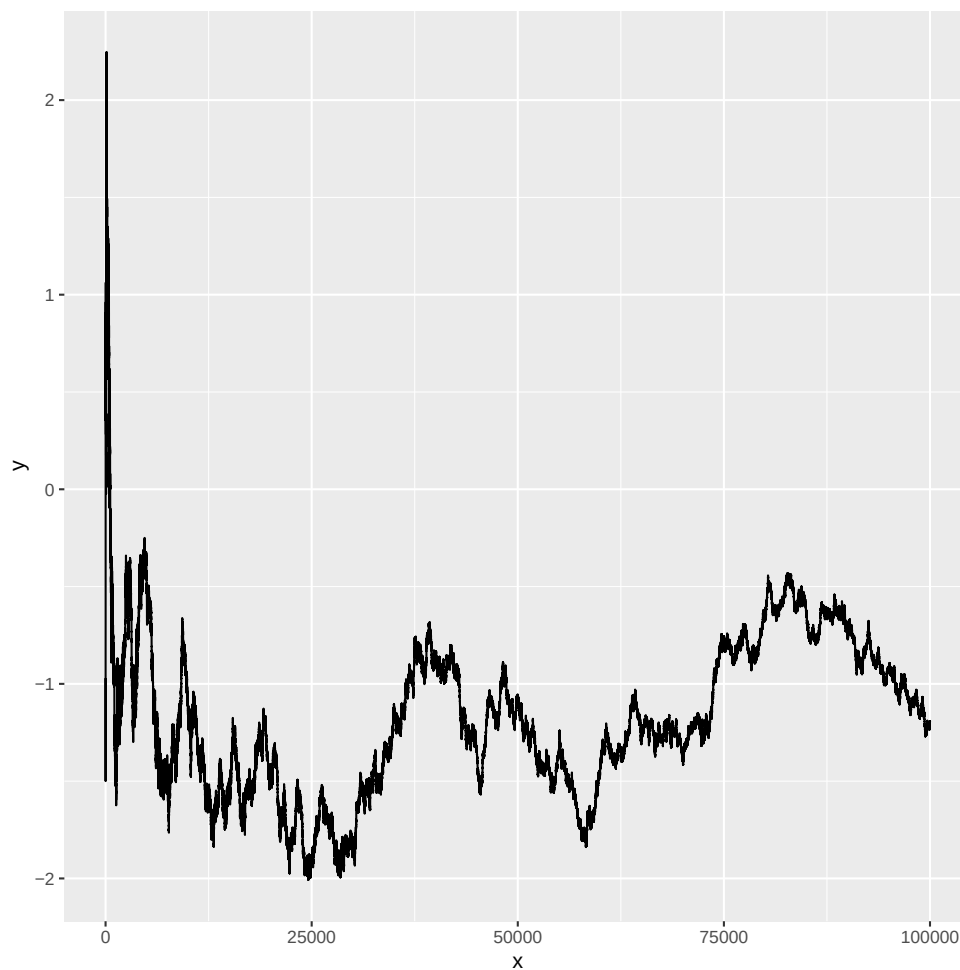
```
## [1] 2.884541
```

Note that in contrast of the Law of Large Numbers, the random quantity (11) converges to a random quantity Z . Z depends, of course, from the sample $\underline{X} = (X_1, \dots)$; however, its distribution does not. In this sense the Gaussians are a universal object and the Central Limit Theorem explains the fact that Gaussians are ubiquitous in science. Note that it is necessary to divide by σ , which has the same units of measure of the X_i , since the statement of the Central Limit Theorem guarantees that Z does not have units of measure,

since the quantity (11) has the same distribution in the limit for any unit of measure we use.

We finish this section by showing how does the quantity (11) depend on the sample size n

```
Sample`size` i- 100000
Sample i- rexp(Sample`size`)
# The exponential distribution has standard deviation 1
sigma i- 1
# and mean 1
mu i- 1
CLT i- sqrt((1:Sample`size`))* ( cumsum(Sample)/1:Sample`size` - mu)/sigma
ggplot( data = tibble(x = 1:Sample`size`, y = CLT )) +
  geom`line`(aes(x,y))
```



as we can see from the graph, as soon as the sample size becomes big enough so that the Central Limit Theorem hold, (11) is a random quantity of order 1.

Ex. 3.9.1 — Rewrite the codes of this Subsection and use it for samples from the exponential random variable, as in the example with the reaction times

If the samples are not independent but the

Ex. 3.9.2 — Think at an example where X_i are all equally distributed but the Law of Large Numbers does not hold. (Hint: what is the case in which the X_i are most dependent?)

The scope of this subsection is to revisit the Eugene Onegin Example and put it in the jargon of testing. Apart from being interesting on its own, we see how to treat the case in which the distribution of the statistics is unknown also under H_0 . Let X_i denote if the i -th alphabet character is either a consonant, 1, or a vowel, 0. The X_i are all equally distributed as Bernoulli random variables and a good guess for the parameter p is (we should take the russian drama corpus, but I want to avoid programming technicalities)

- H_0 : The X_i follow a Bernoulli trial distribution (see Definition ??) and
- H_1 : The X_i form a Markov Chain:

$$\mathbb{P}()$$

for some q and $(1 - q)$

- The statistic that we use is $s = \sqrt{n}(m_n(\bar{\cdot}) - p)$, the one of the Central Limit Theorem

It is very unlikely that the H_1 is true, but the point of testing is not to say whether H_1 is true but to refuse H_0 in favour of H_1 . We can estimate $q_{01} = \mathbb{P}(X_{i+1} = 1|X_i = 0)$ by counting the fraction of vowels followed by consonants, and $q_{10} = \mathbb{P}(X_{i+1} = 0|X_i = 1)$ by counting the relative frequency of consonants followed by vowels. $\hat{q}_{ij} =$

the Central Limit Theorem for the Bernoulli Trial Measure tells us that

(12) e:

3.10 T-TEST

In this section we are going to introduce the z-test and the more conservative t-test, used to prove hypotheses concerning the mean of numerical variables. This section comes right after the Central Limit Theorem 3.6 since we will use it to approximate the distribution of the t statistic, the quantity looked by both of these statistical tests, provided the null hypothesis holds.

3.10.1 Standard Deviation

Recall that from a distribution of random variable we have defined the variance in Definition ?? and the standard deviation in Definition 3.8. Similarly to what happens to the mean of the random variable X and its sample mean associated to a sample $\underline{X} = (X_1, \dots)$, see Definition 3.1, it is possible to define a sample standard deviation

d:sampled'sd

Definition 3.7 (Sampled standard deviation). Let $\underline{X} = (X_1, \dots, X_n)$, be a vector of $n \in \mathbb{N}$ components, and recall the definition of (sample) mean of a vector in Definition 3.2. The sampled standard deviation is

$$sd_n(\underline{X}) = \sqrt{\frac{(X_1 - m_n(\underline{X}))^2 + \dots + (X_n - m_n(\underline{X}))^2}{n - 1}}$$

e:sampled'sd

The fact that in the denominator there is the number $n - 1$ instead of n is due to statistical reasons that we do not study in the present notes. However, for n large, the difference is minimal.

The same remarks as in the definition of standard deviation for a random variable in Definition ?? apply. The sampled standard deviation has the same units of measure as the X_i and it is a measure on how the data contained in the vector $\underline{X}$ distribute around the mean. If the $\underline{X}$ are independent samples from the same distribution, then the relation between the samples standard deviation sd_n and the standard deviation of the distribution is the same one as the relation between the mean of the distribution μ and the sample mean m_n , see Definition ?? and 3.2, and it is given by the Law of Large numbers

lard'deviation

Proposition 3.8. Let $\underline{X} = (X_1, \dots)$ be a sample, see Definition 3.1, from a distribution of standard deviation σ .

$$sd_n(\underline{X}) \rightarrow \sigma$$

e:standard'Deviation

as $n \rightarrow \infty$

Indeed, $(X_1 - m_n(\underline{X}))^2, (X_2 - m_n(\underline{X}))^2, \dots$ are independent samples from the random variable $(X - \mu)^2$, where μ is the mean of the random variable X . The mean of $(X - \mu)^2$ is precisely $\text{Var}(X)$, see Definition ?? and, therefore, the Law of Large Numbers ?? implies that

$$\frac{(X_1 - m_n(\underline{X}))^2 + \dots + (X_n - m_n(\underline{X}))^2}{n} \rightarrow \text{Var}(X)$$

```

sample.size i- 1000
X i- rexp(sample.size)
sqrt(sum(X - mean(X))/(sample.size - 1) )

## Warning in sqrt(sum(X - mean(X))/(sample.size - 1)): Si è prodotto un
NaN

## [1] NaN

sd(X)

## [1] 0.9364483

## The standard deviation of an exponential random variable is

```

Ex. 3.10.1 — Two samples $\underline{X}$ and $\underline{Y}$ of two different numerical variables have been taken and the results visualized in the histogram in Figure ???. Which one corresponds to a higher standard deviation?

3.10.2 Intuition on the t-statistic

The t-test is a test that is used for testing hypotheses on the mean of numerical variables. The example we will follow to introduce it in a practical level is the same one of the heights of the citizens of Moscow. We know that a good estimate for the heights of the citizens of Amsterdam is $\mu_{\text{Amsterdam}} = 178$, and we collect data to prove that the mean height of citizens of Moscow is inferior. To this end we take samples from the population of Moscow and test the following hypotheses

- H_0 : The mean height of the citizens of Moscow is $\mu_{\text{Moscow}} = \mu_{\text{Amsterdam}}$ and we are taking independent samples
- H_1 : The mean height of the citizens of Moscow is $\mu_{\text{Moscow}} < \mu_{\text{Amsterdam}}$ and we are taking independent samples.

Our aim is, once more, to give a rule to decide whether we have enough information to reject H_0 in favour of H_1 . Of course all the discussions in Subsection ?? hold and the principle of the t-test is no different from others

```
sample1 i- c(176, 178, 177, 175, 179)
```

Does the above sample give an hint in favour of H_1 with respect to H_0 ? Is the above sample sufficient to reject H_0 in favour of H_1 ?

```
sample2 i- c(174, 176, 175, 173, 177)
```

Does the above sample give a stronger hint in favour of H_1 with respect to the first sample? If yes, why? The mean of sample2??? is smaller than sample1??, and this is why it provides a stronger hint for H_1 than the hint of sample1

```
sample3 j- c(177, 177.5, 176.5, 178, 176)
```

Does the above sample give an hint in favour of H_1 with respect to H_0 ? Is the above sample sufficient to reject H_0 in favour of H_1 ? Although sample3 has the same mean of sample1, it provides a stronger int with respect to sample1 since there is

```
sample4 j- c(176, 178, 177, 175, 179, 176, 178, 177, 175, 179, 176, 178, 177, 175, 179, 176, 178, 177, 175, 179)
```

All the above considerations are summarised in the *t-statistic*

d:t-statistic

Definition 3.9 (t statistic). Let $\underline{X} = (X_1, \dots, X_n)$ be independent samples from a distribution, μ a guess for the mean of the distribution, m be the sample mean defined in (??) and $sd(\underline{X})$ the standard deviation defined in (??)

$$(??)t \equiv t_n(\underline{X}, \mu) = \sqrt{n} \frac{m(\underline{X}) - \mu}{sd(\underline{X})}$$

We now compute the t-statistic of the above samples

```
# The t-statistic is a function of the sample and of the hypotetic mean $\mu$
t.stat j- function( X, mu) {
  sqrt(n)*(mean(X) - mu)/sd(X)
}
```

If we assume that the null hypothesis H_0 holds, that is, that $\mu = 178$ also for the Moscow population, we can compute the associated t-statistic

```
t.stat( sample1, 178)
t.stat(sample2, 178)
t.stat(sample3, 178)
t.stat(sample4. 178)

## Error in parse(text = input): j;text;4:17: costante numerica inattesa
## 3: t.stat(sample3, 178)
## 4: t.stat(sample4. 178
## ^
```

As we can see, the t-statistic of sample2, sample3 and sample4 is smaller than the one obtained for sample1. The reason is that (??) contains the expressions that en $m_5(\text{sample2}) < m_{\text{sample1}}$, in the first case, in the second case $sd_5(\text{sample3}) < sd_5(\text{sample3})$, in the second case, and in the last case the t-statistic is smaller since n is much bigger for sample4. Both the t-test and the z-test use the t-statistic, and the decision rule is, assuming that under H_1 a smaller t is more probable, is to reject H_0 in favour of H_1 if $t < t_{\text{crit}}$ for a certain t_{crit} . How t_{crit} is decided defined the test Placing the above considerations in the framework for testing hypotheses described in Subsection ??, we have

- A null

- An alternative hypotheses
- A notion of extremality,

The decision

3.10.3 Z-test

If H_0 is true, then the expression (??) resembles the expression (??) in the Central Limit Theorem, with the only difference that the standard deviation is substituted by the sample standard deviation. However, the Law of Large Numbers tells that these two quantities coincide. If the sample size n is big enough, then the Central Limit Theorem guarantees that under H_0 the t -statistic follows a standard normal distribution.

Knowing (approximately) the distribution of t allows us to define a t_{crit} in the usual way. We fix an $\alpha > 0$, usually $\alpha = 0.05$ and we choose t_{crit} such that the probability of making a type 1 error, provided H_0 holds, is α . Since

$$\mathbb{P}(\text{"}H_0 \text{ rejected"}|H_0) = \mathbb{P}(t < t_{\text{crit}}|H_0) \sim \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{t_{\text{crit}}} e^{-\frac{x^2}{2}} dx \quad \boxed{e: t_{\text{crit}}}$$

and the above quantity has to be equal to α . Graphically we are looking for the biggest t_{crit} so that the mass on its left is smaller than α , which is sometimes called the left inverse of the cumulative distribution function and is already implemented in R

Equivalently, we reject H_0 if and only if the p-value of the statistics observed is smaller than α . To this end. The Z-test is already implemented in

3.11 T-TEST

If the X_i are sampled from a Gaussian distribution of mean μ , then the t -statistics associated to μ and n follows a t -student distribution with n degrees of freedom

```
function t_statistic(x, mu, ) {
  return ( sqrt(length(x))*( mean(x) - mu)/sd(x))

sample_size j- 5
n j- 10000
data j- replicate(k, t_statistic(rnorm(sample_size, mean = mu_h0, sd = 1))
gplot(data = tibble( x= t_statistic)) +
  geom_histogram( aes(x , y = after_stats(density)) +
    geom_line(data = tibble(x = -7:7, y = stsudent(-7:7), args = list(df = n-1))aes(x = 1:k , y = ))

## Error in parse(text = input): jtextj:1:18: simbolo inatteso
## 1:      function t_statistic
##      ^
```

The t -test is defined by assuming that . Since the tails of the t -student distribution weight more than the ones of the normal, we have that the critical t of the t -test is greater than the one for the Z -test. In other terms is more conservative.

```
alpha = 0.095
n = 10
qtstudent( df = n- 1)

## Error in qtstudent(df = n - 1): non trovo la funzione "qtstudent"

t.test()

## Error in t.test.default(): argomento "x" assente, senza valore predefinito
```

3.11.1 t-test

The t -test differs from the z -test only by the fact that the distribution of $t = t_n(\underline{X})$, provided H_0 holds, is assumed to be a t -student distribution of $n - 1$ degrees of freedom. This gives rise to different p -values and to a different t_{crit} . For each x , if we observe the event $X = x$ the distribution of Y changes (unless X and Y are independent, see Definition ??). We denote by $\mu(x)$ the mean of Y given X .

```
plot_regression j- function(x, y, m_guess, b_guess) {
  data j- data.frame(x, y)

  # Compute guessed line
  y_guess j- m_guess * x + b_guess
```

```

residuals i- y - y`guess
data$y`guess i- y`guess
data$residuals i- residuals

# Base plot with points
plot i- ggplot(data, aes(x = x, y = y)) +
  geom`point(color = 'blue', size = 3) +
  geom`abline(intercept = b`guess, slope = m`guess, color = 'red', linetype = 'solid', size = 1) +
  labs(title = 'Linear Regression Guess with Residuals', x = 'X', y = 'Y') +
  theme`minimal()

# Add residual lines and labels
plot i- plot + geom`segment(aes(x = x, y = y, xend = x, yend = y`guess), color = 'green') +
  geom`text(aes(label = round(residuals, 2), y = (y + y`guess) / 2), color = 'green', hjust = -0.5)

print(plot)
}

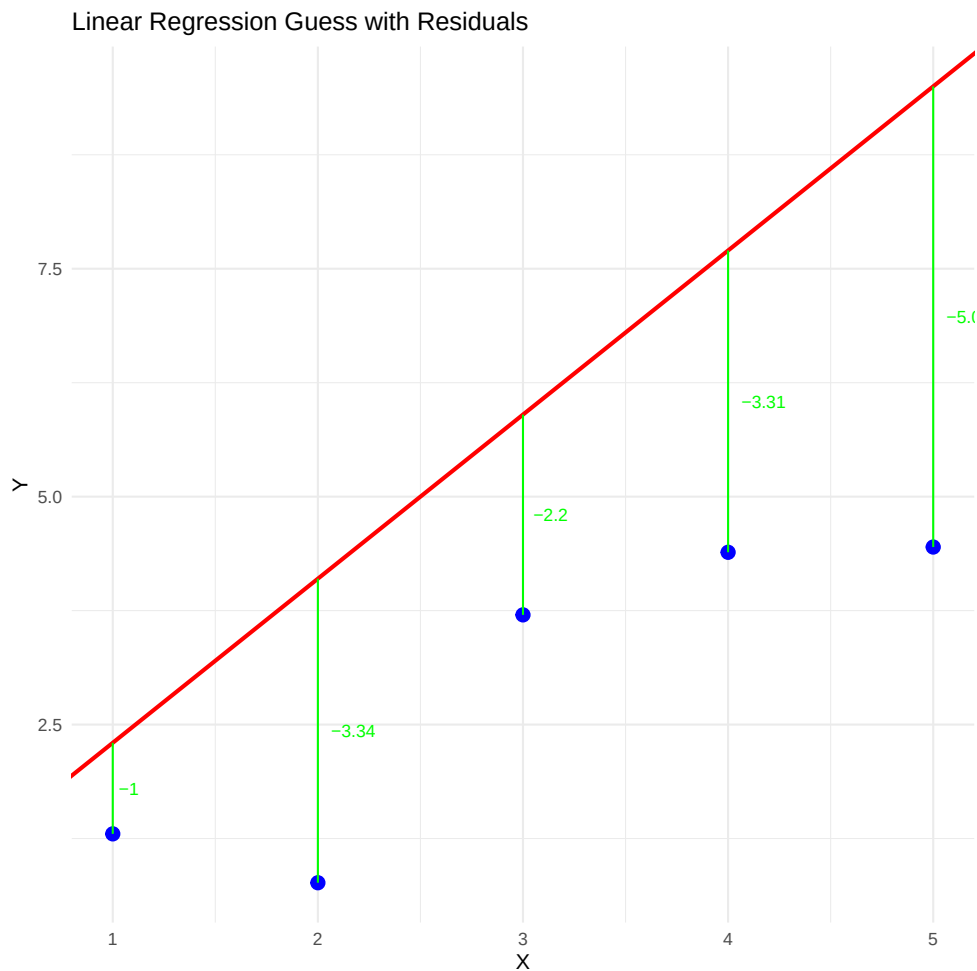
# Example Data
x i- c(1, 2, 3, 4, 5)
y i- x + rnorm(5)

# Example Guess (wrong on purpose)
m`guess i- 1.8 # Slope guess
b`guess i- 0.5 # Intercept guess

plot`regression(x, y, m`guess, b`guess)

## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```



3.11.2 Sample mean as a minimum

$$\mathbb{E}[(X - \mu)^2] \sim \sum_{i=1}^n (X_i - \mu)^2$$

e:sample mean1

and, quite surprisigly, this $\sim$ property commutes with the $\min_{\mu}$ operator!.
 (It is not really surprising since the probability on the right hand side is made by taking the sample mean) Therefore, if

3.11.3 Mean as a predictor

The linear regression assumes you want to predict the value of Y using its mean. If there is no X variable re

3.12 SOME PRECAUTIONS TO TAKE WHEN TESTING

3.12.1 Perform tests after looking at the data

3.12.2 Multiple Comparisons

Another similar issue happens when performing multiple tests. As an example, think of correcting an exam which has a certain number of questions, say 30, at an high school. To decide whether or not two different students have cheated, we perform a simple test and decide that two students have copied from each other if the $k = k(\alpha)$ out of 30 answer coincide (here k depends on the significance level we want. The smaller the α , the closer the k to 30). Now, in the classroom there are 20 students sitting next to each others in pairs, and we decide only to test those pairs since they are most at risk of copying.

The underlying assumption is that the null hypothesis H_0 comes with a model for the possible outcomes of two exams that where made by students that did not copy for each other, so that, under H_0 , we can compute

$$\mathbb{P}(\# > k)$$

and define $k = k(\alpha)$, the critical value for which without copying, whose precise form is not important for illustrating the multiple comparison problem, that allows us to compute As usual, $k(\alpha)$ is de As a practical example, the correlation from a le

3.12.3

Given Y Assume we observe a sample $\underline{Y} = (Y_1, \dots, Y_n)$ of sample size $n \in \mathbb{N}$ from a random variable Y . We want to use the previous information to make a more informed prediction for Y . It is possible to

3.12.4 Multivariate Regression

It is straightforward to extend the previous framework to the multivariate in which for each observation we collect more than two numerical variables. To fix the ideas, let us consider three variables, X , Y and Z , and let $\underline{X} = (X_1, \dots, X_n)$, $\underline{Y} = (Y_1, \dots, Y_n)$ and $\underline{Z} = (Z_1, \dots, Z_n)$ be samples of size $n \in \mathbb{N}$. Using the sample, which is a random quantity, we want to estimate the deterministic quantity

$$\mathbb{E}[Z|X, Y]$$

which is a function of the possible values X and Y might assume. To this end we use the characterization (??)

$$\mathbb{E}[(Z - \mathbb{E}[Z|X, Y])^2|X, Y] = \min_{\mu(x,y)} \mathbb{E}[(Z - \mu(X, Y))^2|X, Y]$$

eq:minimum

so that we choose $\hat{Z}$ so as to minimize the above quantity. This is the real $\hat{Z}$.
How to estimate the fake one???

$$\hat{Z} =$$

https://www.tylervigen.com/spurious/correlation/4018_american-cheese-consumption_correlates-with_blackrocks-stock-price

3.13 SOME OTHER INTERESTING REFERENCES

An article that I haven't read of linear regression in Linguistic Data: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5911484/> A very good answer to the question (if you ever had it) to why p-values that are too small are not reported <https://stats.stackexchange.com/questions/78839/how-should-tiny-p-values-be-reported>
Another book on line

3.13.1 Eugene Onegin

ss:Eugene'Onegin4

We now continue and conclude Example ??, ?? and 3.13. Recall that from a text in russian we have defined a sequence $\underline{X}$ whose i th digit X_i can be 0, indicating that in the i -th place there is a vowel, or 1, indicating a consonant.

- H_0 : $\underline{X}$ is an instance of a Bernoulli trial measure $\mathbb{P}_p$ for some $p \in [0, 1]$, see Definition ??.
- H_1 : $\underline{X}$ is an instance of the two-state Markov model $\mathbb{P}_{a,b}$ for some $a, b \in [0, 1]$, see Definition ??

For the sake of readability, we recall the two aforementioned distributions. For $x_1, \dots, x_n \in \{0, 1\}$ a fixed sequence of digits belonging to 0 and 1, with $n \in \mathbb{N}$, the model H_0 assumes

$$\mathbb{P}_p(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n \mathbb{P}_p(X_i = x_i) = p^{x_1 + \dots + x_n} (1 - p)^{n - (x_1 + \dots + x_n)}$$

and the model H_1 assumes

$$\begin{aligned} \mathbb{P}_{a,b}(X_1 = x_1, \dots, X_n = x_n) &= \mathbb{P}_{a,b}(X_1 = x_1) \mathbb{P}_{a,b}(X_2 = x_2 | X_1 = x_1) \dots \mathbb{P}_{a,b}(X_n = x_n | X_{n-1} = x_{n-1}) \\ &= \mathbb{P}(X_1 = x_1) \prod_{i=1}^n P_{x_{i-1}, x_i} \end{aligned}$$

where the *transition matrix* $P(x, y)$, whose meaning is precisely $\mathbb{P}(X_{i+1} = y | X_i = x) = P_{x,y}$ for every $i \in \mathbb{N}$, is given by the following values

$$P_{0,0} = (1 - a), \quad P_{01} = a, \quad P_{1,0} = b, \quad P_{1,1} = 1 - b$$

Recall also that in Example ?? we have estimated the parameter p for the first model, and the parameters a and b for the second model

```
from j- vec[-length(vec)]
## Error: oggetto 'vec' non trovato
to j- vec[-1]
## Error: oggetto 'vec' non trovato
```

```

# Count transitions
n`01 j- sum(from == 0 & to == 1)

## Error: oggetto 'from' non trovato

n`00 j- sum(from == 0 & to == 0)

## Error: oggetto 'from' non trovato

n`10 j- sum(from == 1 & to == 0)

## Error: oggetto 'from' non trovato

n`11 j- sum(from == 1 & to == 1)

## Error: oggetto 'from' non trovato

# We estimate the jump probability from 0 to 1
# n`00 + n`01 is the total number of jumps from 0, i.e., the total number of 0
a`hat j- n`01 / (n`00 + n`01) # P(1 — 0)

## Error: oggetto 'n`01' non trovato

# We estimate the jump probability from 1 to 0
b`hat j- n`10 / (n`10 + n`11) # P(0 — 1)

## Error: oggetto 'n`10' non trovato

p`1 j- (n`10 + n`11) / (n`00 + n`01 + n`10 + n`11)

## Error: oggetto 'n`10' non trovato

p j- mean(vec)

## Error in mean(vec): oggetto 'vec' non trovato

# Print estimated transition matrix
cat("Estimated transition matrix:\n")

## Estimated transition matrix:

print(matrix(c(1 - a`hat, a`hat, b`hat, 1 - b`hat), nrow = 2, byrow = TRUE,
             dimnames = list(from = c("0", "1"), to = c("0", "1"))))

## Error in matrix(c(1 - a`hat, a`hat, b`hat, 1 - b`hat), nrow = 2, byrow =
TRUE, : oggetto 'a`hat' non trovato

```

We have already seen th

and that we have defined the statistic of the jumps between 0s and 1s to see that it differs significantly from the one of the Bernoulli model H_0 , see (3.2.1). In this last part of the example we compute the statistic that Markov originally computed in his paper [Markov] using the Eugene Onegin poem. The Central Limit Theorem 3.6 states that the fluctuations from the mean of i.i.d. random variables $X_1, \dots, X_n$ with finite variance, that is, the statistic (8), as $n \rightarrow \infty$ approaches to a normal distribution (in a sense that we

have not discussed). In the present case, if H_0 holds true, the X_i are independent Bernoulli random variables with mean p and variance $p(1-p)$, see Example ???. Therefore, the Central Limit Theorem (??) reads in this case

$$\sqrt{n} \frac{\frac{1}{n} \sum_{i=1}^n X_i - p}{\sqrt{p(1-p)}} \rightarrow Z \sim \mathcal{N}(0,1).$$

e:CLT·Bernou

If, instead H_1 were true (and this was the main mathematical motivation for introducing Markov Chains), a CLT holds with $\sqrt{p(1-p)}$ replaced by $\sigma_{\text{Markov}} := \frac{ab(2-a-b)}{(a+b)^3}$, namely,

$$\sqrt{n} \frac{\frac{1}{n} \sum_{i=1}^n X_i - p}{\sigma_{\text{Markov}}} \rightarrow Z \sim \mathcal{N}(0,1)$$

e:CLT·Markov

We will proceed as follows: we first fix n large enough so that the statistic

$$f = \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i - p \right)$$

e:statistic

will be, if H_0 is true, close to a normal of standard deviation $\sqrt{p(1-p)}$, or, if H_1 is true, close to a normal of standard deviation σ_{Markov} . Then we divide the vector $\underline{X}$ in subvectors of length n , and on each we compute the statistic f , so that we have many instances of f . We would like then to use (12) to estimate the distribution of f using the histogram of the values thus obtained, and, indeed, that would be the case if the various instances of f were independent. We therefore need another very plausible approximation, that is, that the values of f obtained on each of the subvectors of $\underline{X}$ are independent. This is indeed very plausible because if X_i is 1 or 0, it influences only its neighbouring digits, but is very unlikely to have consequences on digits that are far apart. Therefore two different f , say f_1 and f_2 depend only through the boundary values X_n and X_{n-1} , and, since each one of the, influences the f statistic only by a factor $1/\sqrt{n}$, the dependence is indeed very small. (Assuming H_1 holds this fact can also be proved) We start by subdividing the vector and computing on each subvector the statistic

```
chunk`size = 400
chunks j- split(vec, ceiling(seq`along(vec) / chunk`size))

## Error in split(vec, ceiling(seq`along(vec)/chunk_size)): oggetto 'vec' non
trovato

Many`f j- vector(mode = 'numeric')
for (i in 1:length(chunks)) {
  Many`f j- c(Many`f , sqrt(chunk`size)*(mean(chunks[[i]]) - p))
}

## Error: oggetto 'chunks' non trovato
```

finally we draw the histogram and compare it with the two different limiting distributions obtained theoretically from the hypothesis H_0 and H_1 , namely

a mean zero normal of standard deviation $\sqrt{p(1-p)}$ and a mean zero norm with standard deviation σ_{Markov} , respectively.

```
clt`iid <- function(a,b){
  return(sqrt((a*b)/(a + b )^2))
}
sigma`iid <- clt`iid(a`hat, b`hat)

## Error in clt_iid(a_hat, b_hat): oggetto 'a_hat' non trovato

clt`Markov <- function(a,b){
  return(sqrt(a*b*(2-a - b)/(a + b )^3))
}
sigma`markov <- clt`Markov(a`hat, b`hat)

## Error in clt_Markov(a_hat, b_hat): oggetto 'a_hat' non trovato

ggplot(data = data.frame(Many`f )) +
  geom`histogram(aes(x = Many`f, after`stat(density) ), bins = 30 ) +
  geom`function(fun = dnorm, args = list(mean= 0, sd = sigma`markov), color = "blue") +
  geom`function(fun = dnorm, args = list(mean = 0, sd = sigma`iid), color = "red")

## Error in list2(na.rm = na.rm, ...): oggetto 'sigma_markov' non trovato
```

We see graphically that H_1 is a much better model, at least for the statistic f . Now a little bit of Bayesian approach, since I don't like testing. In this approach everything is unknown to you can be given a probability and, recall that although both models are probably false, one maybe is better than the other and, therefore, we are more interested in the odds

$$\frac{\mathbb{P}(H_1)}{\mathbb{P}(H_0)}$$

rather than in absolute probabilities. We also recall how the odds between the two models updates after observing the data

$$\frac{\mathbb{P}(H_1|\text{data})}{\mathbb{P}(H_0|\text{data})} = \frac{\mathbb{P}(\text{data}|H_1) \mathbb{P}(H_1)}{\mathbb{P}(\text{data}|H_0) \mathbb{P}(H_0)}$$

Now, for us the observed data will be the f_i , and we have observed a lot of them in the vector $Many.f$ $f_1 = f(X_1, \dots, X_k)$, $f_2 = f(X_{k+1}, \dots, X_{2k})$, ... and we can update the odds many times. Let's fix the prior odds to 1, that is, we start thinking that both models are equally likely (even if I would start from more than one, but let's be conservative)

$$\frac{\mathbb{P}(H_1)}{\mathbb{P}(H_0)} = 1$$

Recalling that $\mathbb{P}(H_0) = \mathbb{P}_p$ (this is not quite correctly since also $\hat{p}$ is simply an estimate and we should take another time a Bayesian approach. But

let's say that we are testing $\tilde{H}_0 : \mathbb{P}_{\hat{p}}$ and $\tilde{H}_1. \mathbb{P}_{\hat{a}, \hat{b}}$ and that $\mathbb{P}(|H_1) = \mathbb{P}_{\hat{a}, \hat{b}}$ (same commentary), and that, since k is large, we replace

$$\mathbb{P}_{\hat{p}}(f_i = x) = \mathbb{P}(\sqrt{p(1-p)}Z \in x + dx) = \frac{1}{\sqrt{2\pi p(1-p)}} e^{-\frac{x^2}{2p(1-p)}} dx$$

$$\mathbb{P}_{\hat{a}, \hat{b}}(f_i = x) = \mathbb{P}(\sigma_{\text{Markov}}Z \in x + dx) = \frac{1}{\sqrt{2\pi\sigma_{\text{Markov}}^2}} e^{-\frac{x^2}{2\sigma_{\text{Markov}}^2}} dx$$

We are now ready to update the odds many many times and to write an article, were we born in the 19th century. i

$$\frac{\mathbb{P}(H_1|f_i)}{\mathbb{P}(H_0|f_i)} = \sqrt{\frac{\sigma_{\text{Markov}}^2}{p(1-p)}} e^{-\left(\frac{1}{\sigma_{\text{Markov}}^2} - \frac{1}{p(1-p)}\right)x^2} \frac{\mathbb{P}(H_1)}{\mathbb{P}(H_0)} \quad (13) \quad \boxed{\text{e:posterior2}}$$

for every $i \in \mathbb{N}$. Notice that (13) takes in input the odds of a probability measure and updates them. Each time we look at the data we update the probability measure with the same function (We are using the independence between the f_i !!!) This is quite good for computational purposes.

```
posterior`odds i- function( data, prior`odds, sigma`1, sigma`2){
  # The posterior odds are P(H`1—data)/P(H`0—data)
  # By the Bayes formula
  # postierior odds = prior odds * ( P(data — H`1 )/P(data — H`0 ))
  return((dnorm(data, mean = 0, sd = sigma`1) /dnorm(data, mean = 0, sd = sigma`2))*prior`odds)
}
odds i- 1
for (x in Many`f[1:20]) {
  odds i- posterior`odds(x, odds, sigma`markov, sigma`iid)
}

## Error in posterior_odds(x, odds, sigma_markov, sigma_iid): oggetto 'sigma_markov'
non trovato

odds

## [1] 1
```

Finally, just for fun, let's just simulate a real markov chain, that is, a stochastic process for which we know that H_0 is true and perform the same procedure

```
simulate`markov`chain i- function(n, a, b, start`state = 0) {
  states i- numeric(n)
  states[1] i- start`state

  for (i in 2:n) {
    if (states[i - 1] == 0) {
      states[i] i- rbinom(1, 1, a) # 0 -> 1 with prob a
    } else {
      states[i] i- rbinom(1, 1, 1 - b) # 1 -> 1 with prob 1 - b
    }
  }
}
```

```

    }
  }

  return(states)
}

binary_2 <- simulate_markov_chain(length(vec), a_hat, b_hat )

## Error in numeric(n): oggetto 'vec' non trovato

chunks_2 <- split(binary_2, ceiling(seq_along(binary_2) / chunk_size))

## Error in split(binary_2, ceiling(seq_along(binary_2)/chunk_size)): oggetto
'binary_2' non trovato

Many_f_with_the_right_distr <- vector(mode = 'numeric')
for (i in 1:length(chunks_2)) {
  Many_f_with_the_right_distr <- c(Many_f_with_the_right_distr , sqrt(chunk_size)*(mean(chunks_2[[i]]) - p))
}

## Error: oggetto 'chunks_2' non trovato

ggplot(data = data.frame(results_2 )) +
  geom_histogram(aes(x = results_2, after_stat(density) ), bins = 30 ) +
  geom_function(fun = dnorm, args = list(mean= 0, sd = sigma_markov), color = "blue") +
  geom_function(fun = dnorm, args = list(mean = 0, sd = sigma_iid), color = "red")

## Error in data.frame(results_2): oggetto 'results_2' non trovato

```

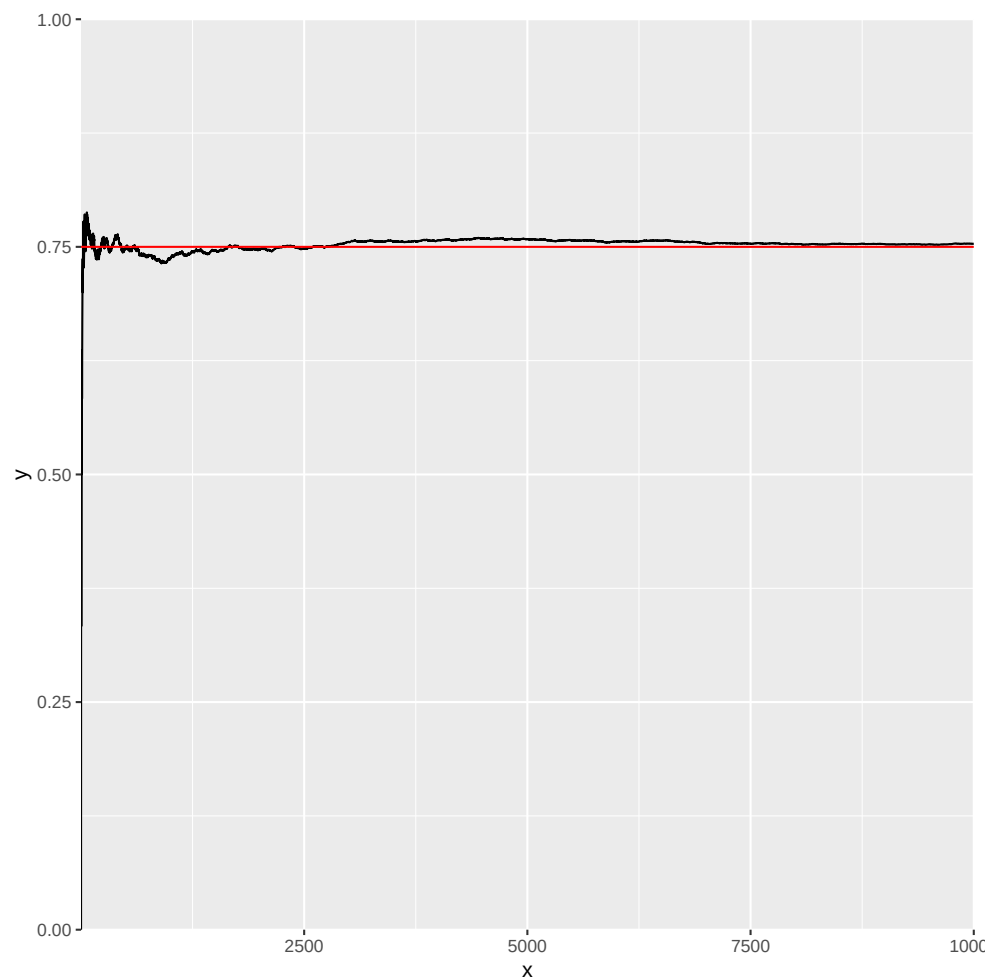
3.14 CENTRAL LIMIT THEOREM FOR MARKOV CHAINS

The LLN for Markov Chains

```

sample.size <- 10000
a <- 0.9
b <- 0.3
MC <- simulate.markov.chain(sample.size, a, b)
# It's mean is b/(a + b)
mu = a/(a + b)
# It's sd is a*b*(2-a-b)/(a + b)^3
sigma.Markov = sqrt(a*b*(2-a-b)/(a + b)^3)
sigma.iid <- sqrt(mu*(1-mu))
sample.mean.MC <- cumsum(MC)/1:sample.size
ggplot(data = tibble(x = 1:sample.size, y = sample.mean.MC), aes(x, y)) +
  geom.line() +
  geom.hline(yintercept = mu, col = 'red') +
  coord.cartesian(ylim = c(0.0,1.0),
                  expand = FALSE)

```

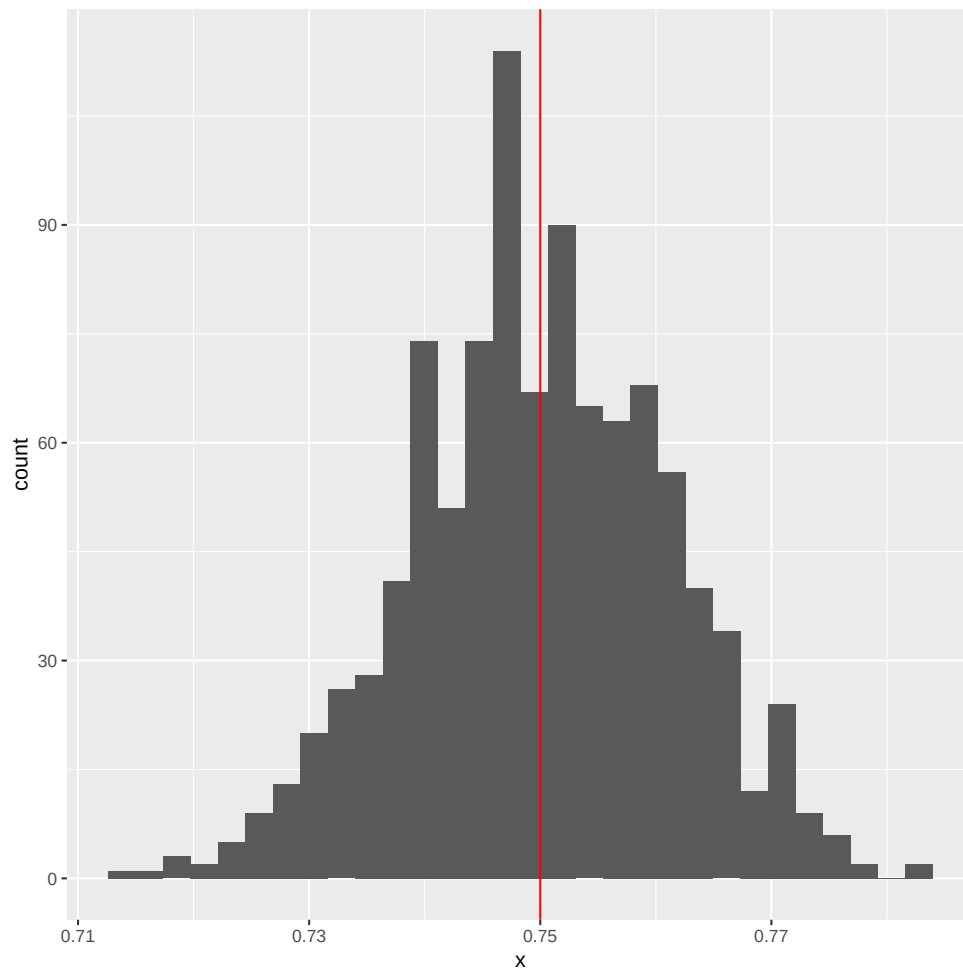


```

sample_size j- 1000 #sample size
k j- 1000 #number of samples
ManySampleMeansMC j- replicate(k, mean(simulate_markov_chain(sample_size, a, b )))
ggplot( data = tibble(x = ManySampleMeansMC )) +
  geom_histogram( aes( x )) +
  geom_vline( xintercept = mu, col = 'red')

## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.

```

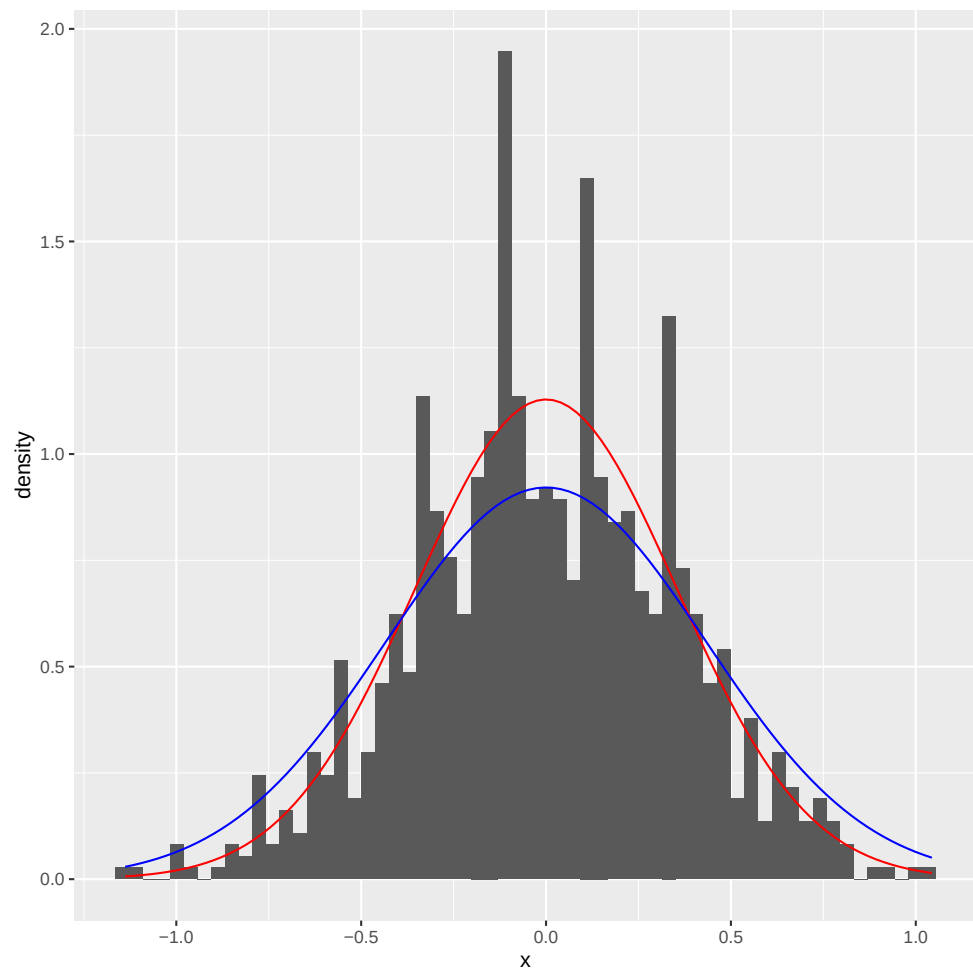


The CLT for Markov Chains

```

CLT j-sqrt(sample_size)*( ManySampleMeansMC - mu)
ggplot( data = tibble(x = CLT )) +
  geom_histogram( aes(x, after_stat(density)), bins = 60) +
  stat_function( fun = dnorm, args = list(mean = 0, sd = sigma_Markov ), col = 'red') +
  stat_function( fun = dnorm, args = list(mean = 0, sd = sigma_iid ), col = 'blue')

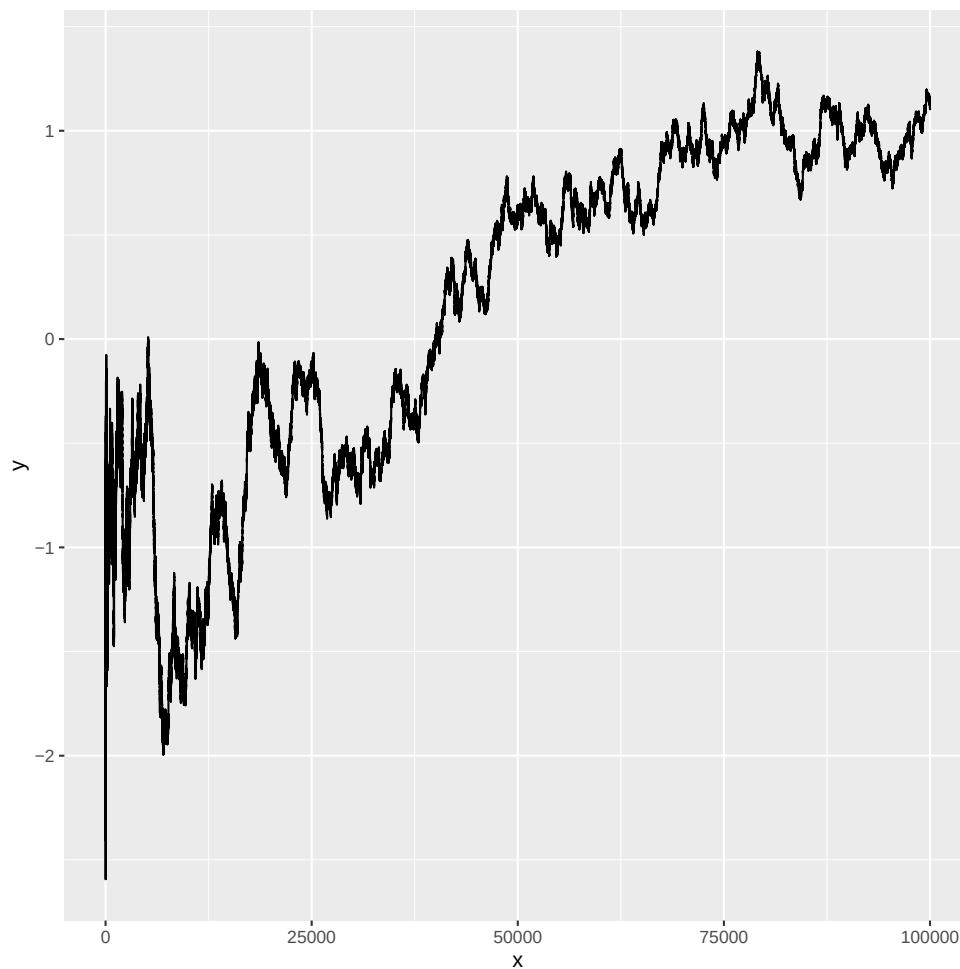
```



```

sample`size` i- 100000
MC i- simulate`markov`chain(sample`size, a, b)
# The exponential distribution has standard deviation 1
# and mean 1
CLT i- sqrt((1:Sample`size`))* ( cumsum(MC)/1:Sample`size` - mu)/sigma`Markov
ggplot( data = tibble(x = 1:Sample`size`, y = CLT )) +
  geom`line`(aes(x,y))

```



Let

$$\begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$$

e:Markoc

be the transition probability of a Markov chain. This means that a and b are the probabilities of transitioning from 0 to 1 and from 1 to 0, respectively. We recall the model (??)

$$\begin{aligned} \mathbb{P}(X_{n+1} = 1 | X_n = 0, X_{n-1} = x_{n-2}, \dots, X_1 = x_1) &= \mathbb{P}(X_{n+1} = 1 | X_n = 0) = a \\ \mathbb{P}(X_{n+1} = 0 | X_n = 1, X_{n-1} = x_{n-2}, \dots, X_1 = x_1) &= \mathbb{P}(X_{n+1} = 0 | X_n = 1) = b \end{aligned}$$

for each $n \in \mathbb{N}$ and $x_{n-1}, \dots, x_1 \in \{0, 1\}$. Given a sequence of 0s and 1s, such as the one coming from the consonants and the vowels in the Eugene Onegin

```
library('rdracor')
ru <- getdracor(corpus = "rus")
draa$name <- ru$playName[10]
text <- gettextchr(spoken(play = draa$name, corpus = "rus"))

## Error in stopifnot(is.character(play) && length(play) == 1): oggetto
'draa$name' non trovato

all$text <- paste(text, collapse = "")
```

```
## Error in paste(text, collapse = ""): cannot coerce type 'closure' to vector
of type 'character'

letters`vector j- unlist(strsplit(all`text, ""))

## Error in strsplit(all_text, ""): oggetto 'all_text' non trovato

spoken`chars j- letters`vector[grepl("[^]", letters`vector)] # Keep only Russian letters

## Error: oggetto 'letters_vector' non trovato

russian`vowels j- strsplit("", "")[[1]]
binary`sequence j- ifelse(spoken`chars %in% russian`vowels, 0, 1)

## Error in spoken_chars %in% russian_vowels: oggetto 'spoken_chars' non
trovato

binary`sequence[1:10]

## Error: oggetto 'binary_sequence' non trovato
```

an estimate for the parameters a and b is given by the functional of the flux (not of the current)

$$\bar{b} = \frac{1}{n} \sum_{i=1}^{n-1} \chi(X_i = 0, X_{i+1} = 1) \bar{a} = \frac{1}{n} \sum_{i=1}^{n-1} \chi(X_i = 1, X_{i+1} = 0)$$

This is for me: The invariant measure of the dynamics depends on the ratio between the transitions from 0 to 1 (b) and from 1 to 0 (a) (the current) $\pi_0 b = \pi_1 a$, so that $(\pi_0, \pi_1) = (a, b)/(a + b)$. In particular, we have two estimates for $p = \pi_1$: the is given by $\bar{b}/\bar{a} + \bar{b}$, and the standard empirical one.

3.15 CENTRAL LIMIT THEOREM FOR THE MARKOV SEQUENCE

$$\bar{b} = \frac{1}{n} \sum_{i=1}^{n-1} \chi(X_i = 0, X_{i+1} = 1) \bar{a} = \frac{1}{n} \sum_{i=1}^{n-1} \chi(X_i = 1, X_{i+1} = 0)$$

$$\begin{pmatrix} -a \\ a \\ b \\ 1-b \end{pmatrix}$$

e:Markoc

Let's compute the variance of $\sum_{i=1}^n \frac{1}{\sqrt{n}} (X_i - \pi_i)$. This is given by

$$\frac{1}{n} \sum_{i,j} E[(X_i - \pi_i)(X_j - \pi_j)]$$

and, assuming we start from the invariant measure, it becomes

$$\text{Var}_\pi(X_0) + \sum_{i=1}^{+\infty} \text{Cov}_\pi(X_i, X_0) = \sigma$$

We need to compute the mean of $P_{i,j}^n f_i f_j$. Since the covariance function that we want to compute is already mean zero, meaning that $f = (-\pi_1, 1 - \pi_1)$ has mean with respect to π given $-\pi_0\pi_1 + \pi_1 - \pi_1^2 = \pi_1(1 - \pi_1 - \pi_2) = 0$ since π is a probability measure. Indeed $f = (-\pi_1, \pi_0)$ and the harmonic $(1, 1)$ function π are an orthonormal basis in $L^2(\pi)$. Therefore

$$\text{Cov}_\pi(X_0, X_n) = \lambda_2^n \langle \cdot - \pi_1, \cdot - \pi_1 \rangle_\pi = \lambda_2^n \text{Var}_\pi(X_0)$$

This gives us that the variance of the variable in the central limit theorem is

$$\text{Var}_\pi(X_0) \frac{1}{1 - \lambda_2}$$

where λ_2 is the second eigenvalue (it is in general like that? no there are the smaller effects of the other eigenvalues) The variance of X_0 is $\pi_0\pi_1^2 + \pi_1(1 - \pi_1)^2 = \pi_1^2 + \pi_1 - 2\pi_1^2 = \pi_1(1 - \pi_1)$ (of course, since a Bernoulli) and $\lambda_2 = 1 - a - b$, so that

$$\sigma^2 = \frac{b/(a+b)a/(a+b)}{a+b} = \frac{ab}{(a+b)^2}$$

which is the sigma for our variable

3.16 EUGENE ONEGIN

3.16.1 Eugene Onegin

We now see that the Markov model We come backk to Example ?? and place it in the statistical setting introduced in Subsection ?? and Section ?. Recall that from the texts of the german drama corpus we have defined a sequence $\underline{X} = X_1 X_2 \dots X_N$ of N digits X_i that are either 1, if the i -th character is a consonant or 0 if it is a vowel, neglecting all the other characters and empty spaces.

The first model H_0 says that H_0 is obtained from the distribution by